Theory and Methods in Causal Inference

Lecture Notes

Jae Kwang Kim

Department of Statistics

Iowa State University

STAT 5900B

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Contents

Preface

These lecture notes accompany the graduate course **STAT 5900B: Theory and Methods in Causal Inference** at the Department of Statistics, Iowa State University.

The course targets graduate students in statistics. It assumes familiarity with probability theory, mathematical statistics, and linear models at the level of a graduate sequence, but no prior exposure to causal inference.

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Organization

The notes are organized in four parts.

Part I — Foundations (Chapters 1–4) builds the graphical and conceptual foundation: the causal trinity (SEM, DAG, potential outcomes), DAGs and d-separation, the do-calculus and identification criteria, and the connection between potential outcomes and the do-calculus.

Part II — Identification (Chapters 5–8) bridges the graphical framework and observed data: randomization and back-door adjustment, propensity score methods, instrumental variables, and mediation and front-door identification.

Part III — Estimation (Chapters 9–12) covers semiparametric efficiency theory: estimating equations and influence functions, doubly robust estimation, flexible nuisance estimation with cross-fitting, and estimation under instrumental variables.

Part IV — Policy Learning and Sequential Decision Making (Chapters 13–15) covers policy evaluation, optimal policy learning, and dynamic treatment regimes.

Notation

Throughout these notes:

- $\mathrm{do}(T = t)$ denotes the do-operator (intervention).
- $Y(t)$ denotes the potential outcome under $T = t$.
- $\mathbb{E}[\,\cdot\,]$ denotes expectation; $\perp\!\!\!\perp$ denotes conditional independence.
- $\mathcal{G}$ denotes a DAG; $\mathcal{G}_{\overline{T}}$ denotes the mutilated graph after intervening on T.

Chapter 1

Introduction

Learning Objectives

By the end of this chapter, students should be able to:

1. Articulate the distinction between an interventional distribution $f(y \mid \mathrm{do}(T{=}t))$ and an observational conditional distribution $f(y \mid T{=}t)$, and explain why they differ whenever T is endogenous.
2. Describe the same causal question in all three languages — SEM, DAG graph surgery, and potential outcomes — and begin to translate between them.
3. Derive the two densities explicitly in the Gaussian linear confounded model, and identify the endogeneity bias $\alpha\gamma\sigma_U^2/\sigma_T^2$ as the gap between them.
4. State the two-step paradigm (identification then estimation) and locate each step in the causal inference pipeline.

This chapter is an orientation, not a complete treatment. It introduces four ideas — the interventional/observational distinction, the causal trinity, the Gaussian confounded model as a running example, and the identification-versus-estimation paradigm — that will be developed rigorously in Chapters 2–4 and beyond.

1.1 Motivation: Why Causal Inference Is Hard

Causal inference studies when and how the observational distribution can be used to recover the distribution that would arise under interventions.

1.1.1 The Central Question

Causal inference is concerned with a deceptively simple question: what would happen to Y if we were to *set* $T = t$? This is fundamentally different from the question that standard statistical procedures answer: what is the distribution of Y among units *observed* to have $T = t$?

Definition: Interventional vs. Observational Distribution

- The *interventional distribution* $f(y \mid \mathrm{do}(T{=}t))$ is the distribution of Y in a hypothetical world where T has been *externally set* to t, regardless of the usual data-generating mechanism for T.
- The *observational conditional distribution* $f(y \mid T{=}t)$ is the distribution of Y among the subpopulation of units for whom $T = t$ was *observed* to hold.

These two distributions coincide only when T is randomized (exogenous). In all other cases they differ, and the difference is *endogeneity bias*.

Definition: Exogeneity and Endogeneity

Suppose the outcome is generated by the structural equation $Y = f_Y(T, X, U)$, where T is the treatment, X are observed covariates, and U represents unobserved common causes of T and Y (confounders).

- The treatment T is *exogenous given* X if $T \perp\!\!\!\perp U \mid X$.
- The treatment T is *endogenous given* X if $T \not\!\perp\!\!\!\perp U \mid X$, so that some unobserved variable U affects both T and Y.

Example: Drug Trial with Unmeasured Severity

Does a new drug ($T = 1$) reduce blood pressure (Y)? Suppose sicker patients are more likely to take the drug, so that unmeasured severity U is a common cause of T and Y. Then:

- *Observational study*: $\mathbb{E}[Y \mid T{=}1] > \mathbb{E}[Y \mid T{=}0]$. Drug users have higher blood pressure on average. A naïve analyst concludes the drug is harmful.
- *Randomized trial*: $\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] < \mathbb{E}[Y \mid \mathrm{do}(T{=}0)]$. Forcing everyone to take the drug reduces blood pressure. The drug is beneficial.

Same data. Different questions. Opposite answers. The entire discrepancy is caused by the back-door path $T \leftarrow U \rightarrow Y$, which the observational quantity $\mathbb{E}[Y \mid T{=}1]$ cannot remove.

The Fundamental Problem of Causal Inference

We observe $f(y \mid T{=}t)$ directly from data. We want $f(y \mid \mathrm{do}(T{=}t))$. The goal of *identification theory* is to find conditions under which the former can be used to recover the latter.

1.2 The Causal Trinity: Three Languages for One Idea

The same causal question — what happens when we set $T = t$? — can be expressed in three distinct but equivalent languages. Collectively, we call these the **causal trinity**: the *structural equation model* (SEM), the *directed acyclic graph* (DAG) with graph surgery, and the *potential outcomes* framework. Figure 1.1 depicts their relationships.

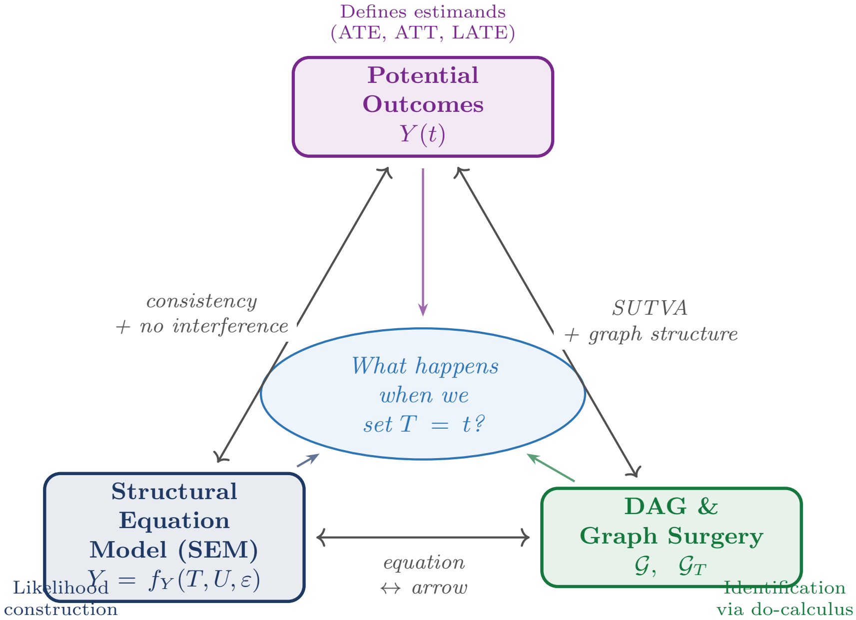


Figure 1.1: The **causal trinity**: three languages that express the same causal question from different vantage points. Potential outcomes define *what* we want to estimate; the DAG with graph surgery specifies *how* to identify it; the SEM provides the *generative mechanism* for likelihood construction.

1.2.1 Language 1: Structural Equation Models

Definition: Structural Equation Model (SEM)

A recursive SEM over variables (U, T, Y) is a system

$$U \sim p(u), \qquad T = f_T(U, \delta), \qquad Y = f_Y(T, U, \varepsilon),$$

where δ and ε are mutually independent exogenous disturbances and U is a confounding factor.

Remark: Error Independence in the NPSEM-IE

The assumption that δ and ε are mutually independent is a substantive restriction. Because U is unobserved and enters both f_T and f_Y, error independence does not remove the dependence between T and $Y(t)$ induced by U; it restricts only the additional channel of dependence running through the equation-specific errors. Richardson and Robins (2014) call SEMs with this structure *Nonparametric Structural Equation Models with Independent Errors* (NPSEM-IEs).
These notes adopt the NPSEM-IE framework throughout: error independence ensures the DAG's Markov condition holds, making the Markov factorization of the joint density immediate.

Observational regime. T is determined by f_T; because U enters both f_T and f_Y, we have $\mathrm{Cov}(T, \varepsilon) \neq 0$ in general: this is endogeneity.

Interventional regime. The do-operator replaces the equation for T with a constant t, yielding the *mutilated system*:

$$U \sim p(u), \qquad T = t \text{ (fixed)}, \qquad Y = f_Y(t, U, \varepsilon).$$

We define the *potential outcome* as $Y(t) := f_Y(t, U, \varepsilon)$; by construction, $Y(t) \sim f(y \mid \mathrm{do}(T{=}t))$.

1.2.2 Language 2: Directed Acyclic Graphs

Definition: Graph Surgery

In the observational DAG, arrows into T encode the mechanisms that determine T. Under $\mathrm{do}(T{=}t)$, all arrows into T are *deleted*. The resulting mutilated graph $\mathcal{G}_{\overline{T}}$ has no back-door paths from T to Y: the causal effect flows only along the direct path $T \to Y$.

1.2.3 Language 3: Potential Outcomes

Definition: Potential Outcome

(Rubin 1974) In the SEM framework, $Y(t) = f_Y(t, U, \varepsilon)$ is the solution for Y in the mutilated system; it is the outcome that *would have been observed* had T been set to t, possibly contrary to fact.
SUTVA (stable unit treatment value assumption): $Y_i = Y_i(T_i)$, i.e., the observed outcome for unit i equals the potential outcome at the actual treatment received. This rules out interference between units and multiple versions of treatment.

1.2.4 Roadmap: How the Trinity Organizes These Notes

Part I — Foundations: the language of DAGs. DAGs and the do-operator are the primary language here because they make the interventional/observational distinction *syntactically explicit*: confounding is visible as an open back-door path, and identifying assumptions are visible as graph criteria.

Part II — Identification: bridging DAGs and observed data. Ignorability $(Y(t) \perp\!\!\!\perp T \mid X)$ is the potential-outcomes counterpart of the back-door criterion; the propensity score $\pi(X) = P(T{=}1 \mid X)$ operationalizes it in observed data.

Part III — Estimation: semiparametric and machine-learning methods. The semiparametric efficiency framework — efficient influence functions, doubly robust estimators, double machine learning — is largely language-neutral.

1.2.5 Historical Roots of the Causal Trinity

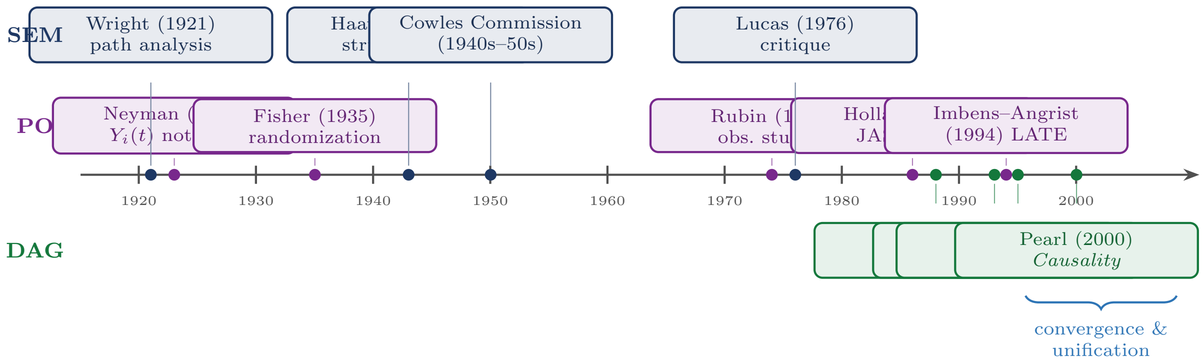


Figure 1.2: Key milestones in the three schools of causal inference. SEM (top, blue) originates in genetics and econometrics; PO (middle, purple) in experimental statistics; DAG (bottom, green) in computer science and philosophy.

1.3 The Core Distinction

We work within the SEM, now augmented with an observed exogenous variable X:

$$T = f_T(X, U, \delta), \qquad Y = f_Y(T, X, U, \varepsilon). \tag{1.1}$$

The single most important inequality of this course is:

$$\underbrace{f(y \mid x,\, \mathrm{do}(T{=}t))}_{\text{interventional}} \neq \underbrace{f(y \mid x,\, T{=}t)}_{\text{observational}} \qquad \text{whenever } T \text{ is endogenous.} \tag{1.2}$$

Definition: Conditional Exchangeability

Given observed covariates X and confounders U, the treatment assignment is *conditionally exchangeable with the intervention* if $Y(t) \perp\!\!\!\perp T \mid X, U$, or equivalently,

$$f(y \mid x,\, T{=}t,\, U{=}u) \;=\; f(y \mid x,\, \mathrm{do}(T{=}t),\, U{=}u).$$

This says that once we condition on all common causes (X, U) of T and Y, observing $T{=}t$ and intervening to set $T{=}t$ produce the same distribution of Y.

Definition: Positivity

The treatment assignment satisfies *positivity* if, for all t in the support of T and almost all (x, u):

$$p(T{=}t \mid X{=}x,\, U{=}u) \;>\; 0.$$

Proposition: Back-Door Adjustment Formula

Under conditional exchangeability and positivity,

$$f(y \mid x,\, \mathrm{do}(T{=}t)) \;=\; \int f(y \mid x,\, T{=}t,\, U{=}u)\; p(u \mid x)\; du.$$

Proof

By positivity, $f(y \mid x,\, T{=}t,\, U{=}u)$ is well-defined. By the law of total probability:

$$f(y \mid x,\, \mathrm{do}(T{=}t)) = \int f(y \mid x,\, U{=}u,\, \mathrm{do}(T{=}t))\, p(u \mid x)\, du.$$

The weight is $p(u \mid x)$, not $p(u \mid x, T{=}t)$, because under $\mathrm{do}(T{=}t)$ the treatment carries no information about U. By conditional exchangeability, $f(y \mid x,\, U{=}u,\, \mathrm{do}(T{=}t)) = f(y \mid x,\, T{=}t,\, U{=}u)$. $\square$

Remark: Confounding Discrepancy

What OLS estimates is the observational density, not the interventional density:

$$f(y \mid x,\, T{=}t) = \int f(y \mid x,\, T{=}t,\, U{=}u)\, p(u \mid x,\, T{=}t)\, du.$$

The kernel is the same as in Equation 1.3, but averaged over the *selection-distorted* weight $p(u \mid x, T{=}t)$. Because U and T are dependent through the back-door path, $p(u \mid x, T{=}t) \neq p(u \mid x)$, and the gap is non-zero whenever T is endogenous.

1.3.1 Two Scenarios: Observed vs. Unobserved Confounder

	Scenario 1: U observed	**Scenario 2: U unobserved**
Confounding type	No unmeasured confounding	Unmeasured confounding
Back-door formula usable?	Yes	No: U latent
Identification route	Back-door adjustment / standardization	IV, front-door, …
Key assumption	Unconfoundedness: $U \perp\!\!\!\perp T \mid X$	Exclusion restriction, monotonicity, …
Chapters in this course	Chapters 2–6	Chapter 7

Scenario 1: U is observed. The back-door adjustment formula Equation 1.3 is usable from data:

$$f(y \mid x,\, \mathrm{do}(T{=}t)) = \int f(y \mid x,\, T{=}t,\, U{=}u)\, p(u \mid x)\, du. \tag{1.3}$$

A special case is *unconfoundedness*: if $U \perp\!\!\!\perp T \mid X$ already holds observationally, Equation 1.3 reduces to the familiar regression $\mathbb{E}[Y \mid X{=}x, T{=}t]$.

Scenario 2: U is unobserved. Equation 1.3 is not usable. Additional structure beyond (Y, T, X) must be imposed: **instrumental variables** (Chapter 7), the **front-door criterion** (Chapter 3), or **sensitivity analysis**.

1.4 The Gaussian Linear Confounded Model

1.4.1 Setup

Example: Gaussian Linear Confounded Model

The structural equations are

$$Y = \beta T + \gamma U + \varepsilon, \qquad T = \alpha U + \delta, \tag{1.4}$$

where U is an unobserved confounder and the three primitive variables are mutually independent: $U \sim \mathcal{N}(0, \sigma_U^2)$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, $\delta \sim \mathcal{N}(0, \sigma_\delta^2)$.
Here β is the causal effect of T on Y, γ is the direct effect of U on Y, and α is the direct effect of U

on T.

The endogeneity of T is immediate: $\mathrm{Cov}(T, \gamma U + \varepsilon) = \gamma\alpha\sigma_U^2 \neq 0$ when $\alpha \neq 0$ and $\gamma \neq 0$. Consequently, OLS regressing Y on T does not estimate β.

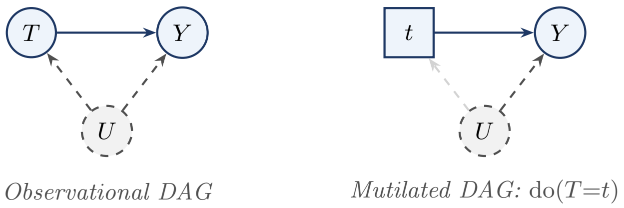


Figure 1.3: Graph surgery on the Gaussian confounded model. The faded arrow in the mutilated DAG has been severed by the intervention $\mathrm{do}(T{=}t)$. The back-door path $T \leftarrow U \rightarrow Y$ is eliminated; only the causal path $T \rightarrow Y$ remains active.

1.4.2 Two Explicit Densities

Write $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$ for the marginal variance of T.

Interventional density. Set $T = t$ externally. U is drawn from its marginal $\mathcal{N}(0, \sigma_U^2)$ independently. Hence

$$f(y \mid \mathrm{do}(T{=}t)) = \mathcal{N}\left(\beta t,\ \ \gamma^2\sigma_U^2 + \sigma_\varepsilon^2\right). \tag{1.5}$$

Observational density. Since (Y, T) is jointly normal with $\mathrm{Cov}(Y, T) = \beta\sigma_T^2 + \gamma\alpha\sigma_U^2$, the conditional normal formula gives:

$$f(y \mid T{=}t) = \mathcal{N}\left(\left(\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}\right) t,\ \ \frac{\gamma^2\sigma_U^2\sigma_\delta^2}{\sigma_T^2} + \sigma_\varepsilon^2\right). \tag{1.6}$$

1.4.3 The Endogeneity Gap

	Interventional	**Observational**
Mean	βt	$\left(\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}\right) t$
Variance	$\gamma^2\sigma_U^2 + \sigma_\varepsilon^2$	$\frac{\gamma^2\sigma_U^2\sigma_\delta^2}{\sigma_T^2} + \sigma_\varepsilon^2$
Residual	$(\gamma U{+}\varepsilon) \perp\!\!\!\perp T$	$(\gamma U{+}\varepsilon)$ correlated with T
Identified by	RCT (or IV, Chapter 7)	OLS

The *endogeneity bias* of OLS is $\gamma\alpha\sigma_U^2/\sigma_T^2$. Numerically, with $\alpha = \gamma = 1$ and $\sigma_U^2 = \sigma_\delta^2 = 1$: bias $= 1/2 = 0.5$.

1.4.4 Lab: Simulating the Two Densities

The analytical results were verified by simulation using $n = 10{,}000$ draws with parameters $\beta = 2$, $\alpha = 1$, $\gamma = 1$, $\sigma_U = \sigma_\varepsilon = \sigma_\delta = 1$.

Experiment 1: OLS bias.

	Simulated	**Theory**	**Formula**
OLS slope	2.5147	2.5000	$\beta + \gamma\alpha\sigma_U^2/\sigma_T^2$
True β	—	2.0000	β
Bias	0.5147	0.5000	$\gamma\alpha\sigma_U^2/\sigma_T^2$

Experiment 2: Two-sample comparison at $t_0 = 1$.

	Mean (Simulated)	Mean (Theory)	SD (Simulated)	SD (Theory)
$\mathbb{E}[Y \mid T{=}1]$	2.497	2.500	1.229	$\sqrt{1.5} = 1.225$
$\mathbb{E}[Y \mid \mathrm{do}(T{=}1)]$	2.001	2.000	1.428	$\sqrt{2} = 1.414$

Experiment 3: Density overlay for varying α. Figure 1.4 shows the interventional density (solid) and the observational density Equation 1.6 (dashed), both evaluated at $t_0 = 1$, for $\alpha \in \{0, 0.3, 0.7, 1.0\}$.

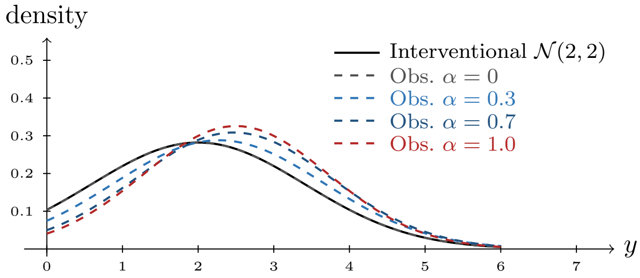


Figure 1.4: Interventional density $\mathcal{N}(2, 2)$ (solid black) versus observational density for $\alpha \in \{0, 0.3, 0.7, 1.0\}$ (dashed), evaluated at $t_0 = 1$. As α increases, the observational curve shifts rightward (growing bias) and narrows (reduced variance).

1.5 The Two-Step Paradigm

Causal inference is a two-step discipline. The two steps are logically distinct and require different tools.

1.5.1 Step 1: Identification

Definition: Identification

A causal quantity $P(y \mid \mathrm{do}(T{=}t))$ is *identified* from the observational distribution P if there exists a functional Ψ such that $P(y \mid \mathrm{do}(T{=}t)) = \Psi(P)$, and $\Psi(P)$ depends only on the observable joint distribution.

Identification is a purely mathematical question about the structure of the causal model: given the assumed graph and assumptions, can the interventional density be expressed as a function of the observed-data distribution? It does not depend on sample size.

The main tools are the back-door criterion (Chapters 2–3, applied in Chapters 5–6), the front-door criterion (Chapter 3), the three rules of the do-calculus (Chapter 3), and IV assumptions (Chapter 7).

1.5.2 Step 2: Estimation

Once $\Psi(P)$ has been identified, the statistical problem is to estimate $\Psi(P)$ from n observations $O_1, \ldots, O_n$ as efficiently as possible. The main tools are efficient influence functions (EIF), inverse probability weighting (IPW), doubly robust estimators, and double machine learning (DML), all covered in Part III.

1.5.3 A Worked Example: Flu Vaccination and Infection

Scenario. A public health agency observes 1,000 individuals over one flu season. Each person either received a flu vaccine ($T = 1$) or did not ($T = 0$), and either became infected ($Y = 1$) or did not ($Y = 0$). Age group X (elderly $= E$, young $= Y$) is recorded for all individuals. Elderly people are more likely to be vaccinated and more susceptible to infection, so X confounds the relationship between T and Y.

Observed data.

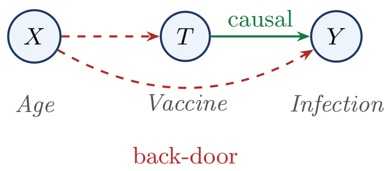


Figure 1.5: Causal DAG for the vaccination example. The green arrow is the causal path $T \to Y$. The red dashed arrows form the back-door path $T \leftarrow X \to Y$.

	Vaccinated ($T = 1$)		**Unvaccinated** ($T = 0$)		Total
	n	Infected	n	Infected	
Elderly ($X = E$)	360	108 (30%)	40	20 (50%)	400
Young ($X = Y$)	60	3 (5%)	540	54 (10%)	600
Total	420	111 (26.4%)	580	74 (12.8%)	1,000

Simpson's paradox.

Stratum	**Vaccinated**	**Unvaccinated**	**Difference**	**Conclusion**
Elderly ($X = E$)	30%	50%	−20 pp	vaccine helps
Young ($X = Y$)	5%	10%	−5 pp	vaccine helps
Aggregate (raw)	**26.4%**	**12.8%**	+13.6 pp	**vaccine harms??**

Within every stratum the vaccine reduces infection. The reversal occurs because the vaccinated group is disproportionately elderly (86% vs. 7%).

Step 1: Identification. The set $\{X\}$ satisfies the back-door criterion. By the back-door formula:

$$P(Y{=}1 \mid \mathrm{do}(T{=}t)) = \sum_{x \in \{E,Y\}} P(Y{=}1 \mid T{=}t, X{=}x)\, P(X{=}x). \tag{1.7}$$

Step 2: Estimation.

$$\hat{P}(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.30 \times 0.4 + 0.05 \times 0.6 = 0.15,$$

$$\hat{P}(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.50 \times 0.4 + 0.10 \times 0.6 = 0.26.$$

The estimated causal risk difference is $\hat{\Delta}_{\text{causal}} = 0.15 - 0.26 = -0.11$: the vaccine causally reduces infection probability by 11 percentage points.

> The Two Steps, Concretely
>
> **Step 1 (identification)** established — from the DAG alone — that the back-door formula Equation 5.3 expresses the causal quantity as a functional of observable data. **Step 2 (estimation)** replaced the population probabilities with sample proportions.
> These two steps are logically separate: Step 1 is a mathematical argument about the causal model that does not depend on sample size; Step 2 is a statistical argument about finite-sample estimation that does not depend on the causal model.

1.6 Summary

1. **Interventional vs. observational distribution.** Causal inference answers questions about interventions $f(y \mid \mathrm{do}(T{=}t))$, not about observations $f(y \mid T{=}t)$. These two distributions coincide only under exogeneity; whenever T is endogenous they differ, and the difference is endogeneity bias.

2. **The causal trinity.** The same causal question can be expressed in three languages — SEM (equation surgery), DAG (graph surgery), and potential outcomes ($Y(t)$). In the SEM framework, $Y(t)$ is defined as the solution for Y in the mutilated system, so $Y(t) \sim f(y \mid \mathrm{do}(T{=}t))$ by construction. Fluency in all three is essential for modern causal reasoning.

3. **Endogeneity bias in the Gaussian confounded model.** In the Gaussian linear confounded model, the coefficient of T in the observational mean $\mathbb{E}[Y \mid T{=}t]$ differs from the causal coefficient β by $\gamma\alpha\sigma_U^2/\sigma_T^2$, where $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$. This is the omitted-variable bias.

4. **Two-step paradigm.** Causal inference is a two-step discipline: *identification* (expressing $\Psi(P)$ as a functional of observable data, a mathematical question about the causal model) followed by *estimation* (constructing $\hat{\Psi}_n$ efficiently from n observations, a statistical question about finite samples). The two steps are logically independent.

Causal inference is the study of when the observational distribution contains enough information to recover the interventional distribution. Everything in this course — the back-door criterion, do-calculus, instrumental variables, doubly robust estimation — is an answer to that question in a particular setting.

1.7 Problems

1. Do-notation fundamentals. For each statement below, decide whether it refers to an interventional or an observational distribution, and rewrite it unambiguously using do-notation where appropriate.

(a) "The probability that a patient recovers given that they took the drug."
(b) "The probability that a patient would recover if we prescribed the drug to everyone."
(c) "Among students who attended tutoring, the average exam score was 85."
(d) "If we enrolled all students in tutoring, the average exam score would be 85."

2. Deriving the observational density. In the Gaussian confounded model, verify Equation 1.6 by carrying out the following steps. Let $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$.

(a) Show that (Y, T) is jointly normal by writing both as linear combinations of the independent normals (U, ε, δ).
(b) Compute $\mathrm{Cov}(Y, T)$ and $\mathrm{Var}(T)$.
(c) Apply the conditional normal formula to derive $\mathbb{E}[Y \mid T{=}t]$ and $\mathrm{Var}[Y \mid T{=}t]$, and hence verify Equation 1.6.
(d) Show that the OLS probability limit is $\beta + \gamma\alpha\sigma_U^2/\sigma_T^2$, and interpret this as an omitted-variable bias formula.

3. The causal trinity. Consider the following verbal causal claim: "Aspirin (T) reduces the risk of heart attack (Y) because it inhibits platelet aggregation (M), but patients with pre-existing cardiovascular disease (U, unobserved) are both more likely to take aspirin and more likely to have a heart attack."

(a) Draw the causal DAG implied by this description.
(b) Write down the recursive SEM with appropriate structural functions f_T, f_M, f_Y.
(c) State the SUTVA assumption and discuss whether it is plausible in this setting.
(d) Explain why $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$ does not identify the causal effect $\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0)]$ in this DAG.

Chapter 2

DAGs and d-Separation

Learning Objectives

By the end of this chapter, students should be able to:

1. Construct a DAG from a verbal description of a causal model and identify parents, descendants, ancestors, and non-descendants of any node.
2. Classify every intermediate node on a path as a chain, fork, or collider, and apply the blocking rule for each.
3. Apply the d-separation criterion to determine all conditional independence relationships implied by a DAG.
4. Recognize and avoid collider bias, including the case where a descendant of a collider is conditioned upon.
5. State the Markov factorization and use it to write the joint density of a DAG in terms of local conditional densities.
6. Interpret d-separation results as conditional independence assumptions on observed variables, and recognize their role as the graphical expression of causal assumptions.

Readers who want a gentler introduction to conditional independence, the three basic path motifs, and the Markov-property interpretation may consult Appendix A before or alongside this chapter.

2.1 Directed Acyclic Graphs

DAGs allow us to translate qualitative causal assumptions into quantitative statistical restrictions. This single idea underlies everything in this chapter: once we draw a graph encoding what we believe about the causal structure of a problem, the graph tells us exactly which variables are independent, which are confounded, and what we need to condition on to identify a causal effect.

Remark

In this chapter, the word *graph* does not mean a plot of data or a graph of a function. It means a collection of nodes and edges used to represent relationships among variables. Our objective is to understand how such graphs encode statistical structure — especially conditional independence, confounding, and path blocking — and how this graphical structure will later support causal identification.

2.1.1 Basic Definition

Definition: Directed Acyclic Graph

A *directed acyclic graph* (DAG) $\mathcal{G} = (V, E)$ consists of a finite set of nodes V and a set of directed edges $E \subseteq V \times V$ such that there is no directed cycle: no node V_i has a directed path to itself.

In a causal DAG each node represents a variable and each directed edge $A \to B$ encodes "A is a direct cause of B, given all other variables in the model." The acyclicity condition is essential: it guarantees that the variables can be ordered topologically so that the joint density admits the recursive factorization

$$p(v_1, \dots, v_k) = \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i)).$$

2.1.2 Structural Relationships

Definition: Structural Relationships in a DAG

For a node $v \in \mathcal{V}$ in $\mathcal{G} = (\mathcal{V}, E)$:

- $\mathrm{Pa}(v) = \{w \in \mathcal{V} : w \to v \in E\}$ are the *parents* of v.
- $\mathrm{De}(v)$ = all nodes reachable from v by directed paths are the *descendants* of v.
- $\mathrm{An}(v)$ = all nodes with a directed path to v are the *ancestors* of v.
- $\mathrm{Nd}(v) = \mathcal{V} \setminus (\mathrm{De}(v) \cup \{v\})$ are the *non-descendants* of v.

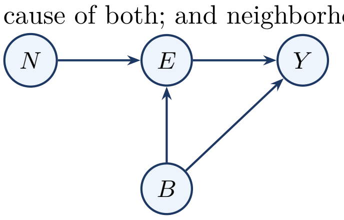
Note that $\mathrm{Nd}(v)$ includes the parents of v; when stating the local Markov property in Section A.4, we will exclude the parents explicitly.

Quick Terminology Summary

A *node* represents a variable, and an *edge* represents a relationship between two variables. If $A \to B$, then A is a *parent* of B and B is a *child* of A. Two nodes are *adjacent* if they are connected by an edge. A *path* is any sequence of connected nodes, regardless of arrow direction; a *directed path* is a path whose arrows all point in the same direction. A *directed acyclic graph* (DAG) is a directed graph with no directed cycles.

Example: Education, Earnings, and Family Background

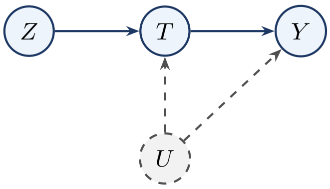
Consider the causal story: education (E) affects earnings (Y); family background (B) is a common cause of both; and neighborhood (N) affects education but has no direct effect on earnings.



The Markov factorization is $p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b)$. Reading off structural relationships: $\mathrm{Pa}(E) = \{N, B\}$, $\mathrm{Pa}(Y) = \{E, B\}$, $\mathrm{Pa}(N) = \mathrm{Pa}(B) = \emptyset$. The descendants of N are $\{E, Y\}$; B and N are non-descendants of each other.

Example: The IV DAG

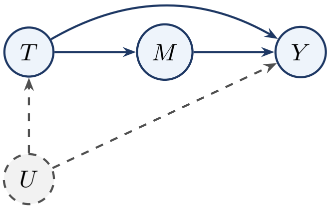
The canonical instrumental variable model involves four nodes: Z (instrument), T (treatment), Y (outcome), and U (unobserved confounder).



$\text{Pa}(T) = \{Z, U\}$, $\text{Pa}(Y) = \{T, U\}$, $\text{Pa}(Z) = \text{Pa}(U) = \emptyset$. The descendants of Z are $\{T, Y\}$; U and Z are non-descendants of each other.

Example: The Mediation DAG

In a mediation model, treatment T affects outcome Y both directly and through an intermediate variable M (the mediator). An unobserved confounder U creates a back-door path $T \leftarrow U \rightarrow Y$, making T endogenous. The mediator M is itself unconfounded.



$\text{Pa}(T) = \{U\}$, $\text{Pa}(M) = \{T\}$, $\text{Pa}(Y) = \{M, T, U\}$, $\text{Pa}(U) = \emptyset$. The total effect of T on Y travels along two paths: the direct path $T \rightarrow Y$ and the indirect path $T \rightarrow M \rightarrow Y$. Because there is a direct path $T \rightarrow Y$ bypassing M, the front-door criterion does not apply here.

2.2 Paths, Blocking, and d-Separation

A DAG encodes direct causal relationships through its edges, but variables can also be statistically related through longer routes that mix directed and reversed edges. The concept of a *path* formalises these routes, and the question of whether a path transmits or blocks statistical dependence — depending on which variables are conditioned upon — is precisely what d-separation answers. The key insight is that conditioning can both *close* channels that were open (by blocking a chain or fork) and *open* channels that were closed (by activating a collider). Figure 2.1 illustrates all three cases.

2.2.1 Path Structures

Definition: Path

A *path* between nodes X and Y in $\mathcal{G}$ is any sequence of distinct nodes $X = V_0, V_1, \ldots, V_k = Y$ such that each consecutive pair is connected by an edge in either direction.

Every intermediate node V_m on a path plays one of three structural roles. In a **chain** ($V_{m-1} \rightarrow V_m \rightarrow V_{m+1}$), V_m is a causal intermediary through which influence is transmitted. In a **fork** ($V_{m-1} \leftarrow V_m \rightarrow V_{m+1}$), V_m is a common cause that creates confounding between the two endpoints. In a **collider** ($V_{m-1} \rightarrow V_m \leftarrow V_{m+1}$), both arrows point *into* V_m; this is the crucial asymmetric case whose behavior under conditioning is the opposite of the other two.

2.2.2 Blocking Rules

Structure	Pattern	Effect of conditioning	Key intuition
Chain	$A \rightarrow M \rightarrow B$	Blocks the path	Causal transmission
Fork	$A \leftarrow M \rightarrow B$	Blocks the path	Common cause
Collider	$A \rightarrow M \leftarrow B$	**Opens** the path	Common effect

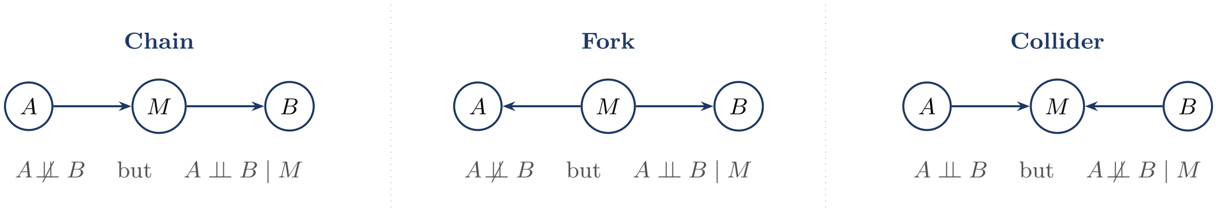


Figure 2.1: The three fundamental path structures in a DAG. Chains and forks are *open by default* and are blocked by conditioning on M. Colliders are *closed by default* and are opened by conditioning on M (or any descendant of M).

The critical asymmetry — colliders are *closed by default* and *opened* by conditioning, while chains and forks are *open by default* and *closed* by conditioning — is what makes collider bias so easy to overlook in practice.

Chain: Smoking → Tar → Cancer. Let A = smoking, M = tar deposits, B = lung cancer. Marginally, heavy smokers have elevated cancer rates — the path is open. Conditioning on tar level blocks the path: once tar level is fixed, the causal channel is fully accounted for.

Fork: Poverty → Poor Diet; Poverty → Lack of Exercise. Let M = poverty, A = poor diet, B = lack of exercise. Unconditionally, diet quality and exercise are correlated through poverty. Conditioning on income level removes this spurious association.

Collider: Accident → Hospitalization ← Cancer. Let A = traffic accident, B = cancer, M = hospitalization. In the general population, $A \perp\!\!\!\perp B$. But among hospitalized patients — conditioning on M — the two become negatively associated. This is Berkson's bias (Berkson 1946).

> Three-Node Motifs at a Glance
>
> In a *chain* $A \to M \to B$, conditioning on M blocks the path. In a *fork* $A \leftarrow M \to B$, conditioning on M removes the spurious association. In a *collider* $A \to M \leftarrow B$, the path is blocked by default, but conditioning on M — or on any descendant of M — opens the path and induces association. These three motifs are the local building blocks of d-separation in arbitrary DAGs.

2.2.3 The d-Separation Criterion

> Definition: d-Blocking and d-Separation
>
> A path π is *d-blocked* by a set $\mathbf{S}$ if:
> 1. π contains a chain $A \to M \to B$ or fork $A \leftarrow M \to B$ with $M \in \mathbf{S}$; or
> 2. π contains a collider $A \to C \leftarrow B$ with $C \notin \mathbf{S}$ and no descendant of C is in $\mathbf{S}$.
>
> Nodes X and Y are *d-separated* by $\mathbf{S}$, written $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$, if every path between X and Y in $\mathcal{G}$ is d-blocked by $\mathbf{S}$.

> Example: Applying the d-Separation Definition to the Three Toy Settings
>
> **(i) Chain.** A = smoking, M = tar, B = cancer. Only path: $A \to M \to B$.
> - $\mathbf{S} = \emptyset$: not d-blocked. $\Rightarrow A \not\perp\!\!\!\perp B$.
> - $\mathbf{S} = \{M\}$: clause (1) applies. $\Rightarrow A \perp\!\!\!\perp B \mid M$.
>
> **(ii) Fork.** M = poverty, A = poor diet, B = lack of exercise. Only path: $A \leftarrow M \to B$.
> - $\mathbf{S} = \emptyset$: not d-blocked. $\Rightarrow A \not\perp\!\!\!\perp B$.
> - $\mathbf{S} = \{M\}$: clause (1) applies. $\Rightarrow A \perp\!\!\!\perp B \mid M$.
>
> **(iii) Collider.** A = accident, M = hospitalization, B = cancer. Only path: $A \to M \leftarrow B$.
> - $\mathbf{S} = \emptyset$: collider $M \notin \mathbf{S}$, clause (2) applies. $\Rightarrow A \perp\!\!\!\perp B$.
> - $\mathbf{S} = \{M\}$: clause (2) no longer blocks. $\Rightarrow A \not\perp\!\!\!\perp B \mid M$ (Berkson's bias).

Example: A Direct Probability Proof for the Chain

Consider the chain $X_1 \to X_2 \to X_3$. By the Markov factorization, $p(x_1, x_2, x_3) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)$. Conditioning on $X_2 = x_2$:

$$p(x_1, x_3 \mid x_2) = \frac{p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)}{p(x_2)} = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Hence $X_1 \perp\!\!\!\perp X_3 \mid X_2$. The fork case follows by an identical calculation.

Theorem: Verma & Pearl (1988)

(Pearl 2009, Theorem 1.2.4) If $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$, then $X \perp\!\!\!\perp Y \mid \mathbf{S}$ in every distribution that is Markov with respect to $\mathcal{G}$.

Remark: Soundness, Not Converse

Theorem 2.1 (Verma & Pearl) states that d-separation implies conditional independence for every distribution Markov with respect to $\mathcal{G}$. The converse need not hold without additional assumptions such as *faithfulness*.

Table 2.1 summarizes the Markov factorization and its independence implication for each of the three path structures.

Table 2.1: Markov factorization and d-separation result for each path structure.

Structure	**Markov factorization** $p(a, m, b)$	**d-separation**
Chain $A \to M \to B$	$p(a)\, p(m \mid a)\, p(b \mid m)$	$(A \perp\!\!\!\perp B \mid M)_{\mathcal{G}}$
Fork $A \leftarrow M \to B$	$p(m)\, p(a \mid m)\, p(b \mid m)$	$(A \perp\!\!\!\perp B \mid M)_{\mathcal{G}}$
Collider $A \to M \leftarrow B$	$p(a)\, p(b)\, p(m \mid a, b)$	$(A \perp\!\!\!\perp B)_{\mathcal{G}}$

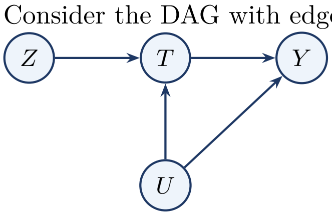
2.2.4 Practical Ways to Check d-Separation

Bayes-Ball Intuition. Imagine releasing a ball from X asking whether it can reach Y, given that nodes in $\mathbf{S}$ are observed. At a chain or fork node not in $\mathbf{S}$: ball passes through. At a chain or fork node in $\mathbf{S}$: ball stops. At a collider not in $\mathbf{S}$ (and no descendant in $\mathbf{S}$): ball stops. At a collider in $\mathbf{S}$ (or with a descendant in $\mathbf{S}$): ball passes through (Shachter 1998).

Moral Graph Transformation. (1) Restrict to the ancestral set $\mathrm{An}(X \cup Y \cup \mathbf{S})$. (2) Moralise: for every collider $A \to C \leftarrow B$, add an undirected edge $A - B$. (3) Remove all edge directions. (4) Delete all nodes in $\mathbf{S}$. If X and Y are disconnected, then $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$ (Lauritzen 1996, Ch. 3).

Example: Checking d-Separation in a Four-Node DAG

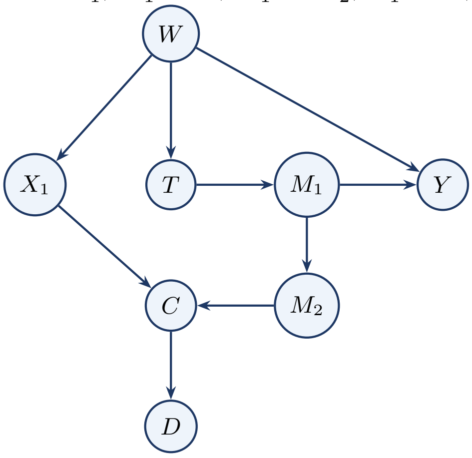
Consider the DAG with edges $Z \to T$, $U \to T$, $T \to Y$, $U \to Y$.



Query 1. Is $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$? The only path $Z \to T \leftarrow U$ has T as a collider with $T \notin \emptyset$ — blocked. $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$.
Query 2. Is $(Z \perp\!\!\!\perp U \mid T)_{\mathcal{G}}$? Now $T \in \mathbf{S}$: the collider is observed, the path is open. $Z \not\perp\!\!\!\perp U \mid T$.
Comparing the two queries: Z and U are marginally independent but become dependent once we condition on T. This is Berkson's bias.

Example: Practice DAG — Eight-Node Graph

Consider the DAG with nodes W, X_1, T, M_1, M_2, Y, C, D and edges $W \to X_1$, $W \to T$, $W \to Y$, $T \to M_1$, $M_1 \to Y$, $M_1 \to M_2$, $X_1 \to C$, $M_2 \to C$, $C \to D$.



(1) Is $X_1 \perp\!\!\!\perp Y$? No. Two open paths: $X_1 \leftarrow W \to Y$ and $X_1 \leftarrow W \to T \to M_1 \to Y$.
(2) Is $X_1 \perp\!\!\!\perp Y \mid W$? Yes. Both open paths pass through the fork W; conditioning on W blocks them.
(3) Is $T \perp\!\!\!\perp M_2 \mid M_1$? Yes. The path $T \to M_1 \to M_2$ is blocked by M_1; all other paths are also blocked.
(4) Does conditioning on C create collider bias? Yes. C is a collider on $X_1 \to C \leftarrow M_2$. Conditioning on C opens this path, inducing a spurious association between X_1 and M_2.
(5) Does conditioning on D open the path $X_1 \to C \leftarrow M_2$? Yes. D is a descendant of C. By clause (2) of Definition 2.1, the collider at C is activated.

2.3 Collider Bias and the IV DAG

2.3.1 Berkson's Bias

Definition: Collider Bias

Collider bias is the spurious association induced between two variables when one conditions on their common effect (a collider), or on a descendant of that collider. In a path $A \to C \leftarrow B$, the variables A and B may be marginally independent, but conditioning on C — or on any descendant of C — can make them statistically dependent.

Collider Bias vs. Confounding

Unlike confounding, which is *removed* by conditioning on a common cause, collider bias is *created* by conditioning on a common effect. This is why conditioning on more variables is not always safer in causal analysis: adding a collider or a descendant of a collider to the adjustment set introduces bias rather than reducing it.

Example: Collider Bias in the IV DAG

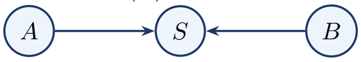
In the IV DAG, T is a collider on the path $Z \to T \leftarrow U$. Marginally, $Z \perp\!\!\!\perp U$ (the instrument is exogenous). However,

$$Z \perp\!\!\!\perp U \qquad \text{but} \qquad Z \not\!\perp\!\!\!\perp U \mid T.$$

Conditioning on T — restricting to the treated subpopulation — makes the instrument Z appear associated with U, invalidating the IV argument within that stratum.

Example: Collider Bias through Selection: Talent and Wealth

Suppose both talent (A) and wealth (B) increase the probability of admission to an elite school, so admission (S) is a collider: $A \to S \leftarrow B$.



In the general population, $A \perp\!\!\!\perp B$. But among admitted students — conditioning on S — lower talent makes higher wealth more likely, and vice versa. This is collider bias: restricting to admitted students is precisely conditioning on a common effect.

2.3.2 Full d-Separation Analysis of the IV DAG

We work through the IV DAG with edges $Z \to T$, $T \to Y$, $U \to T$, $U \to Y$, where U is unobserved.

Endpoints	**Path**	**Type**	**Why**
$Z \leftrightarrow Y$	$Z \to T \to Y$	*Causal*	Directed from Z to Y; carries the IV signal
$Z \leftrightarrow Y$	$Z \to T \leftarrow U \to Y$	*Associational*	Collider at T; activates back-door via U
$Z \leftrightarrow U$	$Z \to T \leftarrow U$	*Associational*	Collider at T; dormant unless T conditioned on

Question 1. Is $Z \perp\!\!\!\perp Y$? There are two paths. The directed path $Z \to T \to Y$ is open, since it is a chain and we are not conditioning on T. The path $Z \to T \leftarrow U \to Y$ contains a collider at T; since we are not conditioning on T, that path is blocked. Because the causal path remains open, $Z \not\perp\!\!\!\perp Y$.

Question 2. Is $Z \perp\!\!\!\perp Y \mid T$? The causal path $Z \to T \to Y$ is blocked because conditioning on T blocks the chain. The path $Z \to T \leftarrow U \to Y$ contains a collider at T, and conditioning on T opens it. Hence $Z \not\perp\!\!\!\perp Y \mid T$. The residual association is entirely spurious — a collider artefact, not a causal effect.

Question 3. Is $Z \perp\!\!\!\perp Y \mid \{T, U\}$? The causal path is blocked by T. The path $Z \to T \leftarrow U \to Y$ has its collider at T opened by conditioning on T, but then blocked at U because we also condition on U. Hence $Z \perp\!\!\!\perp Y \mid \{T, U\}$.

Question 4. Is $Z \perp\!\!\!\perp U$? The only path $Z \to T \leftarrow U$ contains a collider at T. Since we are not conditioning on T or any descendant of T, the path is blocked. Hence $Z \perp\!\!\!\perp U$. This is *exogeneity*.

The four questions give a complete graphical account of the three IV assumptions. Question 1 establishes *relevance*; Question 4 establishes *exogeneity*; Question 3 establishes the *exclusion restriction*. Question 2 is the warning: conditioning on T alone simultaneously closes the causal channel and opens the confounded channel.

2.4 The Markov Property and Factorization

Proposition: Conditional Independence in the Three-Node Chain

Let $X_1 \to X_2 \to X_3$ be a chain with Markov factorization $p(x_1, x_2, x_3) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)$. Then $X_1 \perp\!\!\!\perp X_3 \mid X_2$. The analogous result for the fork $X_1 \leftarrow X_2 \to X_3$ is left as an exercise.

Definition: Local Markov Property

A distribution P satisfies the *local Markov property* with respect to $\mathcal{G}$ if, for every node $V_i \in \mathcal{V}$:

$$V_i \perp\!\!\!\perp (\mathrm{Nd}(V_i) \setminus \mathrm{Pa}(V_i)) \mid \mathrm{Pa}(V_i).$$

That is, once we condition on the direct causes of a node, that node is independent of all variables that are neither its descendants nor its parents.

Remark

The *global Markov property* is the statement that every d-separation in $\mathcal{G}$ implies a conditional independence in P — precisely the content of Theorem 2.1 (Verma & Pearl). For DAGs, the local and global Markov properties are equivalent (Lauritzen 1996, Proposition 3.27).

An immediate consequence is the *Markov factorization*:

$$p(v_1, \dots, v_k) \;=\; \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i))\,. \tag{2.1}$$

Example: Markov Factorization in the Education Example

For the DAG with edges $N \to E$, $B \to E$, $B \to Y$, $E \to Y$:

$$p(n, b, e, y) = p(n)\,p(b)\,p(e \mid n, b)\,p(y \mid e, b).$$

A regression of Y on E alone is confounded by B. Conditioning on B blocks the back-door path $E \leftarrow B \to Y$ and identifies $P(y \mid \mathrm{do}(E{=}e))$ via the back-door formula (Chapter 3).

Example: Markov Factorization in the IV DAG

Including the latent U: $p(z, t, y, u) = p(z)\,p(u)\,p(t \mid z, u)\,p(y \mid t, u)$. Since U is unobserved, the observable factorization is obtained by marginalizing:

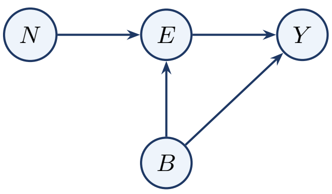
$$p(z, t, y) = \int p(z)\,p(u)\,p(t \mid z, u)\,p(y \mid t, u)\,\mathrm{d}u.$$

This cannot be simplified to $p(z)\,p(t \mid z)\,p(y \mid t)$ when U is a common cause of T and Y: this is precisely the endogeneity problem.

2.5 Worked Example: The Education–Earnings DAG

This section is the template for how we will use DAGs throughout the course: specify the DAG, read off the factorization, use d-separation to identify implied conditional independences, and interpret the result in causal terms.

The causal story. Education (E) affects earnings (Y); family background (B) is a common cause of both; neighborhood (N) affects education but has no direct effect on earnings. All four variables are observed; there are no latent confounders.



Step 1 — Markov factorization. $\mathrm{Pa}(N) = \mathrm{Pa}(B) = \emptyset$, $\mathrm{Pa}(E) = \{N, B\}$, $\mathrm{Pa}(Y) = \{E, B\}$.

$$p(n, b, e, y) \;=\; p(n)\,p(b)\,p(e \mid n, b)\,p(y \mid e, b).$$

Step 2 — d-Separation. Two paths between N and Y: Path 1: $N \to E \to Y$ (chain). Path 2: $N \to E \leftarrow B \to Y$ (collider at E).

- *Query (i):* $(N \perp\!\!\!\perp Y)_{\mathcal{G}}$? Path 1 is a chain with nothing conditioned on — open. $N \not\!\perp\!\!\!\perp Y$.
- *Query (ii):* $(N \perp\!\!\!\perp Y \mid E)_{\mathcal{G}}$? Path 1 blocked by E; Path 2 collider E opened, path continues through B which is open. $N \not\!\perp\!\!\!\perp Y \mid E$. This is a collider trap: conditioning on E opens a spurious association between N and Y through B.

- *Query (iii):* $(N \perp\!\!\!\perp Y \mid \{E, B\})_{\mathcal{G}}$? Path 1 blocked by E; Path 2 opened at E but blocked at B. $(N \perp\!\!\!\perp Y \mid E, B)_{\mathcal{G}}$.
- *Query (iv):* $(N \perp\!\!\!\perp B)_{\mathcal{G}}$? Only path $N \to E \leftarrow B$ has collider E with $E \notin \emptyset$ — blocked. $N \perp\!\!\!\perp B$.

Step 3 — Conditional Independence. By Theorem 2.1 (Verma & Pearl): $N \perp\!\!\!\perp B$, $N \perp\!\!\!\perp Y \mid E, B$, $N \not\!\perp\!\!\!\perp Y$, $N \not\!\perp\!\!\!\perp Y \mid E$.

Step 4 — Identification. The only back-door path is $E \leftarrow B \to Y$. With $\mathbf{S} = \{B\}$:

$$P(y \mid \mathrm{do}(E{=}e)) \;=\; \sum_b P(y \mid e, b)\, P(b).$$

2.6 The Big Picture

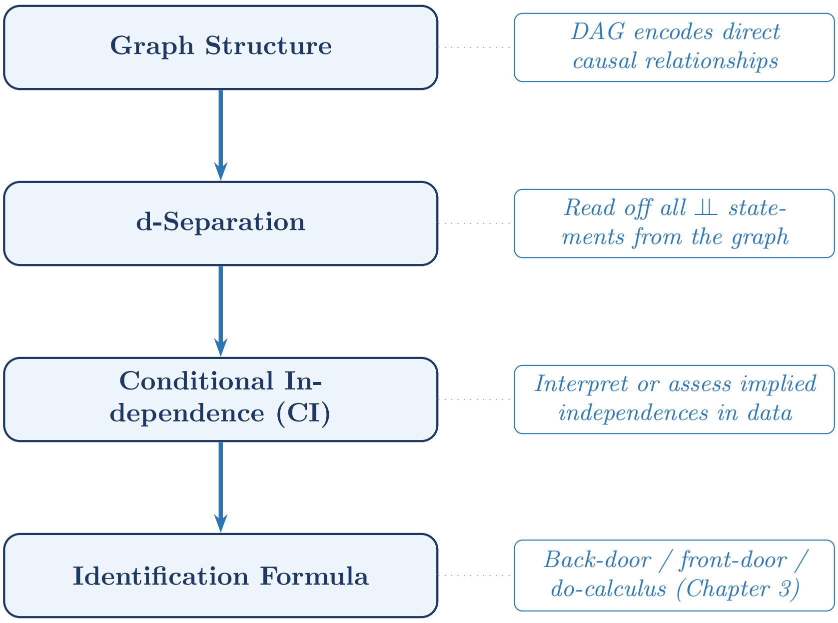


Figure 2.2: The four-step pipeline from causal assumptions to identification formula.

The logical chain in compact form:

structural assumptions $\implies$ DAG $\implies$ d-separation $\implies$ CI under the Markov property $\implies$ identification formulas.

2.7 Summary

1. **DAGs as causal structure.** A DAG is a directed acyclic graph whose nodes represent variables and whose directed edges represent direct causal relationships. Once a DAG is specified, it encodes which variables are causally connected, which are potential confounders, and which may block or open paths.

2. **Three-node motifs.** Every path is built from three local structures: chains, forks, and colliders. Chains and forks are open by default and blocked by conditioning on the middle node. Colliders are blocked by default and opened by conditioning on the collider or one of its descendants.

3. **d-Separation and conditional independence.** The d-separation criterion is a graphical rule: X and Y are d-separated by $\mathbf{S}$ exactly when every path between them is blocked by $\mathbf{S}$. Under the Markov property, this implies the corresponding conditional independence. Some such implications may be assessed empirically, but agreement with data does not by itself verify the graph.

4. **Markov factorization.** The local Markov property yields

$$p(v_1, \ldots, v_k) \;=\; \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i))\,,$$

which expresses the joint distribution in terms of local conditional distributions.

5. **Collider bias.** Conditioning on a collider — or on a descendant of a collider — can induce a spurious association between otherwise independent variables (Berkson 1946). This is the *opposite* of confounding adjustment: conditioning on a common cause removes spurious association, whereas conditioning on a common effect creates it.

6. **A practical workflow.** To analyze a DAG, proceed in four steps: identify the graph structure, determine which paths are open or blocked, translate d-separation statements into conditional independences under the Markov property, and interpret those independences in light of the causal question.

2.8 Problems

1. Warm-up: a single collider. Consider the DAG $X \to Y \leftarrow Z$, where X and Z have no other connections.

(a) Identify the structural role of Y on the path $X \to Y \leftarrow Z$.
(b) Is $X \perp\!\!\!\perp Z$? Apply the d-separation criterion and state which blocking rule applies.
(c) Is $X \perp\!\!\!\perp Z \mid Y$? Explain what happens to the path when Y is conditioned on.

2. d-Separation practice. Consider the DAG: $A \to B \to D$, $A \to C \to D$, $B \to E$, $C \to E$.

(a) List all paths between A and E. (*Hint*: there are four paths; two pass through D.)
(b) For each path, identify the role of each intermediate node.
(c) Does $\{B, C\}$ d-separate A and E?
(d) Does $\{D\}$ d-separate B and C? What type of node is D on the path $B \to D \leftarrow C$?

3. Berkson's bias. Suppose X and Y are independent standard normal variables, and let $S = \mathbf{1}[X+Y > 0]$ (selected into a sample).

(a) Verify analytically that $\mathrm{Cov}(X, Y \mid S{=}1) < 0$.
(b) Draw the DAG for (X, Y, S) and identify S as a collider.
(c) Explain in one sentence why restricting the analysis to the subsample with $S = 1$ biases estimates of any association between X and Y, and name the type of bias this illustrates.

4. Markov factorization and collider activation. Consider the DAG with edges $A \to E$, $A \to W$, $F \to E$, $E \to W$, where A = ability, F = family income, E = education, W = wages.

(a) Write down the Markov factorization $p(a, f, e, w)$.
(b) Is $(F \perp\!\!\!\perp W)_{\mathcal{G}}$? List all paths between F and W.
(c) Is $(F \perp\!\!\!\perp W \mid E)_{\mathcal{G}}$? Identify the role of E on each path.
(d) A researcher regresses W on E and F, omitting A. Is the coefficient on E a causal effect?

5. Theorem for the collider. Consider the collider $A \to M \leftarrow B$ with Markov factorization $p(a, m, b) = p(a)\, p(b)\, p(m \mid a, b)$.

(a) By marginalizing over M, show that $A \perp\!\!\!\perp B$ in the joint distribution.
(b) Show that conditioning on M breaks this independence.
(c) Explain why parts (a) and (b) together are consistent with Theorem 2.1 (Verma & Pearl).

6. Terminology check. Consider the DAG with edges $U \to X$, $U \to Z$, $X \to W$, $Z \to W$, $W \to Y$.

(a) Identify the parents, children, ancestors, descendants, and non-descendants of node W.
(b) List all pairs of adjacent nodes.
(c) Which pairs of nodes are connected by a directed path?
(d) Write the Markov factorization $p(u, x, z, w, y)$.

7. Markov factorization and local Markov property. Consider the DAG with edges $X_1 \to X_2$, $X_1 \to X_3$, $X_2 \to X_4$, $X_3 \to X_4$.

(a) Write the joint density $p(x_1, x_2, x_3, x_4)$ implied by the Markov factorization.
(b) State the local Markov property for each of the four nodes.
(c) Is $(X_2 \perp\!\!\!\perp X_3)_{\mathcal{G}}$? Is $(X_2 \perp\!\!\!\perp X_3 \mid X_1)_{\mathcal{G}}$?

8. Toy proof: conditional independence in the fork. Consider the fork $X_1 \leftarrow X_2 \to X_3$ with Markov factorization $p(x_1, x_2, x_3) = p(x_2)\, p(x_1 \mid x_2)\, p(x_3 \mid x_2)$.

(a) Show directly that $X_1 \perp\!\!\!\perp X_3 \mid X_2$.
(b) Is $X_1 \perp\!\!\!\perp X_3$ marginally? Justify both graphically and algebraically.
(c) Explain why conditioning on the common cause X_2 removes the association.

9. d-Separation in the practice DAG. Refer to the eight-node DAG in the eight-node example above. For each query, state whether it is true or false and justify by listing all relevant paths.

(a) $X_1 \perp\!\!\!\perp Y$
(b) $X_1 \perp\!\!\!\perp Y \mid W$
(c) $T \perp\!\!\!\perp M_2 \mid M_1$
(d) Does conditioning on C induce collider bias between X_1 and M_2? Identify the specific path that is opened and explain why the induced association is spurious.
(e) Does conditioning on D (a descendant of C) open the path $X_1 \to C \leftarrow M_2$? State which clause of Definition 2.1 applies.

Chapter 3

The Do-Calculus and Identification Criteria

Learning Objectives

By the end of this chapter, students should be able to:

1. Define the two intervention graph operations $\mathcal{G}_{\overline{X}}$ and $\mathcal{G}_{\underline{X}}$, explain the causal meaning of $\mathcal{G}_{\overline{X}}$, and describe the technical role of $\mathcal{G}_{\underline{X}}$.
2. Apply the back-door criterion to determine whether a set $\mathbf{S}$ identifies a causal effect, and write the back-door adjustment formula.
3. Apply the front-door criterion in graphs where the confounder is unobserved but a suitable mediator exists.
4. State all three rules of do-calculus, identify the graph condition that licenses each rule, and use them to prove the back-door and front-door formulas.
5. State the completeness theorem of Shpitser and Pearl (2006) and explain its practical implication: if the do-calculus cannot identify a quantity from the observational distribution relative to the assumed graph, then no purely observational method can do so without additional assumptions or new data.

Remark: How to Read This Chapter

This chapter has two pedagogical layers. Sections 1–3 develop the two most important concrete identification criteria — back-door and front-door adjustment — using intervention graphs and direct graphical reasoning. These sections are the core material for a first reading.
Sections 4–5 then place these criteria inside the more general theory of the do-calculus and identifiability. These later sections are conceptually important, but more abstract and may be read as a second pass after the concrete criteria are understood.

3.1 From d-Separation to Intervention: Intervention Graphs

In Chapter 2 we learned to read conditional independence from a DAG using d-separation. This chapter takes the next step: translating the graphical language into *identification formulas* — expressions that write the interventional distribution $P(y \mid \mathrm{do}(T{=}t))$ entirely in terms of quantities observable from data.

3.1.1 What Does Intervention Mean?

Conditioning versus intervening. The conditional distribution $P(Y \mid T{=}t)$ describes the subpopulation of units for whom T was *observed* to equal t. Because treatment assignment may be influenced by confounders, this subpopulation is not representative of the full population. The interventional distribution $P(Y \mid \mathrm{do}(T{=}t))$, by contrast, describes the population that would result if T were *externally set* to t for everyone — removing the natural process that determines T and replacing it with the forced value. The two distributions coincide only when T has no unmeasured common causes with Y.

A simple illustration: suppose temperature Z causes both ice-cream sales T and crime Y. The quantity $P(Y \mid T{=}t)$ is the crime rate on days when ice-cream sales happen to equal t — confounded by Z. The quantity $P(Y \mid \mathrm{do}(T{=}t))$ is the crime rate we would observe if we *fixed* sales at level t by decree. Forcing sales does not change the temperature, so crime remains driven only by Z, not by the sales level we mandated.

The structural basis of the do-operator. In the SEM framework of Chapter 1, an intervention $\mathrm{do}(T{=}t)$ **replaces the entire structural equation** for T with the constant $T = t$. The replacement severs T's dependence on its parents $\mathrm{Pa}(T)$, while all other structural equations remain unchanged.

Graph surgery as the graphical realization. Because $\mathrm{Pa}(T)$ no longer affects T after the intervention, every arrow pointing *into* T in the DAG is removed. The resulting graph — the *intervention graph* $\mathcal{G}_{\overline{T}}$ — represents the post-intervention world. D-separation in $\mathcal{G}_{\overline{T}}$ encodes conditional independence in $P(\cdot \mid \mathrm{do}(T{=}t))$, not in the original observational distribution P.

3.1.2 The Two Graph Operations

Definition: Intervention Graphs $\mathcal{G}_{\overline{X}}$ and $\mathcal{G}_{\underline{X}}$

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a DAG and $X \subseteq \mathcal{V}$ a set of nodes.

- $\mathcal{G}_{\overline{X}}$ (*intervention graph*, or *mutilated graph*) is obtained by *deleting all arrows into* X. This represents the intervention $\mathrm{do}(X{=}x)$, which replaces the structural equation for X by the constant x, severing X's dependence on all its former parents.
- $\mathcal{G}_{\underline{X}}$ (*auxiliary observation graph*) is obtained by *deleting all arrows out of* X. This is a *technical device* used in graphical conditions for certain do-calculus steps. It does *not* represent a probabilistic model of conditioning on X; it is simply the graph in which the relevant d-separation condition must be checked.

Two Graph Operations at a Glance

- **Delete arrows *into*** X ($\mathcal{G}_{\overline{X}}$): models the intervention $\mathrm{do}(X{=}x)$. D-separation in $\mathcal{G}_{\overline{X}}$ encodes independence in the post-intervention distribution $P(\cdot \mid \mathrm{do}(X{=}x))$.
- **Delete arrows *out of*** X ($\mathcal{G}_{\underline{X}}$): a technical device used in graphical conditions for certain do-calculus steps. It is *not* a model of conditioning on X.

Figure 3.1 illustrates both graph operations on the IV DAG $\{Z \to T,\ T \to Y,\ U \to T,\ U \to Y\}$.

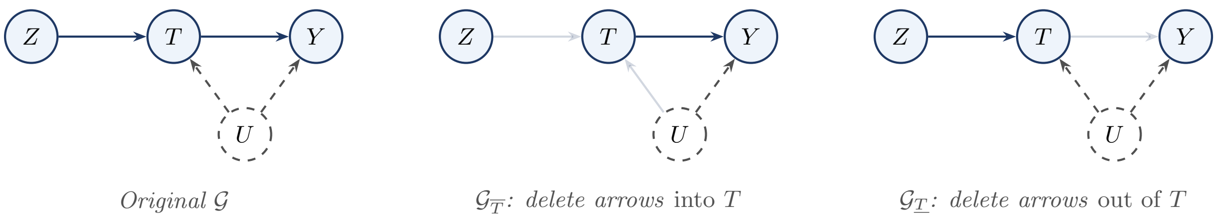


Figure 3.1: The two graph operations on the IV DAG. Faded arrows have been removed. $\mathcal{G}_{\overline{T}}$ models external intervention $\mathrm{do}(T{=}t)$: the back-door path $T \leftarrow U \to Y$ is severed. $\mathcal{G}_{\underline{T}}$ is a technical device used in graphical conditions for certain do-calculus steps; it does not represent conditioning on T.

3.1.3 Why the Intervention Graph Works

The Intervention Principle

In the original DAG $\mathcal{G}$, the node T is determined by its parents $\mathrm{Pa}(T)$ via the structural equation $T = f_T(\mathrm{Pa}(T), \varepsilon_T)$. Intervening on T removes the structural mechanism that normally determines T; therefore the parents of T no longer influence it. The intervention $\mathrm{do}(T{=}t)$ *replaces this equation* with the constant $T = t$, making T independent of all its former parents. Graphically, this severs every arrow into T. The remaining structure of the graph is unchanged: all other structural equations still hold.

A d-separation statement in $\mathcal{G}_{\overline{T}}$ is therefore a conditional independence statement in the *post-intervention* distribution $P(\cdot \mid \mathrm{do}(T{=}t))$, not in the observational distribution P.

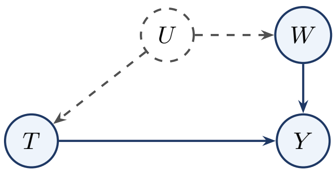
3.2 The Back-Door Criterion

The back-door criterion is the most widely used identification strategy. It gives a simple graphical condition under which conditioning on an observed set $\mathbf{S}$ is sufficient to identify the causal effect of T on Y.

3.2.1 Motivation: When the Confounder Is Unobserved

If all common causes of T and Y were observed, identification would be trivial: adjust for all of them. The interesting case is when some confounders are *unobserved*, yet an observed set $\mathbf{S}$ can still block every spurious path.

Consider the following DAG: a drug T affects recovery Y. Socioeconomic status U is an unobserved confounder ($U \to T$, $U \to Y$), but U also affects an observed proxy W ($U \to W$), which in turn affects Y ($W \to Y$).



The only back-door path is $T \leftarrow U \to W \to Y$. Although U is unobserved, W sits on this path and *is* observed. Conditioning on W blocks the path at the chain node W, and W is not a descendant of T, so $\mathbf{S} = \{W\}$ satisfies the back-door criterion. The causal effect is identified even though U is never measured. The criterion does not require adjusting for the confounders themselves, only for an observed set that *intercepts* every back-door path.

3.2.2 Definition and Theorem

Definition: Back-Door Criterion

A set of observed variables $\mathbf{S}$ satisfies the *back-door criterion* for the causal effect of T on Y in $\mathcal{G}$ if:

1. No node in $\mathbf{S}$ is a descendant of T.
2. $\mathbf{S}$ blocks every *back-door path* from T to Y, i.e., every path that begins with an arrow pointing *into* T.

Remark

The terminology reflects the geometry of the graph relative to T. A *back-door path* enters T from behind — it represents a confounding route through which T and Y become spuriously associated without any causal mechanism from T to Y. The front-door criterion of Section 8.8 is named for the complementary idea: it identifies the causal effect by intercepting the *front-door* (causal) paths at a mediator M, rather than blocking the back-door (confounding) paths directly. Standing at T, confounders sneak in through the back door; the causal effect walks out through the front.

Remark

Condition 2 is equivalent to requiring $(Y \perp\!\!\!\perp T \mid \mathbf{S})_{\mathcal{G}_{\underline{T}}}$: $\mathbf{S}$ d-separates T and Y in the graph $\mathcal{G}_{\underline{T}}$ obtained by *deleting all arrows out of* T. Removing T's outgoing edges eliminates the causal paths $T \to \cdots \to Y$, leaving only back-door paths between T and Y.
The back-door criterion adds two constraints: $\mathbf{S}$ must consist of *observed* variables only, and it must contain no descendant of T. Every valid adjustment set is therefore a valid d-separator in $\mathcal{G}_{\underline{T}}$, but the converse does not hold.

Example: A Single Confounder

Consider the DAG $T \leftarrow C \to Y$ with $T \to Y$, where C is an observed confounder. There is one back-door path: $T \leftarrow C \to Y$.
Does $\mathbf{S} = \{C\}$ satisfy the back-door criterion?

1. C is not a descendant of T.
2. $\{C\}$ blocks $T \leftarrow C \to Y$ (fork at C).

The back-door formula gives: $P(y \mid \mathrm{do}(T{=}t)) = \sum_c P(y \mid t, c)\, P(c)$.

Example: Two Back-Door Paths — Both Must Be Blocked

Consider treatment T, outcome Y, observed variables C_1 and C_2, and an unobserved confounder U: $C_1 \to T$, $C_1 \to Y$, $U \to T$, $U \to C_2$, $C_2 \to Y$, $T \to Y$.
There are exactly two back-door paths from T to Y:

1. $T \leftarrow C_1 \to Y$ (direct confounding by the observed C_1),
2. $T \leftarrow U \to C_2 \to Y$ (indirect confounding via the unobserved U, routing through the observed proxy C_2).

$\mathbf{S} = \{C_1\}$ fails: it blocks path 1 but leaves path 2 open. $\mathbf{S} = \{C_2\}$ fails: it blocks path 2 but leaves path 1 open. $\mathbf{S} = \{C_1, C_2\}$ succeeds: it blocks both paths, and neither C_1 nor C_2 is a descendant of T.

M-Bias: More Variables Can Be Worse

Consider a graph where T and Y each have their own separate unobserved confounders, U_1 and U_2, that happen to share a common observed descendant C: $U_1 \to T$, $U_2 \to Y$, $U_1 \to C$, $U_2 \to C$, $T \to Y$ (with U_1, U_2 unobserved).
Marginally, the only back-door path $T \leftarrow U_1 \to C \leftarrow U_2 \to Y$ is *blocked* by the collider C. The causal effect is already identified without any adjustment!
Conditioning on C *activates* the collider, opening the path. The set $\{C\}$ therefore fails Condition 2: it does not block the back-door path; it creates one.
This phenomenon is called *M-bias* because the graph has an M-shaped skeleton. The practical lesson is that Condition 1 alone ("include all pretreatment variables") is not sufficient; every candidate adjustment variable must also be verified not to open a previously closed back-door path.

Theorem: Back-Door Adjustment Formula

(Pearl 1993) If $\mathbf{S}$ satisfies the back-door criterion for the effect of T on Y in $\mathcal{G}$, and $P(T{=}t \mid \mathbf{S}{=}\mathbf{s}) > 0$ for all $\mathbf{s}$ with $P(\mathbf{S}{=}\mathbf{s}) > 0$, then:

$$P(y \mid \mathrm{do}(T{=}t)) \;=\; \int P(y \mid T{=}t,\ \mathbf{S}{=}\mathbf{s})\, dP(\mathbf{s}). \tag{3.1}$$

Equivalently, for any measurable function h:

$$\mathbb{E}[h(Y) \mid \mathrm{do}(T{=}t)] \;=\; \mathbb{E}_{\mathbf{S}}\Big[\mathbb{E}[h(Y) \mid T{=}t,\ \mathbf{S}]\Big]. \tag{3.2}$$

Remark

The positivity condition requires every value of the treatment $T{=}t$ to be observable within each stratum of $\mathbf{S}$: if some stratum $\mathbf{s}$ never receives treatment t in the data, the conditional distribution $P(y \mid t, \mathbf{s})$ is undefined and Equation 5.3 cannot be computed. Positivity is a *data-support* condition, not a graphical one. When positivity fails on a set of measure zero, the formula still identifies $P(y \mid \mathrm{do}(t))$ because the offending strata contribute zero mass to the outer integral.

3.2.3 The Adjustment Formula: Standardization

Equation 5.3 is also known as the *standardization formula* or, in the epidemiological literature, the *g-formula* (Robins 1986). The interpretation is important:

- $P(y \mid t, \mathbf{s})$ is the stratum-specific conditional distribution of Y given $T{=}t$ and $\mathbf{S}{=}\mathbf{s}$. Within each stratum, T is not confounded by any back-door path (all are blocked by $\mathbf{S}$), so this conditional distribution *is* causal.
- The outer integration $\int \cdots dP(\mathbf{s})$ weights the strata by the *population marginal* distribution of $\mathbf{S}$, rather than the treatment-conditional distribution $dP(\mathbf{s} \mid t)$ that a naïve analysis would use.
- In a randomized experiment, $\mathbf{S} = \emptyset$ satisfies the back-door criterion (no confounding), and Equation 5.3 reduces to $P(y \mid \mathrm{do}(t)) = P(y \mid t)$: the observational and interventional distributions coincide.

Example: Back-Door Adjustment — A Numerical Walkthrough

Use the single-confounder graph ($T \leftarrow C \rightarrow Y$, $T \rightarrow Y$) with all variables binary: C = disease severity, T = treatment, Y = recovery. Severity is equally prevalent, $P(C{=}1) = 0.5$, and sicker patients are preferentially treated.

C	$P(C)$	$P(T{=}1 \mid C)$	$P(Y{=}1 \mid T{=}0, C)$	$P(Y{=}1 \mid T{=}1, C)$
0	0.50	0.20	0.10	0.40
1	0.50	0.80	0.50	0.80

The true ATE within each stratum is 0.30.
Naïve estimand (ignoring confounding): $P(Y{=}1 \mid T{=}1) = 0.72$, $P(Y{=}1 \mid T{=}0) = 0.18$. Naïve difference: **0.54**.
Back-door adjusted estimand:

$$P(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.40 \times 0.5 + 0.80 \times 0.5 = 0.60,$$

$$P(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.10 \times 0.5 + 0.50 \times 0.5 = 0.30.$$

Adjusted difference: **0.30** — matching the true ATE.
The naïve estimate (0.54) overstates the causal effect by 0.24 because treated patients are predominantly severe ($C{=}1$), who recover at higher rates *even without treatment*. The back-door formula removes this confounding by imposing the *same* covariate distribution $P(C)$ on both potential worlds.

3.2.4 Application to the Education–Earnings DAG

Example: Back-Door Identification in the Education Example

This is the Education–Earnings DAG analyzed in full in Chapter 2 (Section 5). The DAG has edges $N \rightarrow E$, $B \rightarrow E$, $B \rightarrow Y$, $E \rightarrow Y$, where N = neighborhood, B = family background, E = education, Y = earnings. We wish to identify $P(y \mid \mathrm{do}(E{=}e))$.
The only back-door path is $E \leftarrow B \rightarrow Y$.

- Does $\{B\}$ satisfy the criterion? B is not a descendant of E ; $\{B\}$ blocks $E \leftarrow B \rightarrow Y$. By

Equation 5.3: $P(y \mid \mathrm{do}(E{=}e)) = \sum_b P(y \mid e, b)\, P(b)$.

- Does $\{N\}$ satisfy the criterion? $\{N\}$ does *not* block $E \leftarrow B \rightarrow Y$ (the fork at B remains open). Fails condition 2.
- Does $\{N, B\}$ satisfy the criterion? Yes: neither N nor B is a descendant of E, and $\{N, B\}$ blocks the only back-door path.

Remark

The Education example illustrates three important points.
(a) Multiple valid sets may exist, and they identify the same quantity. Both $\{B\}$ and $\{N, B\}$ satisfy the back-door criterion here. Any such set yields the same interventional distribution; the identification target is unique even though the adjustment sets are not.
(b) Supersets of a valid adjustment set are not automatically valid. Adding a variable to a valid set can violate Condition 2 if that variable is a collider on some back-door path. In this graph $\{N, B\}$ happens to remain valid, but this must be verified from the graph, not assumed.
(c) Smaller valid sets are often preferable in practice. Adjusting for fewer variables — the minimal sufficient set — is generally more efficient and more robust, because every variable added to $\mathbf{S}$ must also satisfy positivity.

3.3 The Front-Door Criterion

The back-door criterion requires that all confounders be observed. The front-door criterion provides identification in a different configuration: when there exists a measured *mediator* M on the causal path from T to Y with specific independence properties.

3.3.1 The Configuration and Criterion

Definition: Front-Door Criterion

A set of variables $\mathbf{M}$ satisfies the *front-door criterion* for the effect of T on Y in $\mathcal{G}$ if:

1. $\mathbf{M}$ intercepts all directed paths from T to Y (all causal paths pass through $\mathbf{M}$).
2. There are no unblocked back-door paths from T to $\mathbf{M}$ (the effect of T on $\mathbf{M}$ is not confounded).
3. All back-door paths from $\mathbf{M}$ to Y are blocked by T (conditioning on T closes all confounding of the $\mathbf{M} \rightarrow Y$ effect).

Example: A Single Mediator with Unobserved Confounding

Consider the DAG $T \rightarrow M \rightarrow Y$ with $U \rightarrow T$ and $U \rightarrow Y$, where M is observed and U is unobserved. Does M satisfy the front-door criterion?

1. *M intercepts all directed paths from T to Y.* The only directed path is $T \rightarrow M \rightarrow Y$, which passes through M.
2. *No unblocked back-door paths from T to M.* There is no edge $U \rightarrow M$, so no back-door path exists.
3. *All back-door paths from M to Y are blocked by T.* The only back-door path is $M \leftarrow T \leftarrow U \rightarrow Y$. Conditioning on T blocks this path (chain at T).

All three conditions hold, so the front-door formula applies:

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_m P(m \mid t) \sum_{t'} P(y \mid t', m)\, P(t').$$

Example: When the Front-Door Criterion Fails

Each condition of the definition is load-bearing.

Case (a): Condition 2 fails — the $T \to M$ link is confounded. Add the edge $U \to M$ so the unobserved confounder also directly affects the mediator. Condition 1 still holds, but condition 2 now fails: the path $T \leftarrow U \to M$ is an unblocked back-door path from T to M. Stage 1 of the front-door formula uses $P(M{=}m \mid T{=}t)$ as if the $T \to M$ link were unconfounded. But with $U \to M$, the association between T and M is partly due to their shared cause U, not purely the causal effect $T \to M$. Stage 1 no longer estimates $P(m \mid \mathrm{do}(T{=}t))$.
Case (b): Condition 3 fails — the $M \to Y$ link has an extra unobserved confounder. Add a second unobserved variable V with $V \to M$ and $V \to Y$. Conditions 1 and 2 still hold. But condition 3 now fails: the path $M \leftarrow V \to Y$ is a back-door path from M to Y that is *not* blocked by conditioning on T. Stage 2 uses $\sum_{t'} P(y \mid t', m)\, P(t')$ as the back-door-adjusted effect of M on Y, but T is no longer a valid adjustment set for Stage 2.

Modification	**Condition violated**	**Stage broken**
Add $U \to M$	Cond. 2: $T \to M$ confounded	Stage 1
Add $V \to M$ and $V \to Y$	Cond. 3: $M \to Y$ confounded by V	Stage 2

The prototypical graph where the front-door criterion applies is: $T \to M \to Y$, $U \to T$, $U \to Y$, no direct $T \to Y$ edge. Figure 8.2 shows this graph.

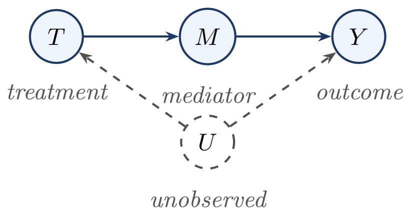


Figure 3.2: The prototypical front-door graph. The confounder U is unobserved, so back-door adjustment on U is infeasible. However, the mediator M is fully observed and satisfies the three front-door conditions, enabling identification via the front-door formula.

3.3.2 The Front-Door Formula

The front-door strategy splits the total effect of T on Y into two identifiable pieces: the effect of T on the mediator M, and the effect of M on Y. Front-door identification does not remove confounding directly; it *routes around it* through an observed causal pathway.

Theorem: Front-Door Adjustment Formula

If M satisfies the front-door criterion for the effect of T on Y (Pearl 1995), and $P(t) > 0$ for all t, then:

$$P(y \mid \mathrm{do}(T{=}t)) \;=\; \sum_m \underbrace{P(m \mid T{=}t)}_{\text{Stage 1}} \; \underbrace{\sum_{t'} P(y \mid T{=}t',\; M{=}m)\, P(t')}_{\text{Stage 2}}. \tag{3.3}$$

Equivalently, writing $\mu(t', m) = \mathbb{E}[Y \mid T{=}t',\; M{=}m]$:

$$\mathbb{E}[Y \mid \mathrm{do}(T{=}t)] \;=\; \mathbb{E}_{M|T=t}\Big[\mathbb{E}_T\big[\mu(T,\, M)\big]\Big]. \tag{3.4}$$

The formula chains two stages: **Stage 1** $P(m \mid T{=}t)$ is the distribution of the mediator given $T{=}t$. Condition 2 guarantees no confounding between T and M, so this is directly identifiable. **Stage 2** $\sum_{t'} P(y \mid T{=}t',\; M{=}m)\, P(t')$ is the back-door-adjusted causal effect of M on Y, with T as the adjustment variable. Condition 3 guarantees that conditioning on T blocks all back-door paths from M to Y.

Derivation Sketch

The formula is the composition of two back-door applications.

1. **Expand over the mediator.** By condition 1, every directed path from T to Y passes through M:
$$P(y \mid \mathrm{do}(T{=}t)) = \sum_m P(y \mid \mathrm{do}(M{=}m))\, P(m \mid \mathrm{do}(T{=}t)).$$
2. **Identify the distribution of M under intervention.** By condition 2 there are no back-door paths from T to M, so T satisfies the (trivial) back-door criterion for its effect on M with empty adjustment set. Hence $P(m \mid \mathrm{do}(T{=}t)) = P(m \mid T{=}t)$.
3. **Identify the effect of M on Y.** By condition 3, T is a valid back-door adjustment set for the $M \to Y$ effect:
$$P(y \mid \mathrm{do}(M{=}m)) \;=\; \sum_{t'} P(y \mid T{=}t',\; M{=}m)\, P(t').$$
4. **Substitute back** to obtain Equation 8.13. □

3.3.3 Interpretation: Two-Stage Causal Accounting

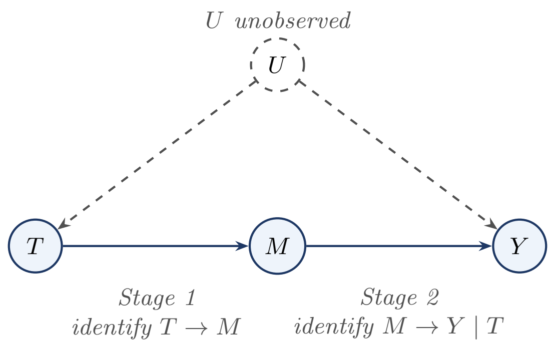


Figure 3.3: Why the front-door criterion works. The total effect of T on Y is not directly identifiable because U confounds T and Y. However, the effect of T on the mediator M is unconfounded (Stage 1), and the effect of M on Y becomes identifiable after conditioning on T, which blocks the back-door path through U (Stage 2).

Example: Smoking, Tar, and Cancer

The classic front-door example (Pearl 1995): T = smoking, M = tar deposits in lungs, Y = lung cancer, U = genetic predisposition (unobserved). The graph is $T \to M \to Y$ with $U \to T$ and $U \to Y$. There is no direct $T \to Y$ path.

Conditions 1–3 are satisfied: all causal paths from T to Y pass through M; no back-door path exists from T to M; and the only back-door path from M to Y is $M \leftarrow T \leftarrow U \to Y$, which is blocked by conditioning on T.

The front-door formula gives a double sum over (m, t') that is entirely computable from the observed joint distribution of (T, M, Y), without any data on U.

Example: Front-Door Formula — Binary Numerical Computation

Let $T, M, Y \in \{0, 1\}$ with U unobserved and the graph $T \to M \to Y$, $U \to T$, $U \to Y$.

T	M	Y	$P(T, M, Y)$
0	0	0	0.360
0	0	1	0.040
0	1	0	0.050

0	1	1	0.050
1	0	0	0.070
1	0	1	0.030
1	1	0	0.120
1	1	1	0.280

The inner sum (Stage 2):

$$S_0 = 0.1 \times 0.5 + 0.3 \times 0.5 = 0.20, \quad S_1 = 0.5 \times 0.5 + 0.7 \times 0.5 = 0.60.$$

The outer sum (Stage 1):

$$P(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.2 \times 0.20 + 0.8 \times 0.60 = 0.52,$$

$$P(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.8 \times 0.20 + 0.2 \times 0.60 = 0.28.$$

The average causal effect is $0.52 - 0.28 = 0.24$. The naïve estimate $P(Y{=}1 \mid T{=}1) - P(Y{=}1 \mid T{=}0) = 0.62 - 0.18 = 0.44$ is almost twice the true effect, because U simultaneously increases T and Y.

Comparing Back-Door and Front-Door

The two criteria are *not* nested. Neither implies the other.

- The back-door criterion requires all confounders to be observed (or blocked by observed variables), but places no restriction on mediators.
- The front-door criterion requires an observed mediator intercepting all causal paths, but the confounder U can be entirely unobserved.

There exist graphs where neither criterion alone applies, but where the full do-calculus (or the ID algorithm) can still identify the causal effect.

3.4 The Three Rules of Do-Calculus

Having seen two concrete identification strategies, we now develop the general algebraic machinery that underlies both. Pearl (1995) introduced the three rules of do-calculus as a system for manipulating expressions involving the do-operator: each rule licenses a transformation that removes or exchanges terms, bringing the expression one step closer to a purely observational formula.

How to Use a Do-Calculus Rule

When applying a do-calculus rule, proceed in three steps:

1. **Choose the relevant modified graph.** Form the graph obtained by deleting the appropriate incoming or outgoing arrows.
2. **Check the d-separation condition in that graph.** Work purely graphically: list the relevant paths and determine whether they are blocked.
3. **Apply the corresponding algebraic transformation.** Once the graphical condition holds, use the rule to remove an observation, insert an intervention, or exchange an observation for an intervention.

The do-calculus is therefore a graph modification → d-separation check → probability transformation procedure.

Remark: First Reading / Second Reading

On a first reading, focus on the qualitative role of the three rules: Rule 1 adds or removes observations, Rule 2 exchanges an action for an observation, and Rule 3 adds or removes actions. The detailed graph subscripts and formal statements can be absorbed gradually.

3.4.1 Statement of the Rules

Each rule's graphical condition uses a specific modified graph. The table below gives the recipe for each subscript that appears.

Subscript	Step 1: delete	Step 2: delete
$\mathcal{G}_{\overline{X}\,\underline{Z}}$	all arrows *into* X	all arrows *out of* Z
$\mathcal{G}_{\overline{X}\,\overline{Z(W)}}$	all arrows *into* X	arrows *into* Z except those from W
$\mathcal{G}_{\overline{X}\,\overline{Z(\overline{W})}}$	all arrows *into* X	arrows *into* Z that are not from ancestors of W in $\mathcal{G}_{\overline{X}}$

Theorem: The Three Rules of Do-Calculus

(Pearl 1995) Let $\mathcal{G}$ be a DAG over variables $\mathcal{V}$, and let $X, Y, Z, W \subseteq \mathcal{V}$ be disjoint sets. The rules hold for any distribution P Markov with respect to $\mathcal{G}$.

Rule 1 (Insertion/Deletion of Observations). *Remove or insert an observation when it becomes irrelevant in the modified graph.*

$$P(y \mid \mathrm{do}(x), z, w) \;=\; P(y \mid \mathrm{do}(x), w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\,\underline{Z}}}.$$

Rule 2 (Action/Observation Exchange). *Replace an intervention by an observation, or vice versa, when the modified graph makes them equivalent.*

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) \;=\; P(y \mid \mathrm{do}(x), z, w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\,\overline{Z(W)}}}.$$

Rule 3 (Insertion/Deletion of Actions). *Remove or insert an intervention when it becomes irrelevant in the modified graph.*

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) \;=\; P(y \mid \mathrm{do}(x), w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\,\overline{Z(\overline{W})}}}.$$

Remark

Each rule is a *conditional independence statement in a specific intervention graph.* The three rules cover all ways of adding or removing terms from a do-expression: Rule 1 handles observations; Rule 2 exchanges actions for observations; Rule 3 deletes actions. Any sequence of such operations that converts a do-expression into a purely observational expression is an *identification proof.*

Remark: Soundness of the Rules

The three rules are *sound*: each holds for every distribution P Markov with respect to $\mathcal{G}$. For Rule 2: the post-intervention distribution $P(\cdot \mid \mathrm{do}(x))$ is Markov with respect to $\mathcal{G}_{\overline{X}}$. Further intervening on Z gives a distribution Markov with respect to $\mathcal{G}_{\overline{X}\,\overline{Z}}$. The d-separation condition $(Y \perp\!\!\!\perp Z \mid X, W)_{\mathcal{G}_{\overline{X}\,\overline{Z(W)}}}$ implies $Y \perp\!\!\!\perp Z \mid W$ in the post-do(x, z) distribution. Since the distribution of Y is the same whether we *intervene* on Z or *observe* Z, the two expressions are equal. Full proofs appear in Pearl (1995).

3.4.2 Intuition for Each Rule

Rule 1 — Adding/Removing Observations. The condition says: in the post-do(x) world, if we further deactivate Z's causal influence (delete its outgoing arrows), Y and Z become independent given W. This means Z carries no information about Y that is not already in W, so it can be freely inserted or removed.

Example: Rule 1 — Deleting a Redundant Observation

Graph: $Z \to T \to Y$, no other edges. Suppose we intervene on T.
In $\mathcal{G}_{\overline{T}\,\underline{Z}}$: delete arrows into T (removes $Z \to T$), then delete arrows out of Z (nothing remains).
Node Z is isolated, so $(Y \perp\!\!\!\perp Z \mid T)$ holds trivially. Rule 1 gives:

$$P(y \mid \mathrm{do}(t), z) \;=\; P(y \mid \mathrm{do}(t)).$$

The intervention $\mathrm{do}(t)$ severs Z's only route to Y. Knowing the former cause Z of T is irrelevant once T has been set externally.

Rule 2 — Action/Observation Exchange. When the condition holds, intervening on Z is equivalent to conditioning on Z (within the scope of $\mathrm{do}(x)$ and W).

Example: Rule 2 — Exchanging Action for Observation

Graph: $X \to Z \to Y$, no other edges. Suppose we intervene on X and wish to show $P(y \mid \mathrm{do}(x), \mathrm{do}(z)) = P(y \mid \mathrm{do}(x), z)$.
In $\mathcal{G}_{\overline{X}\,\overline{Z}}$: delete arrows into X (removes the incoming arrows of X) and arrows into Z (removes $X \to Z$). Node Z has no incoming arrows, so no back-door paths from Z to Y can arise. The only path is the causal $Z \to Y$, and $(Y \perp\!\!\!\perp Z)_{\mathcal{G}_{\overline{X}\,\overline{Z}}}$ holds in the sense that no spurious association arises.
Rule 2 gives:

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z)) \;=\; P(y \mid \mathrm{do}(x), z).$$

Y depends on Z only through Z's value, not through the mechanism that generated Z. Externally fixing Z and merely observing $Z{=}z$ are therefore equivalent for Y.

Rule 3 — Deleting an Intervention. The condition says that Z has no causal path to Y that is not blocked by X or W in the appropriate intervention graph. Setting Z to any value has no effect on Y, so $\mathrm{do}(z)$ can be dropped.

Example: Rule 3 — Deleting an Irrelevant Intervention

Graph: $Z \to X \to Y$. Suppose we intervene on both X and Z.
In $\mathcal{G}_{\overline{X}\,\overline{Z}}$: delete arrows into X (removes $Z \to X$), then delete arrows into Z (none exist). Node Z is now isolated. Since there is no path from Z to Y, $(Y \perp\!\!\!\perp Z)_{\mathcal{G}_{\overline{X}\,\overline{Z}}}$ holds trivially. Rule 3 gives:

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z)) \;=\; P(y \mid \mathrm{do}(x)).$$

Once X is fixed externally, Z's only route to Y (through X) is severed. Setting Z to any value cannot change Y, so $\mathrm{do}(z)$ is superfluous.

Example: A Two-Rule Derivation — the Proxy Variable Graph

T has a direct effect on Y. An unobserved variable U is a common cause of T and an observed proxy S; S in turn causes Y. U does *not* directly cause Y. Edges: $T \to Y$, $U \to T$, $U \to S$, $S \to Y$.
Goal: simplify $P(y \mid \mathrm{do}(t))$ to a purely observational expression.
Step 1. Introduce S by the law of total probability:

$$P(y \mid \mathrm{do}(t)) = \sum_{s} P(y \mid \mathrm{do}(t),\, s)\; P(s \mid \mathrm{do}(t)).$$

Step 2. Rule 3 applied to $P(s \mid \mathrm{do}(t))$. Form $\mathcal{G}_{\overline{T}}$ by deleting the arrow $U \to T$. In this graph, T has no parents and no path connects T to S or to the U–S component. Hence $(S \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$, and Rule 3 gives $P(s \mid \mathrm{do}(t)) = P(s)$.
Step 3. Back-door d-separation argument applied to $P(y \mid \mathrm{do}(t), s)$. Form $\mathcal{G}_{\underline{T}}$ by deleting the arrow $T \to Y$. The only back-door path $T \leftarrow U \to S \to Y$ is blocked by conditioning on S. Hence $\{S\}$ satisfies the back-door criterion, so $P(y \mid \mathrm{do}(t), s) = P(y \mid t, s)$.

Combining:

$$P(y \mid \mathrm{do}(t)) = \sum_s P(y \mid t,\, s)\, P(s).$$

The causal effect is identified by standardizing over the population marginal $P(s)$, even though the confounder U is entirely unobserved.

3.4.3 Proofs of the Identification Formulas

Proof: Back-Door Formula via Rule 3 and the Back-Door d-Separation Argument

Graph structure: $T \to Y$, unobserved $U \to T$, $U \to Y$, and observed $\mathbf{S}$ blocking the back-door path $T \leftarrow U \to Y$.
Step 1: Introduce S by the law of total probability.

$$P(y \mid \mathrm{do}(t)) = \int P(y \mid \mathrm{do}(t),\, \mathbf{s})\, dP(\mathbf{s} \mid \mathrm{do}(t)).$$

Step 2: Simplify $P(\mathbf{s} \mid \mathrm{do}(t))$. Because $\mathbf{S}$ contains no descendants of T (condition 1), the graphical condition for Rule 3 is $(\mathbf{S} \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$: in $\mathcal{G}_{\overline{T}}$, all arrows into T are deleted so no directed path from T reaches any node in $\mathbf{S}$. Therefore $(\mathbf{S} \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$ holds, and Rule 3 gives: $P(\mathbf{s} \mid \mathrm{do}(t)) = P(\mathbf{s})$.
Step 3: Convert $P(y \mid \mathrm{do}(t), \mathbf{s})$ to an observation. Condition 2 ensures that, within each stratum $\mathbf{S} = \mathbf{s}$, all back-door paths from T to Y are blocked. By the truncated factorization formula for $P(\cdot \mid \mathrm{do}(t))$, the post-intervention conditional equals the observational conditional: $P(y \mid \mathrm{do}(t), \mathbf{s}) = P(y \mid t, \mathbf{s})$.
Combining:

$$P(y \mid \mathrm{do}(t)) = \int P(y \mid t, \mathbf{s})\, dP(\mathbf{s}). \quad \square$$

Proof: Front-Door Formula via Three Applications of the Rules

Graph structure: $T \to M \to Y$, unobserved $U \to T$, $U \to Y$, no direct $T \to Y$ edge.
Step 1: Introduce M by the law of total probability.

$$P(y \mid \mathrm{do}(t)) = \int P(y \mid \mathrm{do}(t),\, m)\, dP(m \mid \mathrm{do}(t)).$$

Step 2: Simplify $P(m \mid \mathrm{do}(t))$. In $\mathcal{G}_{\underline{T}}$, there are no back-door paths from T to M by front-door condition 2. Rule 2 applies: $P(m \mid \mathrm{do}(t)) = P(m \mid t)$.
Step 3a: $P(y \mid \mathrm{do}(t), m) = P(y \mid \mathrm{do}(m))$. In $\mathcal{G}_{\overline{T}}$, the arrow $U \to T$ is deleted, so U and T become independent. Since there is no arrow from U into M (condition 2), the conditional of Y given $M{=}m$ integrates over U freely: $P(y \mid \mathrm{do}(t), m) = \int P(y \mid m, u)\, dP(u) = P(y \mid \mathrm{do}(m))$.
Step 3b: By front-door condition 3, $\{T\}$ satisfies the back-door criterion for the effect of M on Y, so $P(y \mid \mathrm{do}(m)) = \int P(y \mid m, t')\, dP(t')$.
Assembling:

$$P(y \mid \mathrm{do}(t)) = \int \left[\int P(y \mid t',\, m)\, dP(t') \right] dP(m \mid t). \quad \square$$

3.5 The Do-Calculus is Complete

A natural question after seeing the three rules is: are they enough? Could there be a graph where the causal effect is identifiable in principle, but none of the rules — applied in any sequence — can derive a purely observational expression? Shpitser and Pearl (2006) answered this definitively.

Throughout Sections 1–4 we have represented latent common causes as explicit unobserved nodes. The completeness theorem uses a more general graph class that encodes latent common causes compactly as *bidirected edges*: a bidirected edge $X \leftrightarrow Y$ stands for an unobserved variable U with $U \to X$ and $U \to Y$. A DAG augmented with such bidirected edges is called a *semi-Markovian DAG*.

Theorem: Completeness of the Do-Calculus

(Shpitser and Pearl 2006) Let $\mathcal{G}$ be a semi-Markovian DAG. A causal quantity $P(y \mid \mathrm{do}(t))$ is identifiable from P relative to $\mathcal{G}$ if and only if the do-calculus can derive a purely observational expression for it.

Example: The Bow Graph — The Simplest Non-Identifiable Structure

The *bow graph* is the semi-Markovian DAG with one directed edge $T \to Y$ and one bidirected edge $T \leftrightarrow Y$ (representing an unobserved U with $U \to T$ and $U \to Y$). The single back-door path is $T \leftarrow U \to Y$; no observed set can block it, so the back-door criterion cannot be satisfied. No observed mediator exists, so the front-door criterion also fails.

We show directly that $P(y \mid \mathrm{do}(t))$ is not determined by $P(T, Y)$ alone, by constructing two models that agree on $P(T, Y)$ but disagree on $P(y \mid \mathrm{do}(t))$.

Let $T, Y, U \in \{0, 1\}$ with $U \sim \text{Bernoulli}(1/2)$.

Model $\mathcal{M}_1$ (pure confounding; no causal effect of T): $T = U$, $Y = U$. Both T and Y are driven entirely by U. The observed joint distribution is $P(T{=}0, Y{=}0) = P(T{=}1, Y{=}1) = 1/2$ and $P(T \neq Y) = 0$. Under $\mathrm{do}(T{=}1)$, $Y = U$ is unchanged, so $P_1(Y{=}1 \mid \mathrm{do}(T{=}1)) = P(U{=}1) = 1/2$.

Model $\mathcal{M}_2$ (full causal effect; U acts on Y only through T): $T = U$, $Y = T$. Since $T = U$, we again have $Y = T = U$, so the observed joint distribution is *identical* to $\mathcal{M}_1$. Under $\mathrm{do}(T{=}1)$, $Y = T = 1$ deterministically, so $P_2(Y{=}1 \mid \mathrm{do}(T{=}1)) = 1$.

Conclusion. The two models produce identical observed distributions $P(T, Y)$, yet $P(Y{=}1 \mid \mathrm{do}(T{=}1))$ equals $1/2$ in $\mathcal{M}_1$ versus 1 in $\mathcal{M}_2$. Therefore $P(y \mid \mathrm{do}(t))$ is *not identified* from $P(T, Y)$ in the bow graph.

Proof Sketch and the ID Algorithm

The *ID algorithm* (Shpitser and Pearl 2006) takes as input a semi-Markovian DAG $\mathcal{G}$, a target intervention distribution $P(y \mid \mathrm{do}(t))$, and the observed joint distribution P, and either returns an explicit observational formula or reports non-identifiability.

Sufficiency (do-calculus identifies whenever identification is possible): proved constructively via the ID algorithm. The key structural concept is the *c-component* (confounded component): a maximal set of nodes connected by bidirected edges. The ID algorithm proceeds recursively: (1) simplification — remove from $\mathcal{G}$ any node that is not an ancestor of Y in $\mathcal{G}_{\overline{T}}$; (2) decomposition by c-components — factorise $P(\mathbf{v} \setminus \mathbf{t} \mid \mathrm{do}(t))$ over the c-components; (3) recursive identification — attempt to identify each c-component factor; (4) failure — if any subproblem cannot be reduced, the algorithm returns FAIL.

Necessity (do-calculus cannot identify non-identifiable quantities): proved by showing that whenever the ID algorithm fails, the causal effect is genuinely non-identifiable. The key graphical obstruction is the *hedge*: a pair of subgraphs (F, F') where F contains a back-door path from T to Y, and F' is a subgraph of $\mathcal{G}_{\overline{T}}$ in which Y remains connected to a latent common cause. The existence of a hedge allows explicit construction of two observationally equivalent models with different causal effects.

Combining: the do-calculus succeeds if and only if no hedge exists, which is if and only if $P(y \mid \mathrm{do}(t))$ is identifiable. □

Remark

The completeness theorem has a crucial practical implication: *if the ID algorithm reports non-identification, then no estimation method — however clever — can recover the causal effect from observational data alone*, given the assumed graph structure. Non-identification is not a limitation of a particular technique; it is a fundamental property of the causal model. Three remedies exist: impose additional structural assumptions (e.g., linearity or monotonicity) that go beyond the graph; collect additional data that break the obstruction (e.g., an instrument or a randomized experiment); or target a different, identified causal quantity instead.

3.6 The Big Picture

This chapter completes the *theoretical* identification machinery of the course. The chain of reasoning begun in Chapter 1 (the interventional/observational distinction) and formalized in Chapter 2 (d-separation) now terminates in a concrete identification formula. Chapters 4–7 then apply this machinery to specific research designs.

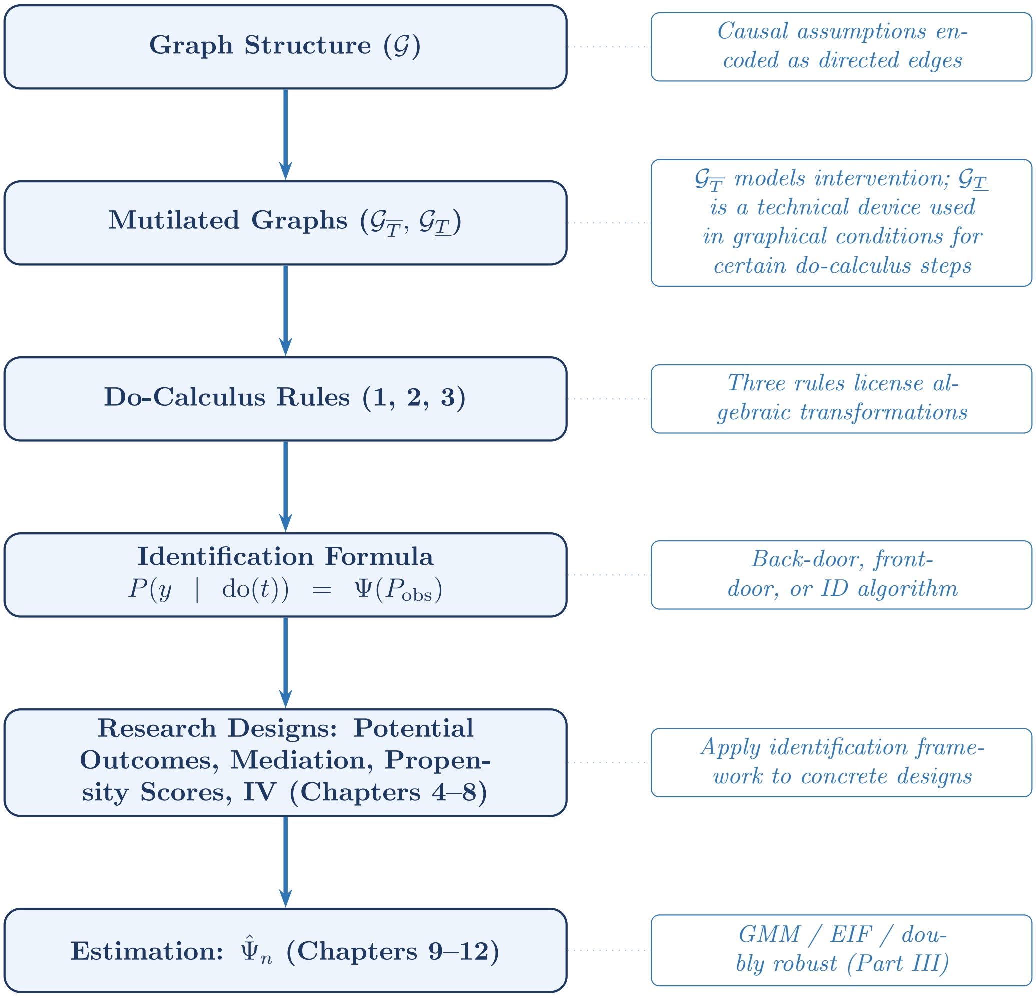


Figure 3.4: The six-step pipeline from causal assumptions to estimation.

The completeness theorem closes the loop: every causal effect that is identifiable from the observational distribution relative to the assumed graph can be derived by the do-calculus, and every effect that is not identifiable in that setting cannot be recovered by any observational method without additional assumptions or new data.

3.7 Summary

1. Two graph operations underpin the do-calculus: $\mathcal{G}_{\overline{X}}$ deletes arrows *into* X and models the intervention $\mathrm{do}(x)$; $\mathcal{G}_{\underline{X}}$ deletes arrows *out of* X and is used as a technical device in graphical conditions for certain do-calculus steps — it does not model conditioning on X.

2. The back-door criterion identifies $P(y \mid \mathrm{do}(t))$ when an observed set $\mathbf{S}$ blocks all back-door paths and contains no descendant of T. The general formula is $\int P(y \mid t, \mathbf{s})\, dP(\mathbf{s})$, the standardization formula (the g-formula for $K = 1$).

3. The front-door criterion identifies $P(y \mid \mathrm{do}(t))$ via an observed mediator M intercepting all causal paths from T to Y, even when the confounder U is entirely unobserved, provided the three front-door conditions hold.

4. The do-calculus has three rules, each licensed by a d-separation condition in an appropriate intervention graph. Rule 1 adds/removes observations; Rule 2 exchanges actions for observations; Rule 3 deletes actions. The back-door and front-door formulas are each derivable as three-step sequences of these rules.

5. **Completeness** (Shpitser and Pearl 2006): the do-calculus is complete for identification from observational data relative to the assumed graph. A causal effect is identifiable if and only if the do-calculus can derive a purely observational expression for it. If the do-calculus cannot identify the quantity in that setting, then no purely observational method can do so without additional assumptions or new data.

3.8 Problems

1. Warm-up: graph operations. Let $\mathcal{G}$ be the IV DAG with edges $Z \to T$, $T \to Y$, $U \to T$, $U \to Y$ (U unobserved).

(a) Draw $\mathcal{G}_{\overline{T}}$ and identify all paths from Z to Y that remain open.
(b) Draw $\mathcal{G}_{\underline{T}}$ and identify all paths from Z to Y that remain. For each path, classify it as a causal chain, fork, or collider path, and state whether it is open or closed by default.
(c) In $\mathcal{G}_{\overline{T}}$, is $Z \perp\!\!\!\perp Y \mid \emptyset$? How does the absence of a direct edge $Z \to Y$ in the DAG relate to the exclusion restriction?

2. Back-door practice. Consider the DAG: $X \to T$, $X \to Y$, $T \to M$, $M \to Y$, $T \to Y$, where X is observed.

(a) List all back-door paths from T to Y.
(b) Does $\{X\}$ satisfy the back-door criterion for the effect of T on Y? Write the resulting adjustment formula.
(c) Does $\{M\}$ satisfy the back-door criterion? Explain why or why not.
(d) Does $\{X, M\}$ satisfy the back-door criterion? Identify which condition of the criterion M violates.
(e) Even if the formal criterion issue in (d) were somehow set aside, explain the substantive bias that arises from conditioning on a mediator M that lies on a causal path $T \to M \to Y$.

3. Front-door identification. Consider the DAG: $T \to M \to Y$, with $U \to T$ and $U \to Y$ (unobserved U, no direct $T \to Y$ edge).

(a) Verify that M satisfies all three front-door conditions.
(b) Derive the front-door formula Equation 8.13 step by step, citing the do-calculus rule used at each step.
(c) Now add a direct edge $T \to Y$ to the graph. Does M still satisfy the front-door criterion? Explain which condition fails.

4. Identification or non-identification? For each of the following DAGs, determine whether $P(y \mid \mathrm{do}(t))$ is identified non-parametrically from observational data. If identified, state the formula; if not, explain the structural obstruction.

(a) $T \to Y$, $U \to T$, $U \to Y$, U unobserved.
(b) Same as (a), with instrument $Z \to T$ and $Z \perp\!\!\!\perp U$ (no direct $Z \to Y$ edge). Show that $P(y \mid \mathrm{do}(t))$ is **not** identified non-parametrically from $P(Z, T, Y)$ by constructing two binary SEMs $\mathcal{M}_1$ and $\mathcal{M}_2$, each compatible with the IV graph, such that (i) $P_{\mathcal{M}_1}(Z, T, Y) = P_{\mathcal{M}_2}(Z, T, Y)$ and (ii) $P_{\mathcal{M}_1}(Y{=}1 \mid \mathrm{do}(T{=}1)) \neq P_{\mathcal{M}_2}(Y{=}1 \mid \mathrm{do}(T{=}1))$. (*Hint*: let $Z, U \sim \mathrm{Bern}(1/2)$ independently and set $T = U$ in both models. Then set $Y = U$ in $\mathcal{M}_1$ and $Y = T$ in $\mathcal{M}_2$.)
(c) $T \to M \to Y$, $U \to T$, $U \to Y$, $U \to M$, U unobserved.
(d) $T \to Y$, $X \to T$, $X \to Y$, X observed.

5. Back-door formula: proof and uniqueness.

(a) Let $\mathcal{M}$ be a structural causal model Markov with respect to $\mathcal{G}$, and let $\mathbf{S}$ satisfy the back-door criterion for (T, Y) in $\mathcal{G}$ with positivity. Prove that $P(y \mid \mathrm{do}(t)) = \int P(y \mid t, \mathbf{s})\, dP(\mathbf{s})$. (*Hint*: begin with $P(y \mid \mathrm{do}(t)) = \int P(y \mid \mathrm{do}(t), \mathbf{s})\, P(\mathbf{s} \mid \mathrm{do}(t))\, d\mathbf{s}$. Apply the Rule 3 graphical condition $(\mathbf{S} \perp\!\!\!\perp T)_{\mathcal{G}_{\overline{T}}}$, using back-door condition 1, to simplify $P(\mathbf{s} \mid \mathrm{do}(t))$. Then invoke the d-separation $(Y \perp\!\!\!\perp T \mid \mathbf{S})_{\mathcal{G}_{\underline{T}}}$, which back-door condition 2 implies, together with the truncated factorization, to argue $P(y \mid \mathrm{do}(t), \mathbf{s}) = P(y \mid t, \mathbf{s})$.)

(b) Suppose $\mathbf{S}_1$ and $\mathbf{S}_2$ both satisfy the back-door criterion for (T, Y) in $\mathcal{G}$ (with positivity). Deduce from part (a) that $\int P(y \mid t, \mathbf{s}_1)\, dP(\mathbf{s}_1) = \int P(y \mid t, \mathbf{s}_2)\, dP(\mathbf{s}_2)$. Interpret this result: the identification target $P(y \mid \mathrm{do}(t))$ is unique even when the adjustment set is not.

6. Rule sequencing on an unseen graph. Consider the graph: $C_1 \to T$, $C_1 \to Y$, $U \to T$, $U \to C_2$, $C_2 \to Y$, $T \to Y$, with C_1 and C_2 observed and U unobserved.

Derive $P(y \mid \mathrm{do}(t)) = \sum_{c_1, c_2} P(y \mid t, c_1, c_2)\, P(c_1, c_2)$ from first principles using the three rules. For each step: (i) name the rule applied, (ii) form the relevant intervention graph by stating which arrows are deleted and what edges remain, and (iii) verify the required d-separation condition by tracing all paths and confirming each is blocked.

7. Non-identifiability by construction. Consider the graph from the front-door failure example: $T \to M \to Y$, $U \to T$, $U \to M$, $U \to Y$ (front-door condition 2 violated). Let $T, M, Y, U \in \{0, 1\}$ with $U \sim \mathrm{Bern}(1/2)$.

(a) Construct two SEMs $\mathcal{M}_1$ and $\mathcal{M}_2$, each compatible with the graph, such that $P_{\mathcal{M}_1}(T, M, Y) = P_{\mathcal{M}_2}(T, M, Y)$ but $P_{\mathcal{M}_1}(Y{=}1 \mid \mathrm{do}(T{=}1)) \neq P_{\mathcal{M}_2}(Y{=}1 \mid \mathrm{do}(T{=}1))$. (*Hint*: set $T = U$ in both models. In $\mathcal{M}_1$ set $M = U$ and $Y = U$; in $\mathcal{M}_2$ set $M = T$ and $Y = M$.)

(b) Explain precisely which identification strategy fails in this graph and why, referencing (i) the back-door criterion, (ii) the front-door criterion (which condition fails and which stage collapses), and (iii) the completeness theorem (what does the existence of your two models imply?).

Chapter 4

Potential Outcomes and Adjustment

Learning Objectives

By the end of this chapter, students should be able to:

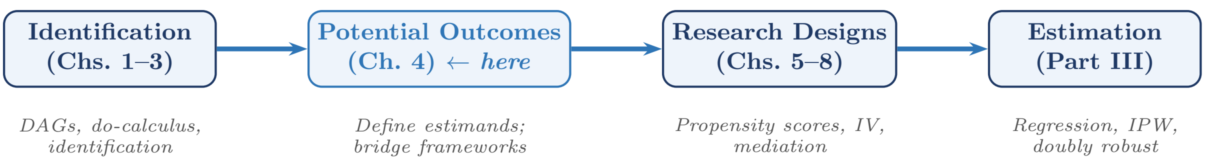
1. Define the potential outcome $Y(t)$, state SUTVA precisely, and derive the consistency equation $Y = Y(T)$.
2. Define the ATE and ATT, explain how they relate to each other, and re-express them as functionals of the structural equation for Y.
3. Show that the linear SEM structural coefficient β equals the ATE under homogeneous effects, and explain why ATE and ATT diverge under treatment effect heterogeneity.
4. State the ignorability assumption, interpret it graphically as a back-door condition, and explain why overlap is a necessary companion assumption.
5. Derive the adjustment formula for the ATE under strong ignorability and positivity, and explain how the back-door criterion provides the graphical justification for conditional exchangeability.

4.1 Motivation: A Third Language for Causality

The first three chapters developed causal inference primarily in the languages of structural equations and directed acyclic graphs. In this chapter we introduce a third language: the potential outcomes framework of Neyman (1923) and Rubin (1974).

Potential outcomes provide a direct language for defining causal estimands. Quantities such as the average treatment effect, the average treatment effect on the treated, and related causal contrasts are most naturally written in terms of counterfactual outcomes $Y(1)$, $Y(0)$, and more generally $Y(t)$. The two frameworks should be viewed as complementary rather than competing: potential outcomes define the causal quantities of interest, while DAGs and the do-calculus clarify identification.

The purpose of this chapter is therefore not to replace the graphical viewpoint developed earlier, but to connect it to the potential-outcomes notation that dominates much of the applied literature. We first define potential outcomes and the standard causal estimands built from them. We then show how ignorability and positivity together identify the average treatment effect, and how the back-door criterion provides the graphical justification for that identification. For readers who want a formal graphical representation of counterfactual variables, Appendix B introduces single world intervention graphs (SWIGs).



4.2 The Neyman–Rubin Potential Outcomes Framework

4.2.1 The Potential Outcome

Definition: Potential Outcome

(Neyman 1923; Rubin 1974) For each unit i and each possible treatment value $t \in \mathcal{T}$, the *potential outcome* $Y_i(t)$ is the value of the outcome that unit i *would have exhibited* had its treatment been set to t, possibly contrary to fact.
The collection $\{Y_i(t) : t \in \mathcal{T}\}$ is called the *schedule of potential outcomes* for unit i. In the binary case $\mathcal{T} = \{0, 1\}$, the schedule is $(Y_i(0), Y_i(1))$.

$Y_i(1)$ is the outcome unit i would achieve *under treatment*; $Y_i(0)$ is the outcome under *control*. Only one of these is ever observed for any given unit. The unobserved potential outcome is called the *counterfactual*.

The Fundamental Problem of Causal Inference

(Holland 1986) For any unit i, at most one potential outcome $Y_i(t)$ is observed. The individual-level causal effect $Y_i(1) - Y_i(0)$ is therefore *never directly observable*. Causal inference is an inference problem precisely because this quantity must be recovered from a population of units rather than from a single unit.

4.2.2 SUTVA and Consistency

Definition: SUTVA

(Rubin 1980) The *stable unit treatment value assumption* has two components:

1. **No interference.** The potential outcome $Y_i(t)$ depends only on unit i's own treatment, not on the treatments of other units: $Y_i(t_1, \dots, t_n) = Y_i(t_i)$.
2. **No hidden versions.** For each treatment level t, there is a single well-defined version.

The consistency equation. Under SUTVA, the observed outcome Y_i equals the potential outcome at the treatment actually received:

$$Y_i \;=\; Y_i(T_i). \tag{4.1}$$

This equation is the bridge between the potential outcome world and the observed data world. It fails whenever SUTVA fails: if treatment spills over between units (e.g., vaccination provides herd immunity), or if the treatment label $T = 1$ covers multiple distinct interventions.

4.2.3 Connection to the Do-Operator

The notation $Y(t)$ and the expression $P(Y \mid \mathrm{do}(T{=}t))$ refer to the same causal idea viewed from two complementary angles. Under the structural causal framework adopted in these notes, together with consistency and no interference, these two descriptions coincide.

Proposition: Potential Outcomes and the Do-Operator

Under the structural causal model of Chapters 1–3, together with consistency and no interference (SUTVA),

$$Y(t) \;\overset{d}{=}\; Y \mid \mathrm{do}(T{=}t), \tag{4.2}$$

so that $P(Y(t) \leq y) = P(Y \leq y \mid \mathrm{do}(T{=}t))$ for every y. Equivalently, $\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid \mathrm{do}(T{=}t)]$.

Proof Sketch

In the mutilated SEM $\mathcal{G}_{\overline{T}}$, the structural equation for T is replaced by $T := t$ for every unit. The structural model then determines the value of Y for unit i from t and unit i's own background variables — which is precisely what the potential outcomes framework defines as $Y_i(t)$. SUTVA's no-interference condition ensures $Y_i(t)$ does not depend on other units' treatment values, so the marginal distribution of Y in $\mathcal{G}_{\overline{T}}$ equals the marginal distribution of $Y(t)$. By definition of the do-operator, the former is $f(y \mid \mathrm{do}(T{=}t))$. $\square$

This equivalence lets us move freely between two notational traditions. When defining estimands such as $\mathbb{E}[Y(1)-Y(0)]$, potential-outcome notation is often most natural. When proving identification results from a graph, the do-operator is often more transparent: $f(y \mid \mathrm{do}(t)) \neq f(y \mid T{=}t)$ whenever T is endogenous, whereas $Y(t)$ carries no such syntactic marker.

4.3 Causal Estimands

4.3.1 The Average Treatment Effect and Its Relatives

Definition: ATE and ATT

For a binary treatment $T \in \{0,1\}$:

- The *average treatment effect* (ATE) is $\tau_{\text{ATE}} = \mathbb{E}[Y(1) - Y(0)]$.
- The *average treatment effect on the treated* (ATT) is $\tau_{\text{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid T{=}1]$.

The ATE and ATT coincide only when the treatment effect does not depend on who selected into treatment. The naïve estimator $\hat{\tau}_{\text{naive}} = \bar{Y}_{T=1} - \bar{Y}_{T=0}$ estimates $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0] \neq \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$, with the gap being the selection bias induced by endogeneity.

Remark

ATE and ATT coincide when $\mathbb{E}[Y(t) \mid T] = \mathbb{E}[Y(t)]$ for $t \in \{0,1\}$, i.e., when treatment assignment is mean-independent of potential outcomes. In a randomized experiment, this holds by design.

4.3.2 Causal Estimands as Functionals of the Structural Equation

From structural equation to potential outcome. In the SEM, the outcome is determined by a structural equation $Y = g(T, \mathbf{X}, U_Y)$. The potential outcome under $\mathrm{do}(T{=}t)$ is:

$$Y_i(t) \;=\; g(t,\, \mathbf{X}_i,\, U_{Y,i}). \tag{4.3}$$

ATE and ATT as structural parameters.

$$\tau_{\text{ATE}} = \mathbb{E}[g(1, \mathbf{X}, U_Y) - g(0, \mathbf{X}, U_Y)]\,, \tag{4.4}$$

$$\tau_{\text{ATT}} = \mathbb{E}[g(1, \mathbf{X}, U_Y) - g(0, \mathbf{X}, U_Y) \mid T{=}1]\,. \tag{4.5}$$

The linear SEM as a special case. In the Gaussian linear SEM $Y = \beta T + \gamma^\top \mathbf{X} + \varepsilon$, the unit-level effect is $Y_i(1) - Y_i(0) = \beta$ for every unit. Consequently, $\tau_{\text{ATE}} = \tau_{\text{ATT}} = \beta$.

Heterogeneous effects. In the nonparametric SEM Equation 4.3, the unit-level effect varies across units. The ATE and ATT differ whenever treatment selection correlates with the individual effect size.

Remark

The SEM representation Equation 4.3 clarifies why the potential outcomes framework and the do-calculus are *equivalent in content but different in emphasis*. The do-calculus works with the interventional distribution $f(y \mid \mathrm{do}(T{=}t))$ as a population-level object. The potential outcomes framework works with unit-level quantities $Y_i(t)$ and asks what population summaries (ATE, ATT) are scientifically meaningful. The SEM ties the two together.

4.4 Ignorability, Positivity, and Adjustment

4.4.1 Strong Ignorability

The central identifying assumption in observational studies is that treatment assignment is as good as random after conditioning on observed covariates X.

Definition: Strong Ignorability

(Rosenbaum and Rubin 1983) The treatment assignment T is *strongly ignorable* given X if:

1. **Unconfoundedness:** $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$.
2. **Overlap (positivity):** $0 < P(T{=}1 \mid X{=}x) < 1$ for all x in the support of X.

4.4.2 Adjustment Formula under Ignorability

Under strong ignorability, the ATE is identified by the standardization formula:

$$\tau_{\text{ATE}} \;=\; \mathbb{E}_X[\mathbb{E}[Y \mid T{=}1, X] - \mathbb{E}[Y \mid T{=}0, X]]\,. \tag{4.6}$$

This is the back-door adjustment formula of Chapter 3 written in potential-outcome notation.

Derivation

For each treatment level $t \in \{0, 1\}$, by the law of iterated expectations:

$$\mathbb{E}[Y(t)] = \mathbb{E}_X[\mathbb{E}[Y(t) \mid X]]\,.$$

Under unconfoundedness, conditioning on X renders the potential outcome mean-independent of treatment assignment:

$$\mathbb{E}[Y(t) \mid X] = \mathbb{E}[Y(t) \mid T{=}t,\, X].$$

By consistency, $\mathbb{E}[Y(t) \mid T{=}t,\, X] = \mathbb{E}[Y \mid T{=}t,\, X]$. Therefore:

$$\mathbb{E}[Y(t)] = \mathbb{E}_X[\mathbb{E}[Y \mid T{=}t,\, X]]\,.$$

Applying this once for $t = 1$ and once for $t = 0$, and taking the difference, yields Equation 4.6. Overlap ensures that the conditional means on the right-hand side are well-defined for all covariate values. □

Example: A Two-Stratum Adjustment Calculation

Suppose $X \in \{0, 1\}$ with $P(X{=}1) = 0.4$ and $P(X{=}0) = 0.6$, and the observed conditional means are $\mathbb{E}[Y \mid T{=}1, X{=}1] = 8$, $\mathbb{E}[Y \mid T{=}0, X{=}1] = 6$, $\mathbb{E}[Y \mid T{=}1, X{=}0] = 5$, $\mathbb{E}[Y \mid T{=}0, X{=}0] = 4$. Under ignorability and positivity, Equation 4.6 gives:

$$\mathbb{E}[Y(1)] = 0.4 \times 8 + 0.6 \times 5 = 6.2, \qquad \mathbb{E}[Y(0)] = 0.4 \times 6 + 0.6 \times 4 = 4.8.$$

Hence $\tau_{\text{ATE}} = 6.2 - 4.8 = 1.4$.

4.4.3 Back-Door Interpretation

The potential-outcomes assumption $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ is the counterfactual expression of the idea that, after conditioning on X, treatment assignment carries no residual information about the outcome.

In the language of DAGs, the closely related idea is that X blocks all back-door paths from T to Y.

Proposition: Back-Door Criterion Implies Ignorability

Under the Markov assumption for $\mathcal{G}$, if X satisfies the back-door criterion for the effect of T on Y — that is, (1) no node in X is a descendant of T, and (2) X blocks every back-door path from T to Y — then unconfoundedness ($Y(t) \perp\!\!\!\perp T \mid X$) holds for all t.

Proof Sketch

In the mutilated graph $\mathcal{G}_{\overline{T}}$, all arrows into T are deleted, severing every back-door path from T to Y. The back-door criterion guarantees that X blocks all remaining non-causal paths between T and Y in $\mathcal{G}_{\overline{T}}$. By the Markov property for the interventional model represented by $\mathcal{G}_{\overline{T}}$, this yields conditional independence between treatment and outcome under intervention. Under the structural causal interpretation adopted in these notes, that interventional independence corresponds to $Y(t) \perp\!\!\!\perp T \mid X$ for all t. A fully explicit graphical version of this translation is given in Appendix B using SWIGs. □

Remark

This is the direction most important for practice: a graphical adjustment set justifies the counterfactual independence needed for standardization, regression adjustment, and propensity-score methods, requiring only the Markov property. The converse — whether every ignorability statement corresponds to a back-door condition — additionally requires faithfulness and is established in Appendix B via the SWIG construction.

Example: Labor Training Program

(LaLonde 1986; Dehejia and Wahba 1999) Let T = receipt of job training, Y = earnings two years later, X = (age, education, prior earnings, race, marital status).
The back-door path $T \leftarrow X \rightarrow Y$ is blocked by conditioning on X, leaving only the causal path $T \rightarrow Y$. Ignorability holds if this DAG is correctly specified. If there is an unobserved variable U (motivation, ability) that affects both training participation and earnings, X fails the back-door criterion and ignorability fails.

Together, ignorability and positivity identify the ATE via Equation 4.6, and the back-door criterion provides the graphical justification for why that formula recovers the causal effect. Appendix B provides a formal graphical representation of this connection through SWIGs.

4.4.4 Overlap and Positivity

Why Overlap Is Non-Negotiable

Without overlap, some covariate strata contain only treated or only control units. In those strata, $\mathbb{E}[Y \mid T{=}0, X{=}x]$ or $\mathbb{E}[Y \mid T{=}1, X{=}x]$ is unobservable, and Equation 4.6 cannot be evaluated at those values of x. Identification of the ATE fails not because the causal structure is wrong, but because the data do not span the support needed. The ATT may remain identified even when full overlap fails, because ATT averages only over the treated population where $P(T{=}1 \mid X) > 0$ by construction.

4.5 Where the Frameworks Agree and Diverge

Task	**Potential Outcomes**	**Do-Calculus / DAG**	**SEM**
Define causal estimands (ATE, ATT)	Especially natural notation	Via $\mathbb{E}[Y \mid \mathrm{do}(t)]$	Via structural equations
Encode causal assumptions	Assumptions typically stated informally; causal graph implicit	Explicit directed edges	Structural equations
Read conditional independence	Requires auxiliary graph or model	d-separation	With graph only
Identification from observational data	Via ignorability assumptions	Back-door, front-door, do-calculus	Via exclusion restrictions
Cross-world assumptions (e.g., monotonicity)	Natural to state	See Appendix B (SWIGs)	Implicit in structural model
Standard in statistics/epidemiology	Dominant	Growing rapidly	Econometrics

Our Position in This Course

Both frameworks are indispensable. We use *potential outcomes* to *define* causal estimands. We use the *do-calculus and DAGs* for *identification*. The back-door criterion is the graphical condition that justifies conditional ignorability and hence the adjustment formula through which causal effects are identified. For a formal graphical representation that hosts both frameworks simultaneously, see Appendix B.

The do-operator language is preferred throughout this course because it makes the interventional/observational distinction *syntactically enforced*: $f(y \mid \mathrm{do}(t))$ cannot be confused with $f(y \mid T{=}t)$, whereas $\mathbb{E}[Y(t)]$ carries no such syntactic marker.

4.6 Summary

1. The potential outcome $Y(t)$ is the outcome that would be observed under the intervention $T = t$. Under consistency, no interference, and the structural causal semantics adopted in these notes, the observed outcome satisfies $Y_i = Y_i(T_i)$, and the potential outcome $Y(t)$ has the same distribution as the outcome under the intervention $\mathrm{do}(T{=}t)$.

2. The ATE and ATT are averages of the unit-level causal effect $Y_i(1) - Y_i(0) = g(1, \mathbf{X}_i, U_{Y,i}) - g(0, \mathbf{X}_i, U_{Y,i})$ over the full population and the treated subpopulation respectively. In the linear SEM, both equal the structural coefficient β. They diverge under heterogeneous effects when treatment selection correlates with individual effect size.

3. Strong ignorability $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ with overlap identifies the ATE via Equation 4.6. The back-door criterion provides a sufficient graphical condition for the conditional exchangeability assumption used in adjustment formulas.

4. SWIGs (Richardson and Robins 2014) provide a formal graphical representation of counterfactual variables, making ignorability a d-separation statement in a single diagram that hosts both potential outcomes and the natural treatment. This material is developed in Appendix B and is not required for a first-pass understanding of adjustment.

5. The frameworks are complementary: potential outcomes define estimands; the do-calculus identifies them. The back-door criterion connects the two by providing the graphical condition under which adjustment recovers the causal effect.

In observational studies, ignorability must be justified by substantive knowledge encoded in a causal graph. Randomized experiments provide a fundamentally different solution: when treatment is assigned randomly, $(Y(0), Y(1)) \perp\!\!\!\perp T$ holds by design, guaranteeing ignorability without any covariate adjustment. Chapter 5 studies randomized experiments as the canonical design where causal effects are identifiable directly from the data-generating mechanism.

4.7 Problems

1. SUTVA and consistency. Suppose $n = 3$ units receive binary treatments (T_1, T_2, T_3) and unit i's outcome may depend on all three treatments: $Y_i = Y_i(T_1, T_2, T_3)$.

(a) How many potential outcomes does unit 1 have? Write them out explicitly.
(b) SUTVA imposes no interference: $Y_i(T_1, T_2, T_3) = Y_i(T_i)$. How many distinct potential outcomes remain?
(c) Give a real-world example where no-interference plausibly holds and one where it plausibly fails. In the latter case, explain what identification strategy (if any) remains available.

2. ATE, ATT, and selection bias. Let $(Y(0), Y(1), T) \sim P$ with $P(T{=}1) = 0.5$. Suppose $\mathbb{E}[Y(1)] = 3$, $\mathbb{E}[Y(0)] = 1$, $\mathbb{E}[Y(1) \mid T{=}1] = 4$, $\mathbb{E}[Y(0) \mid T{=}1] = 2$, $\mathbb{E}[Y(1) \mid T{=}0] = 2$, $\mathbb{E}[Y(0) \mid T{=}0] = 0$.

(a) Compute the ATE and ATT. Are they equal?
(b) Compute the naïve estimator $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$. Decompose the gap between this and the ATE into a selection bias term and an ATT–ATE difference.
(c) Under what graphical condition on the DAG would the naïve estimator equal the ATE? State this condition in both potential outcome language (ignorability) and do-calculus language (back-door criterion).

3. Causal estimands from the structural equation. Consider the nonparametric SEM $Y = g(T, X, U_Y)$ with binary treatment $T \in \{0, 1\}$, observed covariate X, and unobserved error U_Y independent of (T, X) in the mutilated graph.

(a) Write τ_{ATE} and τ_{ATT} as expectations of $g(1, X, U_Y) - g(0, X, U_Y)$ over the appropriate distribution. Under what condition do they coincide?
(b) Specialize to the linear SEM $g(t, x, u) = \alpha + \beta t + \gamma x + u$. Show that the unit-level causal effect $Y_i(1) - Y_i(0)$ is constant across all units, and hence $\tau_{\text{ATE}} = \tau_{\text{ATT}} = \beta$.
(c) Now consider the heterogeneous SEM $g(t, x, u) = (\alpha{+}u)\, t{+}\gamma x$, where $U_Y \sim \mathcal{N}(0, \sigma^2)$, $\text{Cov}(U_Y, T) = \rho$, and $P(T{=}1) = p \in (0, 1)$. Compute τ_{ATE} and τ_{ATT}. Show that they differ when $\rho \neq 0$, and interpret this difference.
(d) In part (c), explain why OLS estimation of Y on T and X does not recover either τ_{ATE} or τ_{ATT} when $\rho \neq 0$.

4. SWIGs and ignorability. *(Requires Appendix B.)* Consider the DAG: $X \to T$, $X \to Y$, $T \to Y$, with X fully observed.

(a) Construct the SWIG $\mathcal{G}(t)$ by splitting T into its random and fixed halves. Draw the result, labeling the random half, fixed half, and the potential outcome $Y(t)$.
(b) In $\mathcal{G}(t)$, identify all paths between the random half T and $Y(t)$. Apply the SWIG d-separation rules to determine which paths are blocked and which are open before conditioning.
(c) Use d-separation in $\mathcal{G}(t)$ to verify that $(Y(t) \perp\!\!\!\perp T \mid X)_{\mathcal{G}(t)}$ holds. Conclude that X satisfies the back-door criterion, confirming the proposition above.
(d) Now add a hidden common cause $U \to T$, $U \to Y$. Draw the revised SWIG. Does the ignorability argument still hold? State the correct conclusion and identify what additional structure (if any) would be needed for identification.

Chapter 5

Randomization and Back-Door Adjustment

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain in do-calculus terms why a randomized experiment makes $f(y \mid \mathrm{do}(T{=}t)) = f(y \mid T{=}t)$, and identify the graphical feature that produces this equality.
2. State the ignorability (unconfoundedness) assumption in both potential-outcomes and graphical language, and distinguish weak from strong ignorability.
3. Apply the regression-adjusted estimator to reduce variance in a randomized experiment, and explain why consistency holds even under model misspecification.
4. Derive and compute the standardization (g-formula) estimator of the ATE from a contingency table or regression output.
5. Explain the method of stratification (subclassification) and describe when and why it removes confounding bias.
6. Identify Simpson's paradox in a numerical example and resolve it using the back-door criterion.
7. Articulate why propensity score methods are needed when X is high-dimensional, motivating Chapter 6.

5.1 Randomized Experiments

5.1.1 The Do-Calculus of Randomization

The defining feature of a randomized experiment is that the analyst *sets* the treatment for each unit, independently of all background variables. In do-calculus terms, the observed data come from the mutilated distribution $f(y \mid \mathrm{do}(T{=}t))$ rather than the observational distribution $f(y \mid T{=}t)$.

The Fundamental Equality of Randomized Experiments

In a completely randomized experiment, T is assigned independently of all pre-treatment variables. The DAG has no arrows into T: every back-door path from T to Y is severed. Consequently:

$$f(y \mid \mathrm{do}(T{=}t)) \;=\; f(y \mid T{=}t)\,. \tag{5.1}$$

The interventional and observational conditional distributions coincide.

In a completely randomized design (CRD), the treatment node T has no parents in the DAG. Graphically, this means:

- No back-door paths from T to Y exist.
- T is independent of all pre-treatment covariates: $T \perp\!\!\!\perp (X, U)$.

- In the mutilated graph $\mathcal{G}_{\overline{T}}$, nothing changes relative to the original DAG because T already had no parents.

Figure 5.1 illustrates the contrast between the observational DAG (with back-door paths) and the randomized DAG (all arrows into T severed).

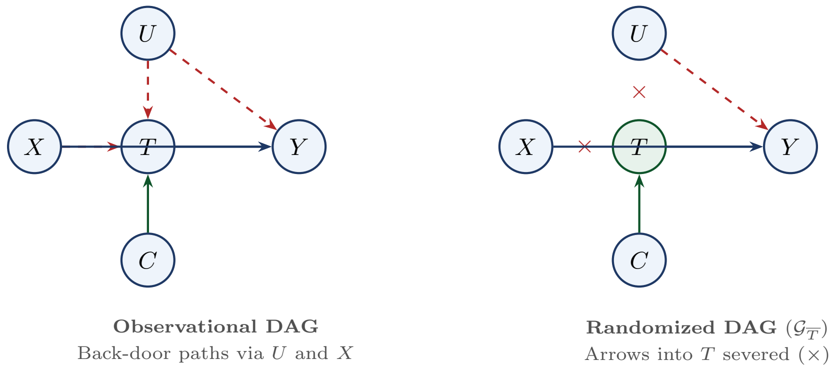


Figure 5.1: Randomization as graph surgery. **Left**: in the observational DAG, the unobserved confounder U and observed covariate X both have arrows into T, creating back-door paths. **Right**: randomization physically severs all arrows into T (marked $\times$) and replaces them with an exogenous coin flip. The resulting graph is the mutilated graph $\mathcal{G}_{\overline{T}}$, in which no back-door paths exist.

Potential outcomes statement. In the potential outcomes language, randomization implies $(Y(0), Y(1)) \perp\!\!\!\perp T$. The potential outcomes are independent of assignment *unconditionally* — no covariate adjustment is needed.

5.1.2 Estimation of the ATE in a Randomized Experiment

Lemma: Identification under Complete Randomization

Under complete randomization,

$$\mathbb{E}[Y(t)] \;=\; \mathbb{E}[Y \mid T{=}t], \qquad t \in \{0,1\}. \tag{5.2}$$

Proof

By Proposition 4.1, $Y(t)$ has the same distribution as Y under $f(y \mid \mathrm{do}(T{=}t))$, so $\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid \mathrm{do}(T{=}t)]$. In a completely randomized experiment, the treatment node has no parents in the DAG, so Equation 5.1 gives $f(y \mid \mathrm{do}(T{=}t)) = f(y \mid T{=}t)$. Therefore $\mathbb{E}[Y(t)] = \mathbb{E}[Y \mid T{=}t]$. $\square$

Under complete randomization, the ATE is estimated by the *difference-in-means* (DIM) estimator:

$$\hat{\tau}_{\mathrm{DIM}} = \bar{Y}_1 - \bar{Y}_0 = \frac{1}{n_1} \sum_{i:\, T_i=1} Y_i - \frac{1}{n_0} \sum_{i:\, T_i=0} Y_i.$$

Theorem: Unbiasedness and Variance of $\hat{\tau}_{\mathrm{DIM}}$

Let $\sigma_t^2 = \mathrm{Var}(Y(t))$ for $t \in \{0,1\}$. Under complete randomization:

$$\mathbb{E}[\hat{\tau}_{\mathrm{DIM}}] = \tau_{\mathrm{ATE}},$$

$$\mathrm{Var}(\hat{\tau}_{\mathrm{DIM}}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.$$

5.2 Ignorability and the Back-Door Formula

5.2.1 Randomization vs. Observational Studies

In a randomized experiment, identification is guaranteed by the design. In an observational study, the analyst has no control over treatment assignment, and back-door paths through unobserved variables may remain open. The question shifts from *does* identification hold to *when* does it hold.

The two routes to identification differ in their logical structure. Randomization *guarantees* identification by design: the analyst removes all arrows into T, so the fundamental equality Equation 5.1 holds automatically. The observational route requires a substantive assumption: the analyst must argue, from subject-matter knowledge encoded in a DAG, that all confounders have been measured and that the back-door criterion is satisfied.

5.2.2 Definition

Definition: Ignorability (Unconfoundedness)

Let X denote a vector of pre-treatment covariates. We say that treatment assignment is **weakly ignorable** given X if:

$$Y(t) \perp\!\!\!\perp T \,\Big|\, X, \qquad t \in \{0,1\}.$$

We say assignment is **strongly ignorable** given X (Rosenbaum and Rubin 1983) if, in addition, the overlap condition holds:

$$0 < P(T{=}1 \mid X) < 1 \quad \text{almost surely.}$$

Ignorability is also called *unconfoundedness*, *selection on observables*, *no unmeasured confounders*, or *conditional exchangeability*. In a randomized experiment, weak ignorability holds *unconditionally* — X is not needed, and the condition reduces to $(Y(0), Y(1)) \perp\!\!\!\perp T$.

5.2.3 Graphical Interpretation

In the DAG language, ignorability corresponds exactly to the back-door criterion. A set X satisfies the back-door criterion for the effect of T on Y if: (1) X blocks every back-door path from T to Y, and (2) X contains no descendant of T. When X satisfies the back-door criterion, $Y(t) \perp\!\!\!\perp T \mid X$ holds and the back-door adjustment formula identifies the causal effect.

Three Languages for Ignorability

Language	Statement of ignorability
SEM	The structural error ε_Y in the outcome equation $Y = f_Y(T, X, \varepsilon_Y)$ is independent of T given X: no unobserved variable enters both the equation that determines T and the equation that determines Y.
DAG / do-calculus	X satisfies the back-door criterion for the effect of T on Y: X blocks all back-door paths from T to Y and contains no descendant of T.
Potential outcomes	$Y(t) \perp\!\!\!\perp T \mid X$ for $t \in \{0,1\}$ (weak ignorability).

The three statements are equivalent under the standard bridge assumptions between frameworks (consistency and no interference).

Ignorability Is Untestable from Observational Data

No statistical test can confirm or refute unconfoundedness using observed data alone. If an unobserved variable U affects both T and Y, the back-door path $T \leftarrow U \rightarrow Y$ is open, and conditioning on any set of observed variables cannot close it. Sensitivity analysis can quantify how large such an unobserved confounder would have to be to overturn a conclusion, but cannot confirm the assumption itself.

5.2.4 From Ignorability to Identification

Under strong ignorability, the ATE is identified:

$$\tau_{\text{ATE}} \;=\; \mathbb{E}_X[\mathbb{E}[Y \mid T{=}1, X] - \mathbb{E}[Y \mid T{=}0, X]]\,. \tag{5.3}$$

This is the back-door adjustment formula of Chapter 3 written in potential-outcome notation. Chapter 4 supplied the proof via the law of iterated expectations, consistency, and unconfoundedness.

5.3 Regression Adjustment

5.3.1 The Outcome Regression (OR) Estimator

Given the identifying formula Equation 5.3, the natural estimator replaces the population conditional means with their sample analogs.

Step 1. Fit a model $\hat{\mu}(t, x) \approx \mathbb{E}[Y \mid T{=}t, X{=}x]$ using the observed data.

Step 2. For each unit i, impute both potential outcome predictions: $\hat{\mu}(1, X_i)$ and $\hat{\mu}(0, X_i)$.

Step 3. Average the imputed treatment effects:

$$\hat{\tau}_{\text{ATE}} = \frac{1}{n}\sum_{i=1}^{n}[\hat{\mu}(1, X_i) - \hat{\mu}(0, X_i)]. \tag{5.4}$$

This three-step recipe is also called the *G-formula estimator* or the *regression-adjusted estimator*.

Example: Worked Outcome Regression

A study compares earnings (Y, in thousands) between job-training participants ($T{=}1$) and non-participants ($T{=}0$), with education level $X \in \{0, 1\}$ as a confounder. Among the 200 treated and 200 control units, 70% of treated units have $X{=}1$ vs. 30% of control units.
The estimated conditional means are $\hat{\mu}(1, 0) = 42$, $\hat{\mu}(1, 1) = 58$, $\hat{\mu}(0, 0) = 35$, $\hat{\mu}(0, 1) = 50$.
The within-stratum effects are $42 - 35 = 7$ (for $X{=}0$ units) and $58 - 50 = 8$ (for $X{=}1$ units). Since 50% of the full sample has $X{=}0$ and 50% has $X{=}1$:

$$\hat{\tau}_{\text{ATE}} = 7 \times 0.5 + 8 \times 0.5 = 7.5.$$

After adjusting for X, the estimated treatment effect is 7.5 — not the naïve difference-in-means, which conflates the treatment effect with the higher baseline earnings of $X{=}1$ units who happen to be treated more often.

5.3.2 Standardization as a Special Case

When X is categorical, no parametric model for $\mu(t, x)$ is needed: the conditional mean $\hat{\mu}(t, x)$ is simply the sample mean of Y within each cell $(T{=}t, X{=}x)$ (a fully saturated outcome model). The resulting estimator is:

$$\hat{\tau}_{\text{STD}} = \sum_{x \in \mathcal{X}} [\hat{\mu}(1, x) - \hat{\mu}(0, x)]\, \hat{p}(x), \tag{5.5}$$

where $\hat{\mu}(t, x) = \bar{Y}_{t,x}$ and $\hat{p}(x) = n_x/n$. This is algebraically identical to Equation 5.4 when X is categorical. The formula is also known as the *g-formula* (Robins 1986) or *direct standardization* in epidemiology.

5.3.3 Regression Adjustment in Randomized Experiments: Lin (2013)

In a *randomized* experiment, outcome regression can be used to reduce variance even though covariate adjustment is not needed for unbiasedness. The *regression-adjusted estimator* of Lin (2013) fits the fully interacted OLS model:

$$Y_i = \alpha + \beta T_i + \gamma^\top \tilde{X}_i + \delta^\top (T_i \cdot \tilde{X}_i) + \varepsilon_i,$$

where $\tilde{X}_i = X_i - \bar{X}$ are mean-centered covariates. The OLS coefficient $\hat{\beta}$ is the ATE estimator.

Theorem: Design-Consistency of the Regression-Adjusted Estimator

(Lin 2013) Under complete randomization, $\hat{\beta}$ from the interacted regression is consistent for τ_{ATE} regardless of whether the linear model is correctly specified.

Proof Sketch

Because $T \perp\!\!\!\perp X$ under randomization, the projection of Y on $(T, X, T \cdot \tilde{X})$ recovers the treatment coefficient without confounding bias even if the true μ is not linear. The interaction terms ensure that the estimated treatment effect is the sample analog of the semiparametric efficient estimator. For the full proof see Lin (2013) or Freedman (2008). □

Remark: The Role of the Regression Model in a Randomized Experiment

In a randomized experiment the regression model serves one purpose only: *variance reduction*. Unbiasedness is guaranteed by the randomization alone, regardless of whether the outcome model is correctly specified. The plain DIM estimator is already design-consistent; the only reason to add covariates is to reduce $\mathrm{Var}(\hat{\tau})$ by absorbing residual variation in Y unrelated to T.
This stands in sharp contrast to the observational setting, where the regression model carries a *double* burden: it must both adjust for confounding (unbiasedness) and fit the outcome surface well (efficiency). Misspecification in an observational study threatens *both* properties simultaneously. In a randomized experiment, it threatens only efficiency.

Remark: Precedence in Survey Sampling

The design-consistency result in the theorem above was established in the survey sampling literature decades before Lin (2013). Isaki and Fuller (1982) introduced the *generalized regression (GREG) estimator* for finite-population means under general probability sampling designs and proved it is design-consistent even when the assisting regression model is misspecified. Deville and Särndal (1992) extended this class to *calibration estimators*, which adjust sampling weights so that weighted sample totals match known population totals for auxiliary variables. Lin (2013) restated and proved this result within the Neyman–Rubin potential-outcomes framework, connected it to semiparametric efficiency theory, and showed that the Huber–White sandwich variance estimator yields asymptotically valid confidence intervals under the randomization distribution.

Remark

Lin's consistency result is specific to randomized experiments. In an observational study, $\hat{\tau}_{\text{OR}}$ is consistent only if $\hat{\mu}$ converges to the true μ. The doubly-robust (AIPW) estimator of Chapter 10 achieves consistency if *either* the outcome model or the propensity score model is correctly specified.

5.4 Stratification

5.4.1 The Idea

Stratification (subclassification) is a non-parametric implementation of the back-door formula. Rather than modeling $\mathbb{E}[Y \mid T, X]$, the analyst divides the sample into *strata* — subgroups of units with the same or similar values of X — and estimates the treatment effect within each stratum.

Within a stratum $\mathcal{S}_k = \{i : X_i \in B_k\}$, if the stratum is narrow enough, ignorability holds approximately and the within-stratum difference in means:

$$\hat{\tau}_k = \bar{Y}_{1,k} - \bar{Y}_{0,k}$$

is an approximately unbiased estimate of the stratum-specific ATE.

5.4.2 The Stratified Estimator

The overall ATE is estimated as a weighted average of within-stratum estimates:

$$\hat{\tau}_{\text{STRAT}} = \sum_{k=1}^{K} \hat{\tau}_k \cdot \frac{n_k}{n}, \tag{5.6}$$

where K is the number of strata, n_k is the number of units in stratum k, and $n = \sum_k n_k$. Cochran (1968) showed that with $K = 5$ strata of equal size, roughly 90% of the bias from a single continuous confounder is removed; with $K = 10$ strata, over 95%.

> Remark: Stratification, Standardization, and Survey Sampling Terminology
>
> When X is categorical, the stratified estimator Equation 5.6 and the standardization estimator Equation 5.5 are algebraically identical. The two names reflect disciplinary tradition rather than a methodological distinction.
> In the survey sampling literature, *stratification* is a design-stage decision while *post-stratification* is an analysis-stage adjustment. Standardization in causal inference corresponds to post-stratification: stratum membership is recorded after sampling, and the estimator is reweighted by marginal stratum counts. Standardization is therefore regression adjustment with a saturated outcome model.

5.4.3 Limitations

Stratification faces the *curse of dimensionality*: with p binary covariates there are 2^p cells, many of which will be empty in practice. This motivates two solutions:

- **Regression adjustment / standardization** imposes parametric structure on $\hat{\mu}(t, x)$, allowing interpolation across cells.
- **Propensity score stratification** (Chapter 6) collapses the multivariate X to a scalar $\pi(X) = P(T{=}1 \mid X)$ and stratifies on the scalar.

5.5 Simpson's Paradox

Simpson's paradox — the reversal of an association when conditioning on a third variable — was introduced in Chapter 1, where a flu-vaccination example showed that the vaccine appeared harmful in the aggregate yet beneficial within every age stratum, because elderly individuals were disproportionately vaccinated. This chapter's development gives that example a second layer of interpretation: the back-door formula used in Chapter 1 is precisely the standardization estimator Equation 5.5 — within-stratum conditional means averaged over the *marginal* distribution of the confounder rather than its treatment-conditional distribution. The weighted average corrects the confounding; the pooled association does not, because it weights strata by $P(X \mid T)$ instead of $P(X)$.

5.5.1 When Should You *Not* Condition on X?

Simpson's paradox also has a mirror image: conditioning on a variable can *create* a spurious association or make a true causal effect disappear. This occurs when X is a *mediator* or a *collider*. The back-door criterion gives the correct prescription: include X in the adjustment set only if it blocks back-door paths *without* opening collider paths.

Conditioning on a Mediator or Collider

Adjusting for a post-treatment variable X that lies on a causal pathway $T \to X \to Y$ blocks part of the causal effect of T, producing a downward-biased estimate of the total effect. Adjusting for a collider X with $T \to X \leftarrow Y$ opens a spurious association between T and Y that does not exist in the population. Neither mistake is detectable from the data alone; the DAG is essential.

The mediator case will be revisited in detail in Chapter 8. The collider case was covered in Chapter 2.

5.6 Lab: Simulation Study of the Outcome Regression Estimator

The identification results in Section 5.2 and the estimation recipes in Section 5.3 rest on two assumptions that pull in opposite directions: the outcome model $\hat{\mu}(t,x)$ must be correctly specified for the linear OR estimator to be consistent in an observational study, yet a more flexible nonparametric estimator pays a variance penalty for that flexibility.

DGP for Lab 5

Let $X \sim \text{Uniform}(0,1)$ be a single pre-treatment covariate and $\varepsilon \sim \mathcal{N}(0,1)$ independent of X. The potential outcome surfaces are:

$$Y(t) \ = \ t \ + \ 3(1+t)X^2 \ + \ \varepsilon, \qquad t \in \{0,1\}. \tag{5.7}$$

The conditional average treatment effect is $\tau(x) = 1 + 3x^2$, so

$$\tau_{\text{ATE}} = \mathbb{E}[1 + 3X^2] = 1 + 1 = 2. \tag{5.8}$$

Two assignment mechanisms are studied:

- **CRD:** $T \sim \text{Bern}(0.5)$, independent of X.
- **Observational:** $T \mid X \sim \text{Bern}(\text{expit}(-2 + 5X))$, giving $\pi(0) \approx 0.12$, $\pi(0.5) \approx 0.62$, $\pi(1) \approx 0.95$. Strong confounding: high-X units are nearly always treated.

Two outcome regression estimators of the form Equation 5.4 are compared: a **Linear OR** (simple linear regression, misspecified because the true μ contains a quadratic term), and a **Local-linear OR** (local linear regression with bandwidth $h = 0.5 \cdot n^{-1/5}$, correctly flexible).

Key Finding: Model Misspecification Is Harmless under Randomization, Harmful under Confounding

Under CRD, the linear OR estimator is consistent despite misspecification (Theorem above): the fitting and evaluation distributions coincide, so pointwise errors cancel on average.
Under the observational design, the fitting and evaluation distributions diverge. High-X units are almost always treated and low-X units almost never. The model for arm $t = 1$ is fit on a sample concentrated at high X, and the model for arm $t = 0$ on a sample concentrated at low X. The fitted lines are tuned to different parts of the covariate space, so when both are evaluated at the full marginal of X, the pointwise errors no longer cancel. Model misspecification is harmless under randomization; it becomes a source of bias precisely when confounding is present.

Remark: What This Lab Does Not Show

The simulation fixes the bandwidth $h = 0.5 \cdot n^{-1/5}$ and does not perform data-driven bandwidth selection. The lab also studies only a scalar X; with a multivariate covariate, the curse of dimensionality would make local linear regression impractical, which is one motivation for the propensity score approach of Chapter 6.

5.7 Where the Propensity Score Fits

The three methods — regression adjustment, standardization, and stratification — all implement the same identifying formula Equation 5.3. They differ in how they deal with the conditioning on the high-dimensional covariate vector X:

Method	**How it handles X**	**Key assumption**
Regression / standardization	Models $\mathbb{E}[Y \mid T, X]$ parametrically	Outcome model correctly specified
Stratification	Groups units by X; estimates effect in each cell	Strata narrow enough for approximate balance
Propensity score (Ch. 6)	Reduces X to scalar $\pi(X) = P(T{=}1 \mid X)$	Propensity model correctly specified

Bridge to Chapter 6

Chapter 6 takes up the propensity score idea formally. It proves the balancing property ($T \perp\!\!\!\perp X \mid \pi(X)$), derives the Propensity Score Theorem (strong ignorability given X implies strong ignorability given $\pi(X)$ alone), and develops three estimation strategies: inverse probability weighting (IPW), matching, and propensity score stratification. Matching need not rely on the propensity score: covariate-distance methods such as Mahalanobis-distance matching and nearest-neighbor matching on X directly are equally valid.

5.8 Summary

1. **Randomization as graph surgery.** A randomized experiment removes all arrows into T, so the fundamental equality $f(y \mid \mathrm{do}(T{=}t)) = f(y \mid T{=}t)$ holds and the difference-in-means estimator is unbiased for the ATE without covariate adjustment.

2. **Ignorability is the key assumption.** Under unconfoundedness $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$, the back-door adjustment formula identifies the ATE. This assumption is substantive, design-determined, and untestable from observational data.

3. **Three estimation strategies.** Regression adjustment (G-formula), standardization, and stratification all implement the same identifying formula via different modeling choices. All require correct specification of either an outcome model (for regression/standardization) or a stratification scheme fine enough to achieve balance.

4. **Simpson's paradox.** Pooled associations can reverse within subgroups when a confounder determines treatment selection. The correct causal estimate requires standardization over the marginal covariate distribution, guided by the back-door criterion.

5. **The propensity score dimension reduction.** When X is high-dimensional, direct stratification or cell-by-cell standardization breaks down. The propensity score $\pi(X)$ provides a scalar sufficient statistic for adjustment. Chapter 6 develops the theory and estimation methods.

5.9 Problems

1. Randomization and the do-calculus. Let the observational DAG be $\{U \to T,\ U \to Y,\ X \to T,\ T \to Y\}$ with U unobserved.

(a) Identify all back-door paths from T to Y.
(b) Does X satisfy the back-door criterion? Does U? Explain.
(c) Now suppose treatment is randomized. Draw the modified DAG and identify all back-door paths. Show that $f(y \mid \mathrm{do}(T{=}t)) = f(y \mid T{=}t)$ holds in the randomized DAG using Rule 2 of the do-calculus.
(d) Under randomization, is covariate adjustment on X necessary for unbiasedness? Is it ever beneficial? Explain.

2. Ignorability and the back-door criterion. Consider the DAG $\{X \to T,\ X \to Y,\ T \to Y,\ U \to Y\}$ with U unobserved.

(a) Does $\{X\}$ satisfy the back-door criterion? Write out the identifying formula for τ_{ATE}.
(b) Now add the edge $U \to T$ to the DAG. Does $\{X\}$ still satisfy the back-door criterion? What does the back-door criterion require when both X and U affect T?
(c) Explain in words what "unconfoundedness given X" means about the role of U in the data-generating process.

3. Standardization. A study of job training (T) and earnings (Y, in thousands) produces the following cell means, with $P(X{=}0) = 0.4$ and $P(X{=}1) = 0.6$:

X	$\hat{\mu}(1,x)$	$\hat{\mu}(0,x)$	n_x/n
0	28	22	0.4
1	35	31	0.6

(a) Compute $\hat{\tau}_{\text{ATE}}$ using the standardization formula.
(b) Compute $\hat{\tau}_{\text{ATT}}$, given that all treated units come from $X{=}1$.
(c) The unadjusted difference in means is $33 - 25 = 8$. Compare to your answers in (a) and (b) and explain the discrepancy.

4. Stratification and Cochran's rule. You have a binary confounder $X \in \{0,1\}$ and form two strata. Within stratum $X{=}0$: $n_0 = 600$, $\bar{Y}_1 = 10$, $\bar{Y}_0 = 8$. Within stratum $X{=}1$: $n_1 = 400$, $\bar{Y}_1 = 15$, $\bar{Y}_0 = 12$.

(a) Compute the stratified ATE estimator $\hat{\tau}_{\text{STRAT}}$.
(b) Suppose an unadjusted difference-in-means gives $\hat{\tau}_{\text{DIM}} = 5.5$. Explain why the two estimates differ and which one is the appropriate causal estimate.
(c) Describe one limitation that would arise if X were a continuous variable with 15 dimensions.

5. Simpson's paradox. A hospital reports that patients who receive intensive care (ICU admission, $T{=}1$) have a higher mortality rate than those who do not ($T{=}0$): $P(Y{=}1 \mid T{=}1) = 0.30$ vs. $P(Y{=}1 \mid T{=}0) = 0.10$.

(a) Construct a numerical example (a $2 \times 2 \times 2$ table with disease severity $X \in \{\text{mild}, \text{severe}\}$ as the confounder) that is consistent with the above pooled numbers and yet shows ICU admission *reduces* mortality within both severity strata.
(b) Compute the standardized ATE using the marginal distribution of X.
(c) Draw the DAG for this example. Identify the back-door path that the pooled comparison fails to block.
(d) A colleague suggests that since the hospital treats patients with both mild and severe illness, the pooled statistic is the right answer. Explain, using potential outcomes notation, why this argument is incorrect.

Chapter 6

Propensity Score Methods

Learning Objectives

By the end of this chapter, students should be able to:

1. Define the propensity score $\pi(X) = P(T{=}1 \mid X)$ and explain why it is a *balancing score*.
2. State and prove the Balancing Property ($T \perp\!\!\!\perp X \mid \pi(X)$) and the Propensity Score Theorem (strong ignorability given X implies strong ignorability given $\pi(X)$).
3. Describe standard methods for estimating the propensity score (logistic regression, machine learning) and the practical pitfalls of each.
4. Derive the IPW identification formula for the ATE and explain the role of the Horvitz–Thompson representation.
5. Distinguish the IPW estimator (targets ATE) from the nearest-neighbor matching estimator presented in this chapter (designed to target ATT).
6. Define strict overlap, describe the practical consequences of near-violations, and apply trimming strategies.
7. Articulate the fundamental limitation of propensity-score methods — the unconfoundedness assumption — and explain why this motivates instrumental variables (Chapter 7).

6.1 Motivation: The Curse of Dimensionality

Chapter 5 established that under strong ignorability, the ATE is identified by the back-door adjustment formula, and developed three estimation strategies: regression adjustment, standardization, and stratification. All three require, in practice, that we condition on a covariate vector X. When X is low-dimensional, this is feasible. When X has many components, it is not.

The problem. Stratification requires cells $\{X = x\}$ to contain both treated and control units. With p binary covariates, there are 2^p cells. With $p = 20$, that is over one million cells — most empty in any realistic sample. Regression adjustment imposes parametric structure to interpolate across cells, but extrapolation bias grows with dimension. The propensity score provides an elegant solution: rather than conditioning on the full vector X, it is sufficient to condition on the scalar $\pi(X) = P(T{=}1 \mid X)$.

6.2 The Propensity Score

6.2.1 Definition

Definition: Propensity Score

The *propensity score* is the conditional probability of receiving treatment given the observed covariates:

$$\pi(X) \;=\; P(T{=}1 \mid X).$$

In a randomized experiment, $\pi(X) = p$ is constant (equal to the allocation probability), so there is nothing to estimate. In an observational study, $\pi(X)$ varies with X and must be estimated from data.

Connection to strong ignorability. Under strong ignorability $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$, the potential outcomes carry no further information about treatment assignment once X is given:

$$P(T{=}1 \mid X,\, Y(0),\, Y(1)) = P(T{=}1 \mid X) = \pi(X).$$

This equality is a *consequence* of the ignorability assumption, not part of the definition of $\pi(X)$.

6.2.2 The Balancing Property

Theorem: The Propensity Score is a Balancing Score

(Rosenbaum and Rubin 1983) $T \perp\!\!\!\perp X \mid \pi(X)$.

Proof

It suffices to show $P(T{=}1 \mid X, \pi(X)) = P(T{=}1 \mid \pi(X))$. Since $\pi(X)$ is a deterministic function of X, conditioning on both X and $\pi(X)$ is the same as conditioning on X alone:

$$P(T{=}1 \mid X, \pi(X)) = P(T{=}1 \mid X) = \pi(X) = P(T{=}1 \mid \pi(X)). \quad \square$$

Remark: Covariate Balance and Design-Before-Analysis

The balancing property implies $f(X \mid T{=}1,\, \pi(x)) = f(X \mid T{=}0,\, \pi(x))$. Stratifying on a discretized $\hat{\pi}(X)$ and comparing covariate distributions within strata via standardized mean differences diagnoses whether the propensity score model achieves adequate balance, *without ever looking at* Y. Because the diagnostic does not involve the outcome, it cannot be used to "fish" for a preferred model specification, making the analysis more objective (Imbens and Rubin 2015).

Remark

The Balancing Property is a statement about the *observed* covariate distribution. It says that within strata of $\pi(X)$, the distribution of X is the same in the treated and control groups. It does *not* say anything about unobserved confounders U: if U affects both T and Y and is not captured by X, the balancing property fails to close the back-door path through U.

6.2.3 The Propensity Score Theorem

Theorem: Propensity Score Theorem

(Rosenbaum and Rubin 1983) Suppose $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$. Then for any balancing score $b(X)$,

$$(Y(0),\, Y(1)) \;\perp\!\!\!\perp\; T \,\Big|\, b(X).$$

In particular, taking $b(X) = \pi(X)$: strong ignorability given X implies strong ignorability given $\pi(X)$ alone, provided $0 < \pi(X) < 1$ almost surely.

Proof

We prove the result for $b(X) = \pi(X)$. By the law of iterated expectations, for any $t \in \{0, 1\}$:

$$P(T{=}1 \mid Y(t),\, \pi(X)) = \mathbb{E}[P(T{=}1 \mid Y(t), X) \mid Y(t), \pi(X)].$$

By ignorability, $T \perp\!\!\!\perp Y(t) \mid X$, so $P(T{=}1 \mid Y(t), X) = P(T{=}1 \mid X) = \pi(X)$. Therefore:

$$P(T{=}1 \mid Y(t),\, \pi(X)) = \mathbb{E}[\pi(X) \mid Y(t), \pi(X)] = \pi(X) = P(T{=}1 \mid \pi(X)).$$

Hence $T \perp\!\!\!\perp Y(t) \mid \pi(X)$. $\square$

Remark

The theorem gives an alternative identification formula:

$$\mathbb{E}[Y(t)] = \mathbb{E}[\mathbb{E}[Y \mid \pi(X),\, T{=}t]].$$

A common misconception is that the two conditional regressions $\mathbb{E}[Y \mid \pi(X), T{=}t]$ and $\mathbb{E}[Y \mid X, T{=}t]$ are equal. They are not: they are different functions and their pointwise values generally differ. The theorem guarantees only that their marginal expectations agree:

$$\mathbb{E}[\mathbb{E}[Y \mid \pi(X),\, T{=}t]] = \mathbb{E}[\mathbb{E}[Y \mid X,\, T{=}t]] = \mathbb{E}[Y(t)].$$

6.2.4 The Propensity Score is the Coarsest Balancing Score

Lemma: The Propensity Score is the Coarsest Balancing Score

(Rosenbaum and Rubin 1983) If $b(X)$ is any balancing score — that is, $T \perp\!\!\!\perp X \mid b(X)$ — then $\pi(X)$ is a function of $b(X)$: $\pi(X) = g(b(X))$ for some measurable function g.

Proof

Since $b(X)$ is a balancing score, $T \perp\!\!\!\perp X \mid b(X)$, so $P(T{=}1 \mid X,\, b(X)) = P(T{=}1 \mid b(X))$. But $b(X)$ is a function of X:

$$\pi(X) = P(T{=}1 \mid X) = P(T{=}1 \mid X,\, b(X)) = P(T{=}1 \mid b(X)).$$

Hence $\pi(X)$ is a measurable function of $b(X)$ alone. $\square$

The lemma says that $\pi(X)$ is the *smallest* sufficient statistic for T among all functions of X: any other balancing score retains at least as much information about X as the propensity score does.

The Dimension Reduction

The Propensity Score Theorem says that instead of adjusting for the full high-dimensional vector X, it is sufficient to adjust for the scalar $\pi(X)$. Identification of the ATE follows immediately:

$$\tau_{\text{ATE}} = \mathbb{E}[\mathbb{E}[Y \mid T{=}1, \pi(X)] - \mathbb{E}[Y \mid T{=}0, \pi(X)]].$$

6.3 Estimation of the Propensity Score

The true propensity score $\pi(X) = P(T{=}1 \mid X)$ is *unknown* in observational studies and must be estimated from the data.

Logistic regression. The classical approach models $\text{logit}(\pi(X)) = X^\top \beta$ and estimates β by maximum likelihood. The fitted values $\hat{\pi}(X_i) = \sigma(X_i^\top \hat{\beta})$ are used in place of the true propensity score.

Machine learning estimators. When X is high-dimensional or the true propensity score is nonlinear, machine learning methods — gradient boosted trees, random forests, regularized logistic regression (LASSO, ridge) — can estimate $\pi(X)$ more flexibly. A critical issue is *overfitting*: an estimator that perfectly separates treated and control units in-sample will produce estimated propensity scores near 0 or 1 everywhere, destroying overlap. Cross-fitting (Chapter 11) addresses this by estimating nuisance functions on held-out folds.

Propensity Score Estimation Is a Nuisance, Not the Goal

The propensity score is estimated in order to construct a good estimator of τ_{ATE}. Balancing tests diagnose whether the estimated score has achieved covariate balance, but they do *not* test whether unobserved confounders are balanced.

6.4 Matching on the Propensity Score

The propensity score also motivates *matching*: for each treated unit i, find a control unit j with $\hat{\pi}(X_j) \approx \hat{\pi}(X_i)$, and use Y_j as a proxy for $Y_i(0)$ (Abadie and Imbens 2006, 2016). The resulting estimator of the ATT is:

$$\hat{\tau}_{\text{ATT}} = \frac{1}{n_1} \sum_{i:\,T_i=1} \Big[Y_i - Y_{\hat{\jmath}(i)}\Big],$$

where $\hat{\jmath}(i)$ is the index of the matched control unit.

One-to-one nearest neighbor matching. For each treated unit i, find:

$$\hat{\jmath}(i) = \arg\min_{j:\,T_j=0} \big|\hat{\pi}(X_i) - \hat{\pi}(X_j)\big|.$$

Matching with replacement reduces bias (each control unit is available to every treated unit) but increases variance.

Caliper matching. Restricts matches to pairs within a maximum distance δ on the propensity score scale. Treated units with no control match within the caliper are discarded, restricting the estimand to the subset of treated units with good overlap.

Remark

Propensity score matching implicitly fits a nonparametric regression of Y on $\pi(X)$ separately within each treatment arm. By the Propensity Score Theorem, $(Y(0), Y(1)) \perp\!\!\!\perp T \mid \pi(X)$, so the conditional mean $\mathbb{E}[Y(t) \mid \pi(X) = p]$ is identified from observed data. Nearest-neighbor matching approximates this conditional mean at each observed $\hat{\pi}(X_i)$ by local constant regression on the scalar propensity score — feasible precisely because of the dimension reduction from X to $\pi(X)$.

6.5 Inverse Probability Weighting

6.5.1 The IPW Identification Formula

Under strong ignorability, the ATE admits an alternative representation as an *inverse probability weighting* (IPW) formula:

$$\tau_{\text{ATE}} = \mathbb{E}\left[\frac{T \cdot Y}{\pi(X)}\right] - \mathbb{E}\left[\frac{(1-T)\cdot Y}{1-\pi(X)}\right].$$

Theorem: IPW Identification

Under strong ignorability ($(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ and $0 < \pi(X) < 1$):

$$\mathbb{E}\left[\frac{T \cdot Y}{\pi(X)}\right] = \mathbb{E}[Y(1)].$$

Proof

$$\mathbb{E}\left[\frac{TY}{\pi(X)}\right] = \mathbb{E}\left[\frac{TY(1)}{\pi(X)}\right] = \mathbb{E}\left[\mathbb{E}\left[\frac{TY(1)}{\pi(X)} \;\Big|\; X\right]\right] = \mathbb{E}\left[\frac{Y(1)}{\pi(X)}\,\mathbb{E}[T \mid X]\right] = \mathbb{E}\left[\frac{Y(1)}{\pi(X)}\,\pi(X)\right] = \mathbb{E}[Y(1)].$$

The first equality uses consistency ($Y = Y(1)$ when $T{=}1$); the third uses ignorability ($Y(1) \perp\!\!\!\perp T \mid X$). $\square$

6.5.2 The Horvitz–Thompson and Hájek Estimators

The *Horvitz–Thompson* (HT) IPW estimator:

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{T_iY_i}{\hat{\pi}(X_i)} - \frac{(1-T_i)Y_i}{1-\hat{\pi}(X_i)}\right].$$

This estimator is consistent when $\hat{\pi}(X)$ converges to $\pi(X)$, but can be *inefficient* due to extreme weights. The *normalized* (Hájek) estimator divides by the sum of weights within each group and is often more stable:

$$\hat{\tau}_{\text{HJ}} = \frac{\sum_i T_iY_i/\hat{\pi}(X_i)}{\sum_i T_i/\hat{\pi}(X_i)} - \frac{\sum_i (1-T_i)Y_i/(1-\hat{\pi}(X_i))}{\sum_i (1-T_i)/(1-\hat{\pi}(X_i))}.$$

The efficient doubly-robust estimator (AIPW) of Chapter 10 combines the outcome model and the propensity score; under correct nuisance specification it achieves higher efficiency than either pure IPW or pure outcome regression alone.

6.6 Overlap and Positivity

6.6.1 The Strict Overlap Condition

Definition: Strict Overlap (Positivity)

The *strict overlap condition* requires:

$$0 < \pi(X) < 1 \quad \text{almost surely.}$$

Every unit has a positive probability of receiving either treatment or control, regardless of its covariate values.

Together with unconfoundedness, strict overlap constitutes *strong ignorability* (Rosenbaum and Rubin 1983).

Why overlap is necessary. The identification formula requires that both $\mathbb{E}[Y \mid T{=}1, X{=}x]$ and $\mathbb{E}[Y \mid T{=}0, X{=}x]$ be defined for all x in the support of X. If $\pi(x) = 0$ for some x, then no treated units exist with $X = x$, so $\mathbb{E}[Y \mid T{=}1, X{=}x]$ is not estimable.

6.6.2 Practical Consequences of Near-Violations

Near-Overlap Problems

When $\hat{\pi}(X_i)$ is close to 0 or 1, the IPW weight $1/\hat{\pi}(X_i)$ or $1/(1-\hat{\pi}(X_i))$ is very large. A small number of units with extreme weights can dominate the estimator, increasing variance dramatically.

6.6.3 Trimming Strategies

Trimming by propensity score. Restrict the analysis to units with estimated propensity scores in $[\eta, 1-\eta]$ for some small $\eta > 0$. The estimand changes to $\tau_{\text{trim}} = \mathbb{E}[Y(1) - Y(0) \mid \eta \le \pi(X) \le 1-\eta]$.

Crump et al. (2009) rule. Crump et al. (2009) derive the optimal trimming threshold minimizing the asymptotic variance of the trimmed ATE estimator. The rule $\eta = 0.1$ — trimming units with $\pi(X) < 0.1$ or $\pi(X) > 0.9$ — is a common default.

ATT instead of ATE. When overlap fails at covariate values where $\pi(X)$ is near 1, re-targeting the estimand to τ_{ATT} avoids the problem, since the ATT requires only $\pi(X) < 1$ on the support of X among the treated.

6.7 Lab: Simulation Study of IPW and Matching Estimators

This lab compares five estimators on a five-covariate data-generating process with treatment-effect heterogeneity. With five continuous confounders, direct stratification is infeasible, making propensity-score methods the natural choice. The estimators span two estimands: the ATE (HT-ATE, HJ-ATE) and the ATT (HT-ATT, HJ-ATT, NNM).

6.7.1 Part 1: Correctly Specified Propensity Score

DGP for Lab 6

Let $X_j \overset{\text{i.i.d.}}{\sim} \text{Normal}(0,1)$ for $j = 1,\dots,5$. The true propensity score is:

$$\pi(X) = \text{expit}(-0.5 + 0.8X_1 + 0.5X_2 - 0.3X_3 + 0.2X_4 + 0.1X_5), \tag{6.1}$$

giving $P(T{=}1) \approx 0.40$. The potential outcomes are:

$$Y(t) = (1 + 0.5X_1)\cdot t + 8 + 0.8X_1 + 0.5X_2 + 0.4X_3 + 0.3X_4 + 0.2X_5 + \varepsilon, \quad \varepsilon \sim \text{N}(0,1). \tag{6.2}$$

The CATE is $\tau(X) = 1 + 0.5X_1$. Because $\mathbb{E}[X_1] = 0$: $\tau_{\text{ATE}} = 1.000$ (exact). Because $\pi(X)$ is increasing in X_1, treated units have higher X_1 on average: $\tau_{\text{ATT}} \approx 1.199$ (oracle run, $n = 10^7$). The gap ≈ 0.199 reflects a selection effect: units most likely to be treated also tend to have larger individual treatment effects. Strong ignorability holds by construction.

Five estimators are compared under both true and estimated propensity scores. Under the estimated score, $\hat{\pi}(X)$ is obtained by correctly specified logistic regression of T on $(X_1,\dots,X_5)$.

Estimator	Formula	Estimand
HT-ATE	$\frac{1}{n}\sum_i \left[\frac{T_iY_i}{\hat{\pi}(X_i)} - \frac{(1-T_i)Y_i}{1-\hat{\pi}(X_i)}\right]$	ATE
HJ-ATE	Hájek-normalized HT	ATE
HT-ATT	$\frac{1}{n_1}\sum_{T_i=1} Y_i - \frac{1}{n_1}\sum_{T_i=0} \frac{\hat{\pi}(X_i)}{1-\hat{\pi}(X_i)} Y_i$	ATT
HJ-ATT	Self-normalized version of HT-ATT (control arm only)	ATT
NNM	$\frac{1}{n_1}\sum_{T_i=1}[Y_i - Y_{\hat{\jmath}(i)}]$, 1:1 with replacement	ATT

Results. $n = 1{,}000$, $B = 2{,}000$ replications, seed 42.

PS	Estimator	Estimand	Mean	Bias	SD	RMSE
Known	HT-ATE	ATE	1.008	+0.008	0.626	0.626
Known	HJ-ATE	ATE	1.003	+0.003	0.145	0.145
Known	HT-ATT	ATT	1.184	−0.016	0.725	0.725
Known	HJ-ATT	ATT	1.203	+0.003	0.131	0.131
Known	NNM	ATT	1.207	+0.008	0.130	0.130
Estimated	HT-ATE	ATE	0.998	−0.002	0.194	0.194
Estimated	HJ-ATE	ATE	1.002	+0.002	0.102	0.103
Estimated	HT-ATT	ATT	1.202	+0.002	0.251	0.251
Estimated	HJ-ATT	ATT	1.202	+0.002	0.101	0.101
Estimated	NNM	ATT	1.205	+0.006	0.121	0.121

Five lessons stand out. **1.** Hájek normalization is essential when outcomes are non-centered: HT-ATE achieves SD = 0.626 vs. HJ-ATE SD = 0.145 — a 4.3-fold reduction. **2.** The same problem afflicts ATT: HT-ATT SD = 0.725 vs. HJ-ATT SD = 0.131 — a 5.5-fold reduction. **3.** HJ-ATT and NNM converge to the same ATT target with nearly identical efficiency (SD 0.131 vs. 0.130 under known PS). **4.** The ATE–ATT gap of ≈ 0.199 is a real difference in estimands, not a bias — declaring the estimand before looking at results is non-negotiable. **5.** Estimated PS collapses HT variance; HJ-ATT with estimated PS (SD = 0.101) beats NNM (SD = 0.121), illustrating the efficiency gain of a well-estimated score.

> The Estimand Must Be Chosen Before the Estimator
>
> In this DGP, $\tau_{\text{ATE}} = 1.000$ and $\tau_{\text{ATT}} \approx 1.199$. The gap arises because treatment-effect heterogeneity is correlated with propensity for treatment (Imbens and Rubin 2015) — individuals often select into treatment partly because they anticipate larger benefits. A researcher who applies NNM and reports its output against the ATE benchmark of 1.000 would conclude the estimator has 20% bias; in fact it is unbiased for the correct target.

6.7.2 Part 2: Robustness to PS Model Misspecification

Part 1 used a correctly specified logistic propensity model. Part 2 introduces a *nonlinear* true propensity score:

$$\pi^*(X) = \text{expit}(-0.5 + 0.8X_1 + 0.5X_1^2 + 0.2X_4 + 0.1X_5), \tag{6.3}$$

with a U-shaped surface in X_1. The true ATE remains 1.000 (exact); the ATT changes to ≈ 1.152. The *estimated* PS is still a linear logistic regression that omits X_1^2 — misspecified.

Results. True ATE = 1.000; true ATT ≈ 1.152. $n = 1{,}000$, $B = 2{,}000$ replications.

PS	Estimator	Estimand	Mean	Bias	SD	RMSE
True	HT-ATE	ATE	1.028	+0.028	0.824	0.825
True	HJ-ATE	ATE	1.013	+0.013	0.152	0.152
True	HT-ATT	ATT	1.186	+0.034	1.333	1.333
True	HJ-ATT	ATT	1.183	+0.031	0.212	0.215
True	NNM	ATT	1.178	+0.026	0.175	0.177
Misspecified	HT-ATE	ATE	1.478	+0.478	0.147	0.501
Misspecified	HJ-ATE	ATE	0.861	−0.139	0.087	0.164
Misspecified	HT-ATT	ATT	1.780	+0.628	0.162	0.649
Misspecified	HJ-ATT	ATT	1.306	+0.154	0.081	0.174
Misspecified	NNM	ATT	1.172	+0.020	0.138	0.139

Three lessons complete the picture. **6.** PS misspecification makes all IPW estimators inconsistent; Hájek normalization is a variance correction only — it cannot fix a structurally wrong model. Note also that HJ-ATE and HT-ATE have biases in *opposite directions* (−0.139 vs. +0.478) because the two estimators respond differently to the systematic over/underestimation of the U-shaped score. **7.** NNM is substantially more robust: its bias under misspecification (+0.020) is far smaller than any IPW estimator, because matching relies on local covariate similarity rather than a globally correct weighting function.

> IPW Needs a Correct PS Model; Matching Needs Covariate Balance
>
> IPW is a *model-dependent* estimator: consistency relies on correct PS specification. Matching is a *design-dependent* estimator: consistency relies on matched pairs being approximately balanced. This condition can hold even when the PS model is wrong, provided the estimated score still separates units well enough that close matches imply closeness on the full covariate vector. The cost is efficiency: when the PS model is correct, matching leaves variance on the table relative to IPW. Neither dominates unconditionally (Yang et al. 2016; Yang and Zhang 2023).

Head-to-head comparison of HJ-ATT and NNM:

Estimator	Bias (Correct PS)	RMSE (Correct PS)	Bias (Misspecified PS)	RMSE (Misspecified PS)
HJ-ATT	+0.002	0.101	+0.154	0.174
NNM	+0.006	0.121	+0.020	0.139

6.8 The Limits of Propensity Score Methods

Every result in this chapter — the Propensity Score Theorem, the IPW identification formula, the matching estimator — rests on the unconfoundedness assumption:

$$(Y(0),\, Y(1)) \;\perp\!\!\!\perp\; T \,\Big|\, X.$$

Design-Based vs. Model-Based Identification

Randomization *guarantees* ignorability: by construction, no back-door paths exist. Propensity-score adjustment *assumes* ignorability: the analyst must argue from subject-matter knowledge that all confounders have been measured. This distinction is fundamental. No statistical test can confirm unconfoundedness from observed data alone (Chapter 5). Sensitivity analysis can quantify how large an unobserved confounder would have to be to overturn a conclusion, but cannot confirm the assumption.

When unobserved confounders are present, propensity score adjustment fails and an instrumental variable is needed to identify the causal effect. Instrumental variables take a different approach: instead of blocking the confounding path $T \leftarrow U \rightarrow Y$, they isolate variation in T induced by an external source Z independent of U. This is derived formally in Chapter 7.

6.9 Summary

1. **The propensity score reduces dimension.** Under unconfoundedness, $\pi(X) = P(T{=}1 \mid X)$ is a *balancing score*: $T \perp\!\!\!\perp X \mid \pi(X)$. Consequently, $(Y(0), Y(1)) \perp\!\!\!\perp T \mid \pi(X)$, so identification requires adjustment for the scalar $\pi(X)$ alone.

2. **IPW identification.** The ATE is identified as $\mathbb{E}[TY/\pi(X)] - \mathbb{E}[(1-T)Y/(1-\pi(X))]$ under strong ignorability. The Horvitz–Thompson IPW estimator is consistent but sensitive to extreme weights; the Hájek variant is often more stable. AIPW (Chapter 10) achieves higher efficiency under correct nuisance specification.

3. **The matching procedure in this chapter targets the ATT.** The nearest-neighbor matching estimator finds control units with similar propensity scores to each treated unit. It differs fundamentally from IPW weighting in estimand and methodology, and is more robust to PS misspecification.

4. **Strict overlap is non-negotiable.** The condition $0 < \pi(X) < 1$ a.s. is necessary for identification of the ATE. Near-violations create extreme IPW weights and large variance; trimming strategies address this at the cost of restricting the estimand.

5. **The fundamental limitation.** Unconfoundedness $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ is a strong, untestable assumption. When unobserved confounders are present, propensity score adjustment fails and an instrumental variable is needed.

Design	**Key assumption**	**Identified estimand**
Randomized experiment	$(Y(0), Y(1)) \perp\!\!\!\perp T$ (by design)	ATE
Propensity-score adjustment	$(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ (unconfoundedness)	ATE or ATT
Instrumental variables	Relevance, exogeneity, exclusion	LATE (compliers)

6.10 Problems

1. Propensity score and balancing. Suppose $X = (X_1, X_2)$, where X_1 is binary and X_2 is continuous. The propensity score model is correctly specified as $\operatorname{logit}(\pi(X)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$.

(a) State and prove the Balancing Property $T \perp\!\!\!\perp X \mid \pi(X)$.

(b) Two units i and j have $X_i = (1, 2.3)$ and $X_j = (0, 3.5)$ but $\hat{\pi}(X_i) = \hat{\pi}(X_j) = 0.4$. If i is treated and j is control, explain why comparing Y_i and Y_j is a valid approximation to the counterfactual comparison, and what assumption is required.

(c) Explain why matching on $\hat{\pi}(X)$ is *not* the same as matching on X directly. Under what conditions do they give the same answer?

2. ATE, ATT, and propensity score weighting. A dataset has $n = 1000$ observations with estimated propensity scores $\hat{\pi}(X_i)$ and $n_1 = 400$ treated units.

(a) Write the Horvitz–Thompson IPW estimator of the ATE as a weighted sum of observed outcomes Y_i. What weights do treated units receive? What weights do control units receive?
(b) Derive an analogous IPW estimator for the ATT. (*Hint*: the ATT averages over the treated distribution; the denominator of the weight for control units should reflect this.)
(c) Show that when the propensity score is constant ($\pi(X) = p$ for all units), the IPW estimator of the ATE reduces to the difference-in-means estimator. Under what experimental design does $\pi(X) = 0.5$ exactly?

3. Overlap. Let $\pi(X)$ be the true propensity score and suppose overlap fails at $X = x_0$: $\pi(x_0) = 1$.

(a) For each of τ_{ATE} and τ_{ATT}, identify the exact formula that fails to be identified when $\pi(x_0) = 1$, and explain why.
(b) Suppose overlap fails only on a set $\mathcal{S}$ with $P(X \in \mathcal{S}) = 0.15$. Define the *trimmed ATE* estimand. How does it differ from the ATE?
(c) A researcher applies the Crump et al. (2009) rule with $\eta = 0.1$, removing all units with $\hat{\pi}(X_i) < 0.1$ or $\hat{\pi}(X_i) > 0.9$, removing 8% of the sample. (i) List two reasons the trimmed estimator may have lower variance than the untrimmed IPW estimator. (ii) What is the cost in terms of external validity?

4. Sensitivity analysis (conceptual). Suppose you estimate $\hat{\tau}_{\text{ATE}} = 3.2$ using propensity-score methods, and a referee questions whether unconfoundedness holds.

(a) Define what it means for U to be a "hidden confounder" in the context of the DAG, and explain why its presence invalidates the IPW identification formula.
(b) Explain in one paragraph what Rosenbaum-style sensitivity analysis would tell you about the robustness of your estimate to hidden confounding.
(c) Suppose a sensitivity analysis shows that a confounder with an odds ratio of 2 for both treatment and outcome selection would be required to explain away your result. How would you interpret this finding? Does it confirm unconfoundedness?

Chapter 7

Instrumental Variables

Learning Objectives

By the end of this chapter, students should be able to:

1. Diagnose the endogeneity problem: explain why an unobserved confounder makes back-door adjustment fail, and describe how an instrumental variable provides an alternative identification route.
2. State the three IV assumptions — relevance, exogeneity, and exclusion — in the DAG/do-calculus, structural-equation, and potential-outcomes languages, and explain which are testable and which require institutional justification.
3. Follow how the IV assumptions identify the Wald estimand, and interpret the reduced form and first stage as its two observable components.
4. Explain what the Wald estimand identifies when treatment effects are heterogeneous — the Local Average Treatment Effect for compliers — why that estimand depends on the choice of instrument, and when it coincides with the ATE.
5. Compare IV and back-door adjustment on the assumptions each requires, the estimand each identifies, and the ways each can fail.

7.1 Why Instrumental Variables?

Chapter 6 showed how causal effects can be identified when all confounders are observed and can be blocked by conditioning on X. When some confounders are unobserved, back-door adjustment fails. This chapter develops instrumental variables (IV) as an alternative identification strategy: rather than blocking the confounding path $T \leftarrow U \rightarrow Y$, IV exploits an external variable Z whose effect on T is free of confounding by U. This chapter asks what IV identifies and under what assumptions; Chapter 12 asks how that estimand is computed and tested in practice.

7.1.1 The Endogeneity Problem

Chapter 6 established that the propensity score provides a powerful surrogate for randomization, but only under *unconfoundedness*: $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$. This requires that every variable affecting both treatment and outcome is observed. In many empirical settings this is implausible:

- In labor economics, unobserved ability or motivation affects both schooling decisions and wages.
- In epidemiology, unobserved health behaviors affect both treatment uptake and outcomes.
- In survey sampling, voluntary participation is driven by factors that also affect the variables of interest.

Whenever an unobserved confounder U creates a back-door path $T \leftarrow U \rightarrow Y$, the adjustment formula fails. The gap is the *endogeneity bias*.

7.1.2 The IV Idea

An *instrumental variable* Z is an observed variable that: (1) *moves* treatment T (relevance); (2) *does so exogenously* — Z is unrelated to the unobserved confounder U (exogeneity); (3) *affects* Y *only through* T — Z has no direct path to Y (exclusion).

Under these three conditions, the variation in T induced by Z is free of confounding by U. The ratio of the Z-induced variation in Y to the Z-induced variation in T identifies the causal effect.

IV does not eliminate the confounding path — it identifies causal effects by isolating a component of treatment variation that is induced by Z and is therefore exogenous. IV *avoids* confounding rather than controlling for it. This distinction matters: IV identifies the LATE for compliers, not the ATE, and the exclusion restriction does the heaviest lifting.

Example: Returns to Schooling

A researcher wants to estimate the causal effect of years of schooling T on log wages Y. Unobserved ability U raises both schooling and wages, so OLS is biased upward. An instrumental variable Z that shifts schooling for reasons unrelated to ability — such as proximity to a school or a policy change in compulsory attendance laws — provides a way to isolate the causal effect of schooling on wages (Angrist and Krueger 1991).

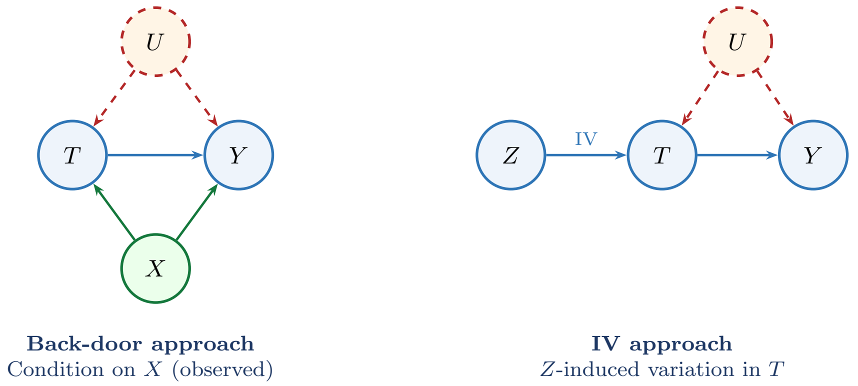


Figure 7.1: Two strategies for identification. **Left**: back-door adjustment conditions on the observed X (green) to block the confounding path. **Right**: IV uses only Z-induced variation in T, which is free of U.

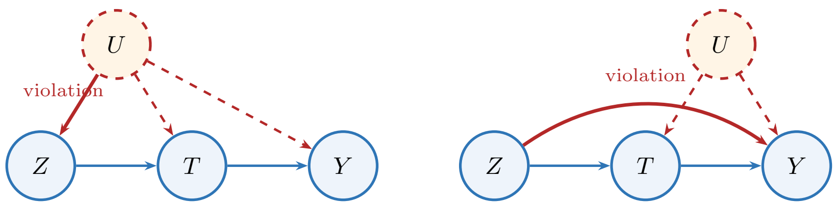
7.2 Graphical Setup and Core Assumptions

7.2.1 The IV DAG

The causal structure for a basic IV model with covariates X has nodes Z, T, Y, X, U with edges $Z \to T$, $T \to Y$, $X \to T$, $X \to Y$, $X \to Z$, $U \to T$, $U \to Y$ (U unobserved). The three IV assumptions correspond to: (1) **Relevance**: directed path $Z \to T$; (2) **Exogeneity**: $Z \perp\!\!\!\perp U \mid X$ by d-separation; (3) **Exclusion**: no direct arrow $Z \to Y$.

The distinction between exogeneity and exclusion is fundamental. Exogeneity says the instrument is not confounded with latent causes of the outcome. Exclusion says it has no causal channel to the outcome except through treatment. A randomized encouragement may be exogenous by design, yet still violate exclusion if the encouragement changes outcomes through information, motivation, or stigma apart from the treatment itself.

7.2.2 The Three Assumptions in Three Languages



Exogeneity failure
$Z \leftarrow U \rightarrow Y$

Exclusion failure
$Z \rightarrow Y$ directly

Figure 7.2: The two main IV assumption failures. **Left**: exogeneity fails when U is a common cause of Z and Y. **Right**: exclusion fails when Z has a direct causal path to Y. The two failures are logically independent.

Assumption	Potential outcomes	Structural / econometric	Do-calculus / DAG
Relevance	$Z \not\perp\!\!\!\perp T$	$\mathrm{Cov}(Z, T) \neq 0$	No d-separation of Z and T in $\mathcal{G}$
Exogeneity	$Z \perp\!\!\!\perp (Y(0), Y(1), T(0), T(1)) \mid X$	$\mathbb{E}[\varepsilon \mid Z, X] = 0$	$Z \perp\!\!\!\perp U \mid X$
Exclusion	$Y_i(t, z) = Y_i(t, z')$ for all z, z'	Z absent from structural equation for Y	$f(y \mid x, \mathrm{do}(T{=}t), z) = f(y \mid x, \mathrm{do}(T{=}t))$

Remark: Three Languages, One Identification Argument

The graphical statement encodes causal structure (which arrows are present). The potential-outcomes statement encodes counterfactual independence. The structural formulation encodes moment orthogonality. They support the same identification argument in a well-specified model, but are not literally the same statement.
The do-calculus column is preferred throughout this course because the exclusion restriction in that language is a statement about the *interventional* density. It makes it impossible to confuse exclusion with the much weaker observational statement $f(y \mid x, T{=}t, z) = f(y \mid x, T{=}t)$.

7.2.3 Relevance

Definition: Relevance

The instrument Z is **relevant** if it has a non-zero causal effect on the treatment T, possibly after conditioning on covariates X: $\mathrm{Cov}(Z, T \mid X) \neq 0$.

Relevance is the population condition that the instrument changes treatment status. The first-stage F-statistic is a sample diagnostic for whether this link is strong enough; it is evidence about instrument strength, not the assumption itself.

7.2.4 Exogeneity

Definition: Exogeneity

The instrument Z is **exogenous** given covariates X if there is no open back-door path from Z to Y through unobserved causes: $Z \perp\!\!\!\perp U \mid X$ in $\mathcal{G}$ by d-separation.

Exogeneity means the instrument carries no information about latent determinants of the outcome other than through treatment. The moment condition $\mathbb{E}[\varepsilon Z \mid X] = 0$ is a *consequence* of this graphical assumption, not a definition. All versions of exogeneity are untestable from observed data alone.

7.2.5 Exclusion

Definition: Exclusion Restriction

The instrument Z satisfies the **exclusion restriction** if it affects the outcome Y *only* through its effect on treatment T. In do-calculus terms:

$$f(y \mid x, \operatorname{do}(T{=}t), z) = f(y \mid x, \operatorname{do}(T{=}t)) \quad \text{for all } z.$$

In potential outcomes terms: $Y_i(t, z) = Y_i(t, z')$ for all $z \neq z'$.

Exclusion is a *causal* restriction, not a conditional-correlation restriction. Adding a direct arrow $Z \to Y$ to the DAG is the formal counterpart of exclusion failing: in the mutilated graph $\mathcal{G}_{\overline{T}}$, the path $Z \to Y$ remains open and Rule 1 of the do-calculus can no longer remove Z.

Why Exclusion Is the Hardest Assumption

Relevance can be tested. Exogeneity can sometimes be defended by design. But exclusion — that Z has *no direct effect* on Y whatsoever — is: **untestable** in just-identified models; **easy to violate in practice** (an instrument that moves treatment often also moves other inputs); and **consequential** — the Wald estimand converges to $\beta + \delta/\pi$, so even a small direct effect δ combined with a weak instrument π can produce large bias.

Remark: Three Assumptions Are Necessary but Not Sufficient

(Hernán and Robins 2006; Levis et al. 2024) The three IV assumptions alone do not yield point identification. A fourth structural assumption is required. Two leading choices: **constant treatment effects** (Framework 1, Section 8.7) under which Wald identifies the ATE; and **monotonicity** (Framework 2, Section 7.7) under which Wald identifies the LATE for compliers. An **additive structural mean model (SMM)** identifies the ATT without either. Strikingly, across many fourth assumptions the same conditional Wald formula $\operatorname{Cov}(Z, Y \mid X)/\operatorname{Cov}(Z, T \mid X)$ serves as the identifying expression — but the causal target changes.

The fourth assumption lies *outside* the graphical language in every case. Monotonicity ($T_i(1) \geq T_i(0)$) is a cross-world statement that cannot be encoded by any arrow in the DAG. IV identification cannot be completed by causal graph analysis alone.

7.3 Identification in the Linear Homogeneous-Effect Model

Framework 1: Constant Treatment Effect and Linear Structure

This section works within a linear structural model in which the treatment effect is the *same* for every unit: $Y_i(t) - Y_i(t') = \beta(t - t')$ for all i, t, t'. Under this assumption, $\beta = \text{ATE} = \text{ATT} = \text{LATE}$.

7.3.1 The Linear Structural Model

Consider the linear structural model:

$$Y = \alpha + \beta T + \gamma^\top X + \varepsilon, \tag{7.1}$$

$$T = \pi Z + \delta^\top X + \eta, \tag{7.2}$$

where $\mathrm{Cov}(\varepsilon, \eta) = \rho\sigma\tau \neq 0$. The *reduced form* substitutes Equation 7.2 into Equation 7.1: the reduced-form coefficient on Z is $\beta\pi$.

7.3.2 The OLS Bias

$$\operatorname{plim} \hat{\beta}_{\mathrm{OLS}} = \beta + \frac{\rho\sigma\tau}{\pi^2 \mathrm{Var}(Z) + \tau^2}.$$

The bias is zero only if $\rho = 0$ (no unobserved confounding). The direction depends on the sign of ρ.

7.3.3 Derivation of the Wald Estimand

The derivation has three ingredients: exogeneity, exclusion, and relevance.

1. **Exogeneity** implies $\mathbb{E}[\varepsilon \mid Z, X] = 0$.
2. **Exclusion** (Z absent from the structural equation for Y) combined with exogeneity gives $\mathbb{E}[\varepsilon \cdot Z \mid X] = 0$. In do-calculus terms: Rule 1 removes Z from $f(y \mid x, \mathrm{do}(T{=}t), z)$ because $Y \perp\!\!\!\perp Z \mid T, X$ in $\mathcal{G}_{\overline{T}}$.
3. Multiplying Equation 7.1 by $(Z - \mathbb{E}[Z \mid X])$ and applying step 2 gives the **reduced-form decomposition** (no-covariate case):
$$\mathrm{Cov}(Y, Z) = \beta\, \mathrm{Cov}(T, Z).$$
4. **Relevance** ensures the denominator is non-zero:
$$\beta = \frac{\mathrm{Cov}(Y,\, Z \mid X)}{\mathrm{Cov}(T,\, Z \mid X)}.$$

For binary $Z \in \{0, 1\}$ without covariates, this is the **Wald estimand**:

The Wald Estimand

$$\beta = \frac{\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]}{\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]}.$$

The numerator is the *reduced form*: the total effect of Z on Y. The denominator is the *first stage*: the effect of Z on T.

The key lesson is that the Wald ratio identifies β in this section *only* because treatment effects are assumed to be constant across units. In Section 7.7, the same ratio will survive, but its interpretation changes from a common causal effect to a complier-specific average effect.

7.4 Why the IV Assumptions Matter

Each assumption failure produces a distinct, quantifiable distortion in the Wald estimand, amplified by a weak first stage.

When relevance fails. If $\pi = 0$, the Wald estimand is undefined. When π is small, IV variance diverges and finite-sample bias pulls the estimate toward OLS at a rate $\propto 1/F$.

When exclusion fails. If $Y = \alpha + \beta T + \delta Z + \gamma^\top X + \varepsilon$, the Wald estimand converges to $\beta + \delta/\pi$. A small direct effect combined with a weak instrument can produce large bias.

When exogeneity fails. If $\mathrm{Cov}(Z, \varepsilon \mid X) \neq 0$, the Wald estimand converges to $\beta + \mathrm{Cov}(\varepsilon, Z)/\mathrm{Cov}(T, Z)$ — amplified by a weak first stage.

Assumption	Directly testable?	Basis for assessment
Relevance	Yes	First-stage F-statistic
Exogeneity	No	Institutional knowledge; randomization; placebo regressions
Exclusion	No (just-ID); partially (overID)	Institutional argument; J-test checks consistency

7.5 Lab: OLS vs. IV Across Instrument Strengths

This lab verifies the OLS bias formula numerically and traces the bias–variance tradeoff across the full range of instrument strength.

> DGP for Lab 7
>
> $Y_i = \beta T_i + \varepsilon_i$, $T_i = \pi Z_i + \eta_i$, with $Z_i \sim \mathrm{N}(0,1)$, $\eta_i = U_i$, $\varepsilon_i = \rho U_i + \sqrt{1-\rho^2}\,\xi_i$, $U_i, \xi_i \sim \mathrm{N}(0,1)$ i.i.d. Fixed parameters: $\beta = 1$, $\rho = 0.8$. Grid: $\pi \in \{0, 0.10, 0.15, 0.20, 0.30, 0.50, 1.00, 2.00\}$. Expected $F \approx 500\pi^2$. All three IV assumptions hold by construction.

π	F	Theory bias	OLS Bias	OLS RMSE	IV Bias	IV SD	IV RMSE
0.00	1	+0.800	+0.801	0.802	—	—	—
0.10	6	+0.792	+0.791	0.792	−0.128	4.391	4.392
0.15	13	+0.782	+0.782	0.783	−0.236	6.487	6.491
0.20	21	+0.769	+0.769	0.770	−0.060	0.380	0.385
0.30	46	+0.734	+0.735	0.735	−0.025	0.163	0.165
0.50	127	+0.640	+0.640	0.640	−0.004	0.091	0.091
1.00	500	+0.400	+0.401	0.402	−0.001	0.045	0.045
2.00	2006	+0.160	+0.160	0.161	0.000	0.022	0.022

True $\beta = 1.000$. $n = 500$, $B = 2{,}000$ replications. IV mean and SD at $\pi = 0.10$ and 0.15 are dominated by extreme outliers; the IV median is ≈ 1.01 at both values.

Four lessons stand out.

1. The OLS bias formula is exact. The theory bias column ($0.8/(\pi^2+1)$) matches the simulated OLS bias to four decimal places. OLS bias decreases as π grows but is never zero.

2. IV is consistent but has catastrophically heavy tails when the instrument is weak. At $F \approx 6$–13, the IV *median* is ≈ 1.01 (confirming consistency), but a small fraction of replications in which the first stage happens to be near zero inflates the SD to 4.4–6.5. The weak instrument problem is uncontrolled tail behavior, not inconsistency.

3. The RMSE crossover occurs near $F \approx 20$. IV first beats OLS at $\pi = 0.20$ ($F \approx 21$): IV RMSE $= 0.385 <$ OLS RMSE $= 0.770$. The Staiger–Stock rule of thumb ($F \geq 10$) is therefore slightly too lenient; $F \geq 20$–25 is more appropriate in this DGP.

4. A strong instrument eliminates both problems. At $\pi = 1.00$ ($F \approx 500$): IV RMSE $= 0.045$ vs. OLS RMSE $= 0.402$ — a 9-fold improvement from IV.

> RMSE Cannot Be the Only Criterion
>
> OLS has lower RMSE than IV for $\pi < 0.20$ in this DGP. This does not vindicate OLS. An estimator with RMSE $= 0.78$ because it is biased by 0.80 is useless for causal inference: the bias is systematic and does not shrink with sample size. As $n \to \infty$, IV RMSE shrinks to zero while OLS RMSE stays at 0.80. The RMSE comparison is meaningful only in finite samples; it quantifies when a weak instrument is not yet practically useful, not when OLS is acceptable.

Remark: The First-Stage F-Statistic as a Diagnostic

(Bound et al. 1995) The Staiger–Stock threshold $F \geq 10$ corresponds to IV RMSE within a factor of roughly two of the strong-instrument limit. The simulation here suggests this threshold may understate the problem when endogeneity is strong ($\rho = 0.8$): at $F = 13$ the IV RMSE is 6.5, far above any useful threshold. A researcher should report the first-stage F alongside any IV estimate, and treat values below 20–25 with particular caution.

7.6 Multiple Instruments and Overidentification

When there are exactly as many instruments as endogenous variables ($q = p$), the model is *just-identified*. When $q > p$, the model is *overidentified*: the extra instruments impose testable moment restrictions. The Sargan–Hansen J-test (Sargan 1958; Hansen 1982) checks mutual consistency of instruments; rejection indicates at least one instrument violates exogeneity or exclusion, but does not identify which one. The test statistic and distribution are derived in Chapter 12.

What Overidentification Does Not Do

Passing the J-test does not confirm any instrument is valid — only that sample moment conditions are mutually compatible. Multiple invalid instruments can agree if they share the same violation. Overidentification is an opportunity for a consistency check, not a substitute for the institutional argument.

7.7 Heterogeneous Treatment Effects and the LATE Framework

Framework 2: Heterogeneous Treatment Effects

We now drop homogeneity and allow $\tau_i = Y_i(1) - Y_i(0)$ to vary across units, in the binary instrument, binary treatment setting ($Z, T \in \{0, 1\}$). Under an additional assumption of monotonicity, the Wald ratio identifies the *Local Average Treatment Effect* (LATE): the average treatment effect for the subpopulation whose treatment status is changed by the instrument.

7.7.1 Compliance Types

Definition: Compliance Types

(Angrist et al. 1996) For binary $Z, T \in \{0, 1\}$, define $T_i(z)$ as the treatment unit i would take under instrument value z. The four compliance types are:

Type	$T_i(0)$	$T_i(1)$
Complier	0	1
Always-taker	1	1
Never-taker	0	0
Defier	1	0

The instrument Z only shifts treatment for *compliers*: always-takers and never-takers have the same treatment status regardless of Z, contributing nothing to the denominator.

7.7.2 The Monotonicity Assumption

Definition: Monotonicity

(Angrist and Imbens 1994) The treatment assignment is **monotone in** Z if: $T_i(1) \geq T_i(0)$ for all i (no defiers).

7.7.3 The LATE Theorem

Theorem: LATE

(Angrist and Imbens 1994) Under the three core IV assumptions and monotonicity, the Wald estimand identifies the **Local Average Treatment Effect**:

$$\frac{\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]}{\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]} = \mathbb{E}[Y(1) - Y(0) \mid T_i(1) > T_i(0)] \equiv \tau_{\text{LATE}}.$$

Proof

Denominator. By exogeneity ($Z \perp\!\!\!\perp T(0), T(1)$):

$$\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0] = \mathbb{E}[T(1) - T(0)] = P(\text{complier}),$$

since always-takers contribute $1 - 1 = 0$, never-takers $0 - 0 = 0$, and no defiers exist.
Numerator. By exogeneity and law of total expectation over compliance types:

$$\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0] = \mathbb{E}[Y(T(1))] - \mathbb{E}[Y(T(0))].$$

Always-takers and never-takers contribute 0; compliers contribute $Y(1) - Y(0)$. Therefore:

$$\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0] = P(\text{complier})\, \mathbb{E}[Y(1) - Y(0) \mid \text{complier}].$$

Ratio: Dividing numerator by denominator yields τ_{LATE}. □

7.8 Interpreting IV Estimands

	Framework 1 (homogeneous)	**Framework 2** (heterogeneous)
Key assumption	$\tau_i = \beta$ for all i	Monotonicity; no defiers
What IV identifies	$\beta = \text{ATE} = \text{ATT} = \text{LATE}$	$\tau_{\text{LATE}} = \mathbb{E}[\tau_i \mid \text{complier}]$
Estimand depends on Z?	No	Yes (different $Z \Rightarrow$ different compliers $\Rightarrow$ different LATE)
Identifies ATE?	Yes	Only under additional homogeneity

7.8.1 When Does LATE Equal ATE?

ATE decomposes by compliance type:

$$\text{ATE} = P(\text{co})\, \mathbb{E}[\tau_i \mid \text{co}] + P(\text{at})\, \mathbb{E}[\tau_i \mid \text{at}] + P(\text{nt})\, \mathbb{E}[\tau_i \mid \text{nt}].$$

LATE equals only the first term divided by its probability weight. ATE and LATE coincide if and only if: (1) effects are homogeneous, or (2) everyone is a complier, or (3) effects for always-takers and never-takers happen to equal the LATE. In practice, none of these is likely to hold exactly.

7.8.2 Different Instruments, Different Estimands

Because the LATE is specific to the complier population, two valid instruments for the same treatment can legitimately identify different LATEs. This is informative about treatment effect heterogeneity, not a contradiction.

Example: Compulsory Schooling Laws vs. Distance to College

Two classic instruments for years of schooling: compulsory schooling laws (Angrist and Krueger 1991) select compliers at the margin of dropping out (typically lower-income students); distance to the nearest college (Card 1995) selects students deterred by geography. Both identify the return to schooling for their respective complier populations, and the LATEs can legitimately differ even if both instruments are valid.

7.8.3 The Policy Relevance of LATE

For many policy questions, the LATE is exactly the right estimand — when the contemplated intervention resembles the instrument, the complier population under the study design is close to the policy-relevant margin. When the ATE over the full population is required, IV alone is insufficient under heterogeneous effects; additional structure or a second instrument is needed.

7.9 IV versus Back-Door Adjustment

Dimension	Back-door / propensity score	Instrumental variables
Core assumption	All confounders observed	Valid instrument: relevance, exogeneity, exclusion
Unobserved confounders	Fatal	Permitted
Estimand	ATE, ATT, or ATC	LATE (compliers only)
Testability	Unconfoundedness untestable; overlap testable	Relevance testable; exogeneity/exclusion untestable (just-ID)
Main threat	Unmeasured confounder	Exclusion restriction violation
Identifies ATE?	Yes, under strong ignorability	Only under homogeneous effects

The two strategies have complementary failure modes. Back-door adjustment fails when X does not capture all confounders. IV fails when the exclusion restriction is violated, with bias δ/π amplified by a weak first stage. They are tools for different identification problems, not competitors.

Five questions for any IV application: (1) What exactly is the instrument? (2) Why does it shift treatment? (3) Why is it as-if random relative to latent outcome determinants? (4) Why can it affect the outcome only through treatment? (5) What population margin does it shift?

7.10 Applied Example: Charter School Lotteries and the KIPP Lynn Study

This example illustrates a canonical randomized-encouragement IV design (Angrist et al. 2012). The lottery randomizes *offer status*, not actual treatment. Z = lottery offer; S = years of KIPP attendance (endogenous); Y = standardized MCAS test score.

Relevance. First-stage coefficient ≈ 1.2: lottery winners had spent 1.2 more years at KIPP at the time of each MCAS exam. First-stage F is far above conventional thresholds.

Exogeneity. Guaranteed by the randomization protocol. Balance test: p-value $= 0.615$ for demographic characteristics and baseline test scores.

Exclusion. The lottery offer merely provides access — it does not deliver instruction. One potential violation is a discouragement effect on losers; the authors note that lottery losers' scores are typical of demographically comparable students in Lynn.

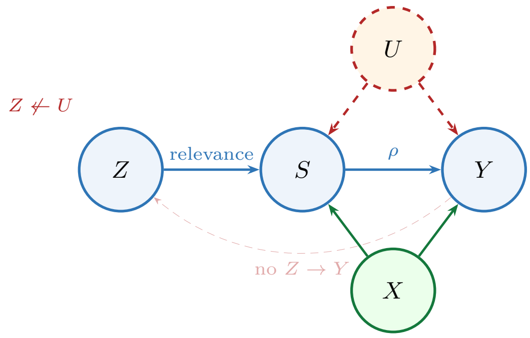


Figure 7.3: Causal DAG for the KIPP Lynn lottery IV design. Exogeneity is guaranteed by the randomization protocol; exclusion rests on the institutional argument that the lottery offer affects achievement only through attendance.

Results ((Angrist et al. 2012), Table 4, specification with demographic and baseline-score controls):

Subject	First stage	Reduced form	2SLS
Math	1.221 (0.068)	0.430 (0.067)	0.352 (0.053)
ELA	1.228 (0.068)	0.164 (0.073)	0.133 (0.059)

SE clustered at the student level. $N = 833$ *student-by-test observations.*

Each year at KIPP raises math scores by approximately 0.35σ and ELA scores by approximately 0.13σ. The 2SLS estimand identifies the LATE for *lottery compliers* only. Reading gains are driven almost entirely by LEP students ($\approx 0.43\sigma$) and SPED students ($\approx 0.27\sigma$).

7.11 Summary

1. **IV identifies effects from exogenous treatment variation.** When back-door adjustment fails because U creates a path $T \leftarrow U \rightarrow Y$, a valid instrument Z identifies the causal effect by exploiting only the component of treatment variation that Z induces. IV avoids confounding rather than controlling for it.

2. **Three assumptions, ordered by testability.** Relevance is assessable with the first-stage F. Exogeneity and exclusion must be defended by institutional knowledge and causal structure. Each violated assumption produces a distinct, quantifiable bias in the Wald estimand, amplified by weak instruments.

3. **Framework 1 (homogeneous effects):** Under constant treatment effects and the linear structural model, the Wald estimand identifies $\beta = \text{ATE} = \text{ATT} = \text{LATE}$.

4. **Framework 2 (heterogeneous effects + monotonicity):** The Wald estimand identifies $\tau_{\text{LATE}} = \mathbb{E}[\tau_i \mid \text{complier}]$ — the effect for the subpopulation whose treatment status is changed by the instrument.

5. **Different instruments identify different effects.** The LATE depends on the instrument through the complier population it selects. LATE equals ATE only under additional structure, most notably treatment-effect homogeneity.

6. **IV versus back-door adjustment.** Complementary failure modes: back-door fails when confounders are unobserved; IV fails when exclusion is violated. They are tools for different identification problems.

7. **Applied example: KIPP Lynn lottery.** The lottery guarantees exogeneity by design; exclusion is defended institutionally; first-stage coefficient ≈ 1.2. Estimates identify the LATE for lottery compliers: 0.35σ per year in math, 0.13σ in ELA.

8. **Estimation deferred to Chapter 12.** This chapter establishes what IV identifies and under what assumptions. How the Wald ratio is estimated — 2SLS, asymptotic inference, overidentification tests — is the subject of Chapter 12.

7.12 Problems

1. The three IV assumptions in three languages. Consider the DAG $\{Z \to T,\ T \to Y,\ U \to T,\ U \to Y,\ X \to T,\ X \to Y,\ X \to Z\}$ with U unobserved.

(a) List all back-door paths from T to Y. Does X alone satisfy the back-door criterion? Explain.
(b) Verify the three IV assumptions using d-separation: (i) Relevance: show Z and T are not d-separated in $\mathcal{G}$. (ii) Exogeneity: show $Z \perp\!\!\!\perp U \mid X$ in $\mathcal{G}$. (iii) Exclusion: show $Y \perp\!\!\!\perp Z \mid T, X$ in $\mathcal{G}_{\overline{T}}$.
(c) Now add the arrow $Z \to Y$ to the DAG. Which IV assumption is violated? Show explicitly which step of the Wald derivation breaks down.
(d) Translate each of the three IV assumptions into the structural language: write the equations for T and Y and identify which coefficient restriction corresponds to each assumption.

2. Bias under assumption violations. Let $Y = \beta T + \varepsilon$ and $T = \pi Z + \eta$ with $\mathbb{E}[\varepsilon \mid Z] = 0$ and $\pi \neq 0$.

(a) Starting from $\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]$, substitute the structural equation for Y and simplify. What role does exogeneity play?
(b) Show that $\mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0] = \pi$ in the linear first-stage model.
(c) Derive the Wald estimand and confirm it equals β.
(d) Now suppose exclusion fails: $Y = \beta T + \delta Z + \varepsilon$ with $\delta \neq 0$. Derive the probability limit of the Wald estimator and confirm the bias formula from Section 7.4.
(e) Suppose instead exogeneity fails: $\mathbb{E}[\varepsilon \mid Z] = cZ$ for some constant $c \neq 0$. Derive the probability limit of the Wald estimator and express the bias in terms of c and π.

3. Order, rank, and the limits of overidentification. Consider a model with one endogenous variable T and two instruments Z_1 and Z_2, both satisfying exogeneity and exclusion.

(a) State the order condition and verify it is satisfied.
(b) State the rank condition. What would it mean geometrically if Z_1 and Z_2 were perfectly collinear in the first-stage regression?
(c) Explain intuitively why having two valid instruments rather than one should improve estimation precision.
(d) Now suppose Z_1 is valid but Z_2 violates exclusion. Under what conditions does the Sargan–Hansen J-test have power to detect Z_2's invalidity? Under what conditions does the test fail?
(e) Why does passing the J-test not confirm that both instruments are valid? Give a concrete example where both are invalid and the J-test has no power.

4. Compliance types and the LATE. In a binary instrument, binary treatment study: 30% compliers with $\tau_c = 6$; 25% always-takers with $\tau_a = 3$; 45% never-takers with $\tau_n = 1$; no defiers.

(a) Compute $P(\text{complier}) = \mathbb{E}[T \mid Z{=}1] - \mathbb{E}[T \mid Z{=}0]$.
(b) Compute the ATE as a weighted average of τ_c, τ_a, τ_n with appropriate weights.
(c) The Wald estimand equals $\tau_c = 6$. By how much does this overstate the ATE, and why?
(d) A second study uses a different binary instrument Z' with a complier population of 50% and a LATE of 2. Is this contradictory? What can you infer about relative treatment effects in the two complier populations?
(e) Explain, using compliance type language, why the denominator of the Wald estimand equals $P(\text{complier})$.

5. The exclusion restriction: plausibility and violations. For each proposed instrument, (i) state whether the restriction is plausible and why; (ii) describe a specific mechanism by which it could be violated; (iii) assess whether the violation would bias the IV estimate upward or downward.

(a) **Instrument**: rainfall in the home region of a politician, used as an instrument for government infrastructure spending. **Outcome**: local economic growth.
(b) **Instrument**: distance to the nearest hospital, used as an instrument for hospital admission. **Outcome**: 30-day mortality.

(c) **Instrument**: a randomly assigned financial incentive to enroll in a health screening program. **Outcome**: health status two years later.
(d) **Instrument**: lottery number in the Vietnam-era draft lottery, used as an instrument for military service. **Outcome**: lifetime earnings. *(This is the Angrist (1990) study; discuss why this instrument is widely regarded as satisfying the exclusion restriction.)*

6. IV versus back-door adjustment. A researcher studies the effect of job training (T) on earnings (Y). Two strategies: (A) a rich set of pre-treatment covariates X and a propensity-score estimator; (B) a lottery that randomly selected units to be offered training, used as an instrument Z.

(a) Under what assumption does strategy (A) identify the ATE? What specific unobserved variable would most plausibly violate this assumption?
(b) Strategy (B) identifies a LATE. Describe the complier population in words. Is the LATE likely to be larger or smaller than the ATE in this setting?
(c) Both strategies yield estimates of $1,800 and $2,400 per year. Describe a Hausman-type test. Under what null hypothesis does the test have an approximate χ^2 distribution?
(d) If the two estimates differ significantly, which strategy would you trust more and why? What additional evidence would help distinguish endogeneity bias in (A) versus LATE $\neq$ ATE in (B)?

Chapter 8

Mediation and Front-Door Identification

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain the distinction between the total causal effect of T on Y and a pathway-specific effect that operates through a mediator M, and describe why this distinction matters scientifically.
2. Draw the prototype mediation DAG, write its structural equations, and identify the direct and indirect pathways.
3. Define the controlled direct effect (CDE) using the do-operator, identify it via the back-door formula for the joint intervention (T, M), and explain why it depends on the fixed level m.
4. Define the natural direct and indirect effects (NDE, NIE) using the potential outcomes notation $Y(t, M(t'))$, state the NDE+NIE=TE decomposition, and explain why these quantities involve cross-world counterfactuals.
5. State the four sequential ignorability assumptions for identification of natural effects, write the mediation formula, and identify which assumption is violated when there is an unmeasured treatment-induced mediator–outcome confounder.
6. Set up the Baron–Kenny three-equation system, derive the product and difference formulas for the indirect effect, and explain why the decomposition fails in nonlinear or interaction models.
7. State the three front-door conditions, derive the front-door formula using the do-calculus, and explain why the front-door graph enables identification despite unobserved T–Y confounding.
8. Contrast mediation analysis and instrumental variables on the dimensions of variable position, identification goal, and key assumption.

8.1 Motivation: Mechanisms

The identification results of Chapters 5–7 all answer the same question: *what is the total causal effect of T on Y?* Mediation analysis asks a finer question: *through what mechanism does that effect operate?*

More concretely, the total effect of T on Y may flow along multiple causal pathways. Some of this effect passes through an intermediate variable M — the *mediator* — along the path $T \to M \to Y$. The remainder flows directly along $T \to Y$, bypassing the mediator entirely. Decomposing the total effect into these two components is the goal of mediation analysis.

This decomposition matters for scientific and policy reasons. In a clinical trial of a behavioral intervention (T) on depression (Y), a researcher may want to know how much of the benefit operates through improved sleep quality (M) versus other pathways — because if sleep is the main channel, targeting sleep directly may be a more efficient intervention. In an economics study of education (T) on wages (Y), how much operates through occupation (M) versus cognitive skills?

Running example. Throughout this chapter we use the depression intervention as the primary example: T is a randomized behavioral intervention, M is self-reported sleep quality measured mid-trial, and Y is a depression score (e.g., PHQ-9 scale).

The challenge is that mediators are *post-treatment variables*: they are affected by the treatment, and may themselves be confounded with the outcome. Conditioning on a post-treatment variable creates exactly the collider and selection-bias problems studied in Chapters 2 and 3. A naive approach — simply including M as a covariate in a regression of Y on T — conflates adjustment with mediation and can introduce bias even in a randomized experiment.

This chapter also develops the *front-door criterion*, a distinct identification strategy that uses the mediation structure of the DAG to identify causal effects even when treatment and outcome are confounded by an unobserved variable.

8.2 The Mediation DAG

8.2.1 The Prototype Graph

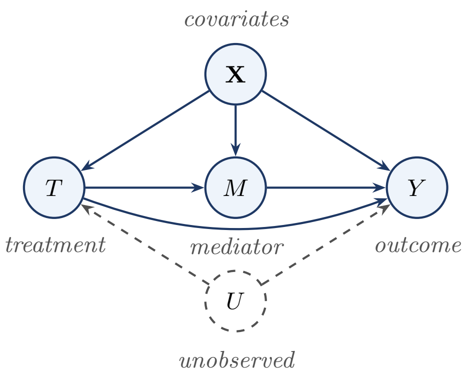


Figure 8.1: The prototype mediation DAG. The causal effect of T on Y operates through two pathways: the direct path $T \to Y$ and the indirect path $T \to M \to Y$ via the mediator. The unobserved variable U confounds the treatment–outcome relationship; $\mathbf{X}$ denotes observed pre-treatment covariates.

The graph encodes two causal pathways: the **direct pathway** $T \to Y$ (treatment affects the outcome without passing through the mediator) and the **indirect pathway** $T \to M \to Y$ (treatment first shifts the mediator, which shifts the outcome).

The prototype graph corresponds to the nonparametric SEM:

$$T = f_T(\mathbf{X}, U, \varepsilon_T), \qquad M = f_M(T, \mathbf{X}, \varepsilon_M), \qquad Y = f_Y(T, M, \mathbf{X}, U, \varepsilon_Y).$$

8.2.2 What Makes Mediation Harder Than Total Effect Estimation

The key difficulty is that M is a *post-treatment* variable. This creates two interrelated problems.

Collider bias. Conditioning on M can open collider paths. If $V \to M$ and $V \to Y$ with V unobserved, the path $T \to M \leftarrow V \to Y$ is blocked when M is not conditioned on, but opens as soon as M is included as a covariate. This is precisely why naively regressing Y on (T, M) does not isolate the direct effect.

Mediator–outcome confounding. Even when treatment is randomized, the mediator M is never randomized. An unobserved V with $V \to M$ and $V \to Y$ creates a back-door path from M to Y that randomization of T does not close.

Post-Treatment Variables Are Dangerous to Condition On

The mediator M is caused by the treatment T. This single fact makes every operation involving M structurally different from operations involving pre-treatment covariates $\mathbf{X}$.
Conditioning on a post-treatment variable in a regression or matching procedure is *not* a neutral act. It can open collider paths that were previously blocked, introduce selection bias, and produce estimates of neither the total effect nor any well-defined direct effect. This chapter studies three regimes in which conditioning on or marginalizing over M is justified: (1) **CDE**: both T and M

are intervened on simultaneously via $\mathrm{do}(T, M)$; (2) **NDE/NIE**: M is marginalized over using a distribution from a different treatment arm, justified by sequential ignorability; (3) **Front-door**: M is summed over in the front-door formula, justified by graphical conditions. Outside these three regimes, conditioning on a post-treatment variable should be treated as an error until proven otherwise.

8.2.3 Working Assumption for §§ 3–6

Throughout §§ Section 8.3–Section 8.7, we work with the *reduced prototype graph* (obtained from Figure 8.1 by assuming all T–Y and T–M confounding is captured by the observed $\mathbf{X}$). The unobserved confounder U is restored in Section 8.8.

8.3 Total Causal Effect

Definition: Total Effect

The **total effect** (TE) of T on Y is:

$$\mathrm{TE} = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y \mid \mathrm{do}(T{=}1)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0)].$$

The total effect captures the combined impact of all causal pathways — both $T \to Y$ and $T \to M \to Y$. It is identified by the back-door formula whenever $\mathbf{X}$ satisfies the back-door criterion for the effect of T on Y.

Running example. For the depression trial, the TE is the expected change in depression score if the entire study population were assigned to the intervention versus control. It combines the effect operating through sleep improvement with every other pathway. Mediation analysis asks: of this total, how much is due to sleep?

8.4 Controlled Direct Effect

8.4.1 Definition

The do-calculus provides the cleanest definition of the direct effect: intervene on both T *and* M simultaneously, fixing M at a specified level m. Any remaining effect of T on Y must then flow exclusively through the direct edge $T \to Y$.

Definition: Controlled Direct Effect (CDE)

The **controlled direct effect** of changing T from 0 to 1 while fixing $M = m$ by intervention is:

$$\mathrm{CDE}(m) \;=\; \mathbb{E}[Y \mid \mathrm{do}(T{=}1), \mathrm{do}(M{=}m)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0), \mathrm{do}(M{=}m)]\,. \tag{8.1}$$

This corresponds to the mutilated graph $\mathcal{G}_{\overline{TM}}$ in which all edges into both T and M are deleted. Fixing $M = m$ for everyone shuts down the indirect pathway $T \to M \to Y$.

Remark

The CDE depends on the level m at which M is fixed. In a linear additive model, $\mathrm{CDE}(m) = \tau'$ for all m, a special property of linearity. When T and M interact in their effect on Y, the CDEs at different values of m differ, and no single number summarizes the direct effect.

8.4.2 Identification of the CDE

Theorem: Identification of the CDE

Suppose $\mathbf{Z}$ satisfies the back-door criterion for the joint intervention $\mathrm{do}(T, M)$ on Y: (i) $\mathbf{Z}$ contains no descendant of T or M, and (ii) $\mathbf{Z}$ blocks every back-door path from $\{T, M\}$ to Y. Suppose further (iii) joint positivity: $P(T{=}t,\ M{=}m \mid \mathbf{Z}{=}\mathbf{z}) > 0$ for P-almost every $\mathbf{z}$. Then:

$$\mathbb{E}[Y \mid \mathrm{do}(T{=}t),\ \mathrm{do}(M{=}m)] = \sum_{\mathbf{z}} \mathbb{E}[Y \mid t,\ m,\ \mathbf{z}]\, P(\mathbf{z}). \tag{8.2}$$

Under the working assumption, $\mathbf{Z} = \mathbf{X}$ blocks every back-door path from $\{T, M\}$ to Y, so:

$$\mathbb{E}[Y \mid \mathrm{do}(T{=}t),\ \mathrm{do}(M{=}m)] = \sum_{\mathbf{x}} \mathbb{E}[Y \mid t,\ m,\ \mathbf{x}]\, P(\mathbf{x}).$$

8.4.3 The CDE Does Not Have a Complementary Indirect Effect

A common misconception is that the CDE and some “controlled indirect effect” sum to the total effect. This is not generally true. The residual quantity $\mathrm{TE} - \mathrm{CDE}(m)$ depends on m, has no clean do-calculus expression corresponding to “the indirect pathway,” and cannot be interpreted as the effect that flows through M. The correct complementary decomposition uses the natural effects of Section 8.5.

Natural Effects Are a Strictly Harder Estimand Than the CDE

The CDE is a single-world estimand. The quantity $\mathbb{E}[Y \mid \mathrm{do}(T{=}t), \mathrm{do}(M{=}m)]$ involves a single joint intervention. Identification reduces to the standard back-door criterion for the pair (T, M).
The NDE and NIE are cross-world estimands. The nested counterfactual $Y(t, M(t'))$ requires an individual to simultaneously inhabit two worlds: the world where $T = t$ (which determines Y) and the world where $T = t'$ (which determines what M would *naturally* be). When $t \neq t'$, these are logically distinct states of the world that no single experiment can realize. The do-operator alone cannot express this quantity.
The identification price is steep. The CDE requires only the back-door criterion. The NDE and NIE require four sequential ignorability assumptions, including Assumption 4 (no treatment-induced mediator–outcome confounder), which cannot be satisfied by randomizing T and cannot be verified from data.

8.5 Natural Direct and Indirect Effects

8.5.1 Motivation: Cross-World Counterfactuals

The CDE fixes the mediator by external intervention. A more scientifically natural question is: what is the effect of T on Y that bypasses M when M is held at the value it *would naturally take* under the reference treatment $T = 0$?

This requires the *nested potential outcomes* notation $Y(t, M(t'))$, which denotes the outcome that would be observed if T were set to t and M were simultaneously set to the value it would naturally take if T were t'. These are called *cross-world counterfactuals* because t and t' may differ, referring to two different hypothetical states of the world simultaneously.

8.5.2 Definitions

Definition: Natural Direct and Indirect Effects

(Pearl 2001) For binary $T \in \{0, 1\}$:

$$\mathrm{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))]\,, \tag{8.3}$$

$$\mathrm{NIE} = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))]\,. \tag{8.4}$$

The **natural direct effect** (NDE) is the expected change in the outcome when T shifts from 0 to 1, holding the mediator at the value it would naturally take under $T = 0$. The **natural indirect**

effect (NIE) is the expected change in the outcome due to the shift in the mediator from $M(0)$ to $M(1)$, holding $T = 1$ fixed.

Running example. The NDE is the expected reduction in depression score when the intervention is delivered but each participant's sleep is held at the value it *would have had* without the intervention — the part of the benefit from cognitive, behavioral, or therapeutic-alliance pathways, not from sleep. The NIE is the complementary piece: how much of the depression reduction is due to the sleep improvement that the intervention itself causes.

Decomposition. The total effect decomposes as:

$$\text{TE} \;=\; \text{NDE} \;+\; \text{NIE}. \tag{8.5}$$

Proof

$$\begin{aligned}\text{NDE} + \text{NIE} &= \mathbb{E}[Y(1, M(0)) - Y(0, M(0))] + \mathbb{E}[Y(1, M(1)) - Y(1, M(0))] \\ &= \mathbb{E}[Y(1, M(1)) - Y(0, M(0))] = \mathbb{E}[Y(1) - Y(0)] = \text{TE}. \quad \square\end{aligned}$$

Remark

In the linear additive Baron–Kenny SEM with no $T \times M$ interaction, the CDE and NDE coincide: $\text{NDE} = \tau'$ and $\text{NIE} = ab$. When there is a $T \times M$ interaction, $\text{NDE} \neq \text{CDE}$ and the linear decomposition breaks down.

8.6 Identification of Natural Effects

8.6.1 Sequential Ignorability

Identification of the NDE and NIE requires four conditions, known collectively as *sequential ignorability* (Imai et al. 2010). These conditions are substantially stronger than the ignorability assumptions used for the total effect or the CDE.

Sequential Ignorability Assumptions

1. **No unmeasured treatment–outcome confounding.** $\{Y(t', m),\, M(t)\} \perp\!\!\!\perp T \mid \mathbf{X}$ for all t, t', m. Graphically: $\mathbf{X}$ blocks all back-door paths from T to Y and from T to M.
2. **No unmeasured treatment–mediator confounding.** Implied by Assumption 1 for the $T \to M$ sub-problem.
3. **No unmeasured mediator–outcome confounding given T.** $Y(t', m) \perp\!\!\!\perp M \mid T, \mathbf{X}$ for all t', m. Graphically: $(T, \mathbf{X})$ blocks all back-door paths from M to Y.
4. **No treatment-induced mediator–outcome confounder.** There is no post-treatment variable L such that $T \to L$, $L \to M$, and $L \to Y$.

Positivity (Overlap) Conditions

- **(P1) Treatment overlap.** $P(T = t \mid \mathbf{X} = \mathbf{x}) > 0$ for all $t \in \{0, 1\}$ and for P-almost every $\mathbf{x}$.
- **(P2) Mediator overlap across treatment arms.** $P(M = m \mid T = t, \mathbf{X} = \mathbf{x}) > 0$ whenever $P(M = m \mid T = t', \mathbf{X} = \mathbf{x}) > 0$.

What randomization of T does and does not provide. Randomizing T satisfies Assumptions 1 and 2 by design. It does *not* satisfy Assumptions 3 or 4. The mediator M is a post-treatment variable that is never randomized; any unobserved V with $V \to M$ and $V \to Y$ violates Assumption 3 regardless of how T was assigned. Assumption 4 can be violated by a variable L that is itself *caused* by the treatment, so randomizing T actually creates the conditions under which Assumption 4 violations can arise.

Randomization of T Does Not Secure Assumption 4

Unlike unmeasured T–Y confounders (which randomization of T can eliminate), a treatment-induced mediator–outcome confounder L is itself caused by the treatment: the path $T \to L$ is set in motion by the intervention. The absence of such an L must be argued from design, timing, measurement, or subject-matter knowledge. This is why identification of natural effects is considered strictly harder than identification of the total effect or the CDE.

8.6.2 The Mediation Formula

Theorem: Mediation Formula

(Pearl 2001) Under sequential ignorability, positivity (P1)–(P2), and with discrete M and $\mathbf{X}$:

$$\mathbb{E}[Y(t, M(t'))] = \sum_m \sum_{\mathbf{x}} \mathbb{E}[Y \mid T{=}t,\ M{=}m,\ \mathbf{X}{=}\mathbf{x}]\, P(M{=}m \mid T{=}t',\ \mathbf{X}{=}\mathbf{x})\, P(\mathbf{X}{=}\mathbf{x}). \tag{8.6}$$

Proof Sketch (Five Steps)

Step 1 (law of total expectation): $\mathbb{E}[Y(t, M(t'))] = \sum_{m,\mathbf{x}} \mathbb{E}[Y(t,m) \mid M(t'){=}m,\ \mathbf{X}{=}\mathbf{x}]\, P(M(t'){=}m \mid \mathbf{X}{=}\mathbf{x})\, P(\mathbf{X}{=}\mathbf{x})$.
Step 2 (Assumption 4: cross-world independence): Within the NPSEM-IE framework, no-treatment-induced-confounder implies $Y(t,m) \perp\!\!\!\perp M(t') \mid \mathbf{X}$, so conditioning on $M(t')$ can be dropped: $\mathbb{E}[Y(t,m) \mid M(t'){=}m,\ \mathbf{X}{=}\mathbf{x}] = \mathbb{E}[Y(t,m) \mid \mathbf{X}{=}\mathbf{x}]$.
Step 3 (Assumptions 1 and 3): By $Y(t,m) \perp\!\!\!\perp T \mid \mathbf{X}$ and $Y(t,m) \perp\!\!\!\perp M \mid T, \mathbf{X}$: $\mathbb{E}[Y(t,m) \mid \mathbf{X}{=}\mathbf{x}] = \mathbb{E}[Y(t,m) \mid T{=}t,\ M{=}m,\ \mathbf{X}{=}\mathbf{x}]$.
Step 4 (consistency for Y): $\mathbb{E}[Y(t,m) \mid T{=}t,\ M{=}m,\ \mathbf{X}{=}\mathbf{x}] = \mathbb{E}[Y \mid T{=}t,\ M{=}m,\ \mathbf{X}{=}\mathbf{x}]$.
Step 5 (Assumption 2 and consistency for M): $P(M(t'){=}m \mid \mathbf{X}{=}\mathbf{x}) = P(M{=}m \mid T{=}t',\ \mathbf{X}{=}\mathbf{x})$.
Substituting Steps 2–5 into Step 1 yields Equation 8.6. □

Interpretation. The mediation formula "mixes" the outcome regression under $T = t$ with the mediator distribution under $T = t'$. To compute the NDE, set $t = 1$ and $t' = 0$: take the conditional mean of Y evaluated at treatment 1, but weight the mediator by its distribution under treatment 0. The presence of the second treatment index t' on the right-hand side is the visible trace of the cross-world step.

The NDE and NIE from the formula:

$$\text{NDE} = \sum_{m,\mathbf{x}} \big[\mathbb{E}[Y \mid T{=}1, M{=}m, \mathbf{X}{=}\mathbf{x}] - \mathbb{E}[Y \mid T{=}0, M{=}m, \mathbf{X}{=}\mathbf{x}]\big] P(M{=}m \mid T{=}0, \mathbf{X}{=}\mathbf{x})\, P(\mathbf{X}{=}\mathbf{x}),$$

$$\text{NIE} = \sum_{m,\mathbf{x}} \mathbb{E}[Y \mid T{=}1, M{=}m, \mathbf{X}{=}\mathbf{x}] \big[P(M{=}m \mid T{=}1, \mathbf{X}{=}\mathbf{x}) - P(M{=}m \mid T{=}0, \mathbf{X}{=}\mathbf{x})\big] P(\mathbf{X}{=}\mathbf{x}).$$

Estimation Recipe: Plug-In for the Mediation Formula

Given data $\{(Y_i, T_i, M_i, \mathbf{X}_i)\}_{i=1}^n$, a plug-in estimator requires:

1. **Fit an outcome model.** Regress Y on $(T, M, \mathbf{X})$ to obtain $\hat{\mu}(t, m, \mathbf{x})$.
2. **Fit a mediator model.** Regress M on $(T, \mathbf{X})$ to obtain $\hat{p}(m \mid t, \mathbf{x})$.
3. **Predict outcomes at the evaluation arm.** For each unit i, compute $\hat{\mu}(t, m, \mathbf{X}_i)$.
4. **Reweight the mediator at the reference arm.** For discrete M: weight $\hat{\mu}(t, m, \mathbf{X}_i)$ by $\hat{p}(m \mid t', \mathbf{X}_i)$.
5. **Average over the empirical distribution of X:**

$$\frac{1}{n} \sum_{i=1}^{n} \sum_m \hat{\mu}(t, m, \mathbf{X}_i)\, \hat{p}(m \mid t', \mathbf{X}_i). \tag{8.7}$$

6. **Quantify uncertainty.** Nonparametric bootstrap or influence-function-based estimators

(Chapters 9–10).
The plug-in estimator is consistent only when both $\hat{\mu}$ and $\hat{p}$ are correctly specified.

8.7 The Linear Mediation Model: Baron–Kenny

8.7.1 The Three-Equation System

The regression-based approach of Baron and Kenny (1986) restricts the reduced prototype graph to a linear SEM:

$$Y = \alpha_1 + \tau T + \gamma_1^\top \mathbf{X} + \varepsilon_1, \tag{8.8}$$

$$M = \alpha_2 + aT + \gamma_2^\top \mathbf{X} + \varepsilon_2, \tag{8.9}$$

$$Y = \alpha_3 + \tau' T + bM + \gamma_3^\top \mathbf{X} + \varepsilon_3. \tag{8.10}$$

The four coefficients of interest: τ (total effect of T on Y), a (first-stage: T on M), τ' (direct: T on Y controlling for M), and b (second-stage: M on Y controlling for T).

8.7.2 The Product and Difference Formulas

Proposition: Mediation Decomposition in the Linear Model

Under Equation 8.8–Equation 8.10:

$$\tau \;=\; \tau' \;+\; ab. \tag{8.11}$$

The indirect effect is $\tau_{\text{ind}} = ab$ (product method) and the direct effect is $\tau_{\text{dir}} = \tau - ab = \tau'$ (difference method).

Proof

Substituting Equation 8.9 into Equation 8.10: $Y = (\alpha_3 + b\alpha_2) + (\tau' + ab)\,T + (\gamma_3 + b\gamma_2)^\top \mathbf{X} + (b\varepsilon_2 + \varepsilon_3)$. Comparing with Equation 8.8 gives $\tau = \tau' + ab$. $\square$

Effect	**Formula**	**Path(s)**
Total	τ	$T \to Y$ and $T \to M \to Y$ combined
Direct	$\tau' = \tau - ab$	$T \to Y$ only
Indirect	ab	$T \to M \to Y$ only
Proportion mediated	ab/τ	Share of total effect via M

Running example. Suppose the fitted coefficients are $\hat{\tau} = 0.50$, $\hat{a} = 0.40$, $\hat{b} = 0.60$, $\hat{\tau}' = 0.26$. Then: indirect effect via sleep: $\hat{a}\hat{b} = 0.40 \times 0.60 = 0.24$; direct effect: $\hat{\tau} - \hat{a}\hat{b} = 0.50 - 0.24 = 0.26 = \hat{\tau}'$; proportion mediated: $0.24/0.50 = 0.48$.

The Decomposition $\tau = \tau' + ab$ Is an Algebraic Identity, Not a Causal Theorem

The equality $\tau - \tau' = ab$ is a purely algebraic consequence of substituting one linear equation into another. It holds because linearity makes the indirect effect separable and additive. It does *not* hold in general: in **nonlinear models**, the product and difference methods yield numerically different estimates; when **T and M interact**, the CDE depends on m and the indirect effect cannot be summarized by a single number ab; for **non-continuous mediators**, the product ab has no simple causal interpretation.
The correct generalization is the mediation formula (Theorem above), which reduces to ab and τ' only in the linear, no-interaction special case.

8.7.3 Inference: The Sobel Test and Bootstrap

The delta method gives an approximate variance for the product $\hat{a}\hat{b}$. The *Sobel test* (Sobel 1982) uses the simplified approximation:

$$\widehat{\mathrm{Var}}_{\mathrm{Sobel}}(\hat{a}\hat{b}) = \hat{b}^2\,\widehat{\mathrm{Var}}(\hat{a}) + \hat{a}^2\,\widehat{\mathrm{Var}}(\hat{b}), \tag{8.12}$$

yielding $z = \hat{a}\hat{b}/\sqrt{\widehat{\mathrm{Var}}_{\mathrm{Sobel}}(\hat{a}\hat{b})}$.

Statistical validity versus causal interpretation. The Sobel test is a statistical inference procedure for the product of two fitted regression coefficients. Interpreting the tested product as an *indirect causal effect* additionally requires the causal identification conditions and the structural restrictions (linearity and no $T \times M$ interaction). In practice, bootstrap confidence intervals for ab are preferred over the Sobel test because the distribution of a product of estimates is skewed in finite samples.

The "Significance of Both Paths" Criterion Is Not a Test for Mediation

A common misuse of the Baron–Kenny framework declares mediation when $T \to Y$, $T \to M$, and $M \to Y$ are each individually significant. This has three serious defects: (1) statistical significance $\neq$ mediation; (2) the Sobel test tests a regression product, not a causal quantity; (3) zero total effect does not preclude indirect effects — direct and indirect effects of opposite sign can cancel. The modern alternative is to estimate ab (or the NIE via the mediation formula) and construct bootstrap confidence intervals.

8.7.4 The Baron–Kenny Assumptions: Two Distinct Categories

Causal ignorability conditions (identification assumptions with graphical interpretations): (1) No unmeasured T–Y confounding; (2) No unmeasured T–M confounding; (3) No unmeasured M–Y confounding given T — this is *not* implied by randomization of T.

Structural modeling restrictions (parametric assumptions, not identification conditions): (4) Linearity and additivity; (5) No $T \times M$ interaction. Restrictions 4 and 5 can in principle be tested; Conditions 1–3 cannot.

Under Conditions 1–3 and Restrictions 4–5, Baron–Kenny is a valid estimator of the NDE and NIE. Outside those conditions, τ' and ab estimate regression coefficients in misspecified structural equations, not causal effects.

Estimand	**Framework needed**	**Section**
Total effect $P(y \mid \mathrm{do}(t))$	Do-calculus	Ch. 5 (back-door)
Controlled direct effect (CDE)	Do-calculus	Section 8.4
Natural direct effect (NDE)	Potential outcomes	Section 8.5
Natural indirect effect (NIE)	Potential outcomes	Section 8.5
Front-door identification	Do-calculus	Section 8.8

8.8 Front-Door Identification

8.8.1 The Front-Door DAG

Every identification strategy in §§ Section 8.4–Section 8.7 assumed that an observed covariate set $\mathbf{X}$ blocks the back-door paths from T to Y through U. What if U is wholly unobserved and no such adjustment set exists?

The front-door criterion turns this obstacle into an opportunity: under two additional restrictions on the prototype mediation graph, the mediation structure itself provides identification of the total effect without conditioning on U. The two restrictions are: (1) remove the direct $T \to Y$ edge, so that M fully mediates the effect of T on Y; and (2) require that U has no arrow into M, so that the $T \to M$ sub-effect is unconfounded.

Why the running example does not apply. The depression/sleep scenario fails Condition 1 (full mediation): a behavioral intervention on depression plausibly operates through several non-sleep channels, so the direct edge $T \to Y$ is present. The canonical front-door example is Pearl's smoking–tar–cancer graph: T = smoking, M = tar deposited in lungs, Y = lung cancer, U = unobserved genetic susceptibility.

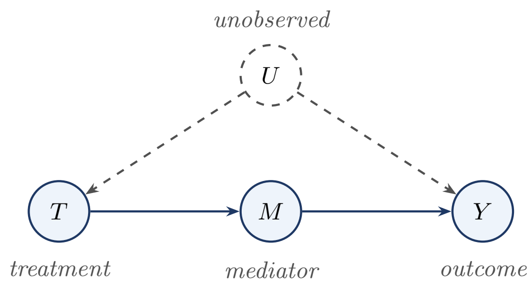


Figure 8.2: The front-door graph. Compared with the prototype mediation DAG, two edges are absent: the direct $T \to Y$ edge is removed, and U has no arrow into M. The front-door formula identifies $P(y \mid \mathrm{do}(t))$ without observing or conditioning on U.

Two Different Questions About a Mediator-Like Variable

	Ordinary mediation	**Front-door identification**
Question	How much of the total effect of T on Y flows through M?	Can M identify the total effect of T on Y despite unmeasured T–Y confounding?
Role of M	Pathway: carries part of the causal effect	Instrument-like relay: routes around confounding
Target estimand	NDE, NIE (decomposition)	$P(y \mid \mathrm{do}(t))$ (total effect itself)
Direct $T \to Y$ edge	Present	Absent (required)

8.8.2 The Three Front-Door Conditions

Front-Door Conditions

A set $\{M\}$ satisfies the **front-door criterion** for the effect of T on Y if:

1. **Full mediation.** All directed paths from T to Y pass through M (no direct $T \to Y$ edge).
2. **No unblocked back-door path from T to M.** All back-door paths from T to M are blocked (no unobserved variables affect both T and M).
3. **No unblocked back-door path from M to Y given T.** All back-door paths from M to Y are blocked by conditioning on T.

In Figure 8.2: Condition 1 holds (no $T \to Y$ edge). Condition 2 holds (U has no arrow into M). Condition 3 holds (the only back-door path from M to Y is $M \leftarrow T \leftarrow U \to Y$, blocked by conditioning on T).

8.8.3 Derivation of the Front-Door Formula

Theorem: Front-Door Formula

(Pearl 1995) Suppose M satisfies the front-door criterion for the effect of T on Y, and positivity conditions (F1)–(F2) hold. Then:

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_m P(m \mid t) \sum_{t'} P(y \mid m, t')\, P(t'). \tag{8.13}$$

Proof via Three Do-Calculus Steps

Step 1: Identify the effect of T on M. By Condition 2, there are no unblocked back-door paths from T to M, so T satisfies the back-door criterion for M with the empty adjustment set:

$$P(m \mid \mathrm{do}(T{=}t)) = P(m \mid t).$$

Step 2: Identify the effect of M on Y. By Condition 3, conditioning on T blocks all back-door paths from M to Y, so $\{T\}$ is a valid back-door adjustment set:

$$P(y \mid \mathrm{do}(M{=}m)) = \sum_{t'} P(y \mid m, t')\, P(t').$$

Step 3: Combine via full mediation. By the law of total probability under $\mathrm{do}(T{=}t)$:

$$P(y \mid \mathrm{do}(T{=}t)) = \sum_{m} P(m \mid \mathrm{do}(T{=}t))\; P(y \mid \mathrm{do}(T{=}t), \mathrm{do}(M{=}m)) .$$

By Condition 1 (no direct $T \to Y$ edge), once M is set to m by intervention, T is d-separated from Y in $\mathcal{G}_{\overline{T}\,\overline{M}}$; Rule 3 of the do-calculus removes $\mathrm{do}(T{=}t)$ from the second factor: $P(y \mid \mathrm{do}(T{=}t), \mathrm{do}(M{=}m)) = P(y \mid \mathrm{do}(M{=}m))$.
Substituting Steps 1 and 2 yields Equation 8.13. □

8.8.4 Intuition: Routing Around Confounding

The front-door formula achieves identification in two steps that each use only unconfounded variation.

1. **T to M:** There is no confounding on the $T \to M$ edge (U does not affect M), so $P(m \mid t)$ is the causal effect of T on M.
2. **M to Y:** There is back-door confounding on $M \to Y$ through $T \leftarrow U \to Y$, but it is blocked by summing over T: $\sum_{t'} P(y \mid m, t')\, P(t')$ marginalizes out T, removing the confounding path.

The key insight: U confounds T and Y but *not* the $T \to M$ edge. The front-door formula exploits this asymmetry.

Feature	**Prototype mediation DAG**	**Front-door DAG**
Direct $T \to Y$ edge	Present	Absent
U confounds T–Y	Yes	Yes
Goal	Decompose total effect into direct/indirect	Identify total effect despite T–Y confounding
Requires Assumption 3	Yes	Satisfied by Condition 3, not by randomization
Identified by front-door formula	No	Yes

8.9 Mediation vs. Instrumental Variables

Mediation analysis and instrumental variables both involve a third variable connected to the treatment–outcome relationship, but the causal role of that variable is fundamentally different.

Feature	**Instrumental Variables**	**Mediation Analysis**
Position of third variable	Pre-treatment (Z precedes T)	Post-treatment (M follows T)
Causal role	Exogenous source of variation in T	Pathway through which T affects Y
Primary goal	Identification of $T \to Y$ effect under confounding	Decomposition of $T \to Y$ effect into pathways
Key assumption	Exclusion: Z affects Y only through T	Sequential ignorability: no unmeasured M–Y confounding given T

Feature	Instrumental Variables	Mediation Analysis
Estimand	LATE (Wald) or ATE (2SLS under homogeneity)	NDE, NIE, or CDE
Unobserved T–Y confounders	Permitted	Must be addressed separately (Assumption 1)

The Key Conceptual Distinction

	IV	Mediation
What the third variable does	Generates clean variation in T (exogenous source)	Carries part of T's causal effect (pathway)
Exclusion vs. inclusion	Z excluded from Y's structural equation	M included in Y's structural equation
Question answered	Does T cause Y?	How does T cause Y?

Can the same variable be both? Not in the same analysis. As a mediator, M is on the causal path from T to Y (inclusion required). As an instrument, M would require the exclusion restriction — that it affects Y only through T — which contradicts M being a mediator. The two frameworks make mutually incompatible structural assumptions about the intermediate variable. However, the front-door identification formula is IV's closest relative within mediation analysis: it uses the mediator M to achieve identification of the total effect of T on Y even when T is confounded.

8.10 Summary

1. **Mediation studies mechanisms.** Mediation analysis decomposes the total effect of T on Y into a direct component ($T \to Y$) and an indirect component ($T \to M \to Y$). The challenge is that the mediator is a post-treatment variable that may be confounded.

2. **The total effect is the baseline.** The TE $= \mathbb{E}[Y(1) - Y(0)]$ captures all pathways combined. Mediation analysis asks for a finer decomposition.

3. **The CDE uses do-calculus.** The controlled direct effect fixes $M = m$ by joint intervention $\text{do}(T, M)$ and is identified by the back-door formula applied to the pair (T, M). No assumptions beyond the back-door criterion are needed, but the CDE depends on m and does not have a natural "indirect" complement.

4. **Natural effects require potential outcomes.** The NDE and NIE involve cross-world counterfactuals $Y(t, M(t'))$ that cannot be expressed with the do-operator alone. They decompose the total effect as TE = NDE + NIE, and are identified by the mediation formula under sequential ignorability. The critical Assumption 4 — no treatment-induced mediator–outcome confounder — is not secured by randomization of T.

5. **The linear model simplifies but restricts.** The Baron–Kenny three-equation system gives the clean formulas $\tau_{\text{ind}} = ab$ and $\tau_{\text{dir}} = \tau'$, with $\tau = \tau' + ab$. This decomposition is purely algebraic and holds only under linearity and no interaction.

6. **Front-door identification uses mediation structure.** When M fully mediates $T \to Y$, no $T \to M$ confounding exists, and T blocks $M \to Y$ confounding, the front-door formula identifies the total effect despite unobserved T–Y confounding.

7. **Mediation and IV are complementary, not equivalent.** IV uses a pre-treatment variable to generate exogenous variation in T; mediation uses a post-treatment variable to decompose the causal effect. The exclusion restriction of IV and the sequential ignorability of mediation are mutually incompatible for the same intermediate variable.

8.11 Problems

1. Identifying the CDE. Consider the DAG: $T \to M$, $T \to Y$, $M \to Y$, $X \to T$, $X \to M$, $X \to Y$, with all variables observed.

(a) Write the identification formula for $\mathbb{E}[Y \mid \mathrm{do}(T{=}1), \mathrm{do}(M{=}m)]$ using the back-door criterion for the joint intervention (T, M).
(b) Add an unobserved U with $U \to T$ and $U \to Y$. Does the back-door formula still identify the CDE? Explain which condition, if any, fails.
(c) Instead add U with $U \to M$ and $U \to Y$. Does the back-door formula still identify the CDE? Explain.

2. CDE vs. total effect. In the reduced prototype graph, let $\mathbf{Z}$ satisfy the back-door criterion for both the total effect and the joint intervention (T, M).

(a) Write expressions for the total effect and the CDE(m) using the back-door formula.
(b) Under what graphical condition does CDE(m) equal the total effect for all m? Interpret this condition.

3. Natural direct and indirect effects. Verify the NDE + NIE = TE decomposition algebraically for the linear SEM $M = \alpha T + \eta$, $Y = \beta T + \gamma M + \varepsilon$ (no interaction).

(a) Compute $Y(t, M(t'))$ in the linear model. Show that $Y(t, M(t')) = \beta t + \gamma(\alpha t') + \text{noise}$.
(b) Derive NDE $= \beta$ and NIE $= \alpha\gamma$ from the definitions Equation 8.3–Equation 8.4.
(c) Confirm $\text{NDE} + \text{NIE} = \beta + \alpha\gamma = \text{TE}$.
(d) Now suppose a $T \times M$ interaction is added: $Y = \beta T + \gamma M + \delta(T \cdot M) + \varepsilon$. Compute CDE$(m)$ and NDE. Show that $\text{NDE} \neq \text{CDE}(m)$ when $\delta \neq 0$.

4. The Baron–Kenny three-equation system. In the reduced prototype graph with the linear SEM Equation 8.8–Equation 8.10:

(a) State the three identification assumptions. For each, give the graphical condition in terms of back-door paths.
(b) Derive the equality $\tau = \tau' + ab$ algebraically.
(c) Suppose estimated coefficients are $\hat{\tau} = 0.50$, $\hat{a} = 0.40$, $\hat{b} = 0.60$, $\hat{\tau}' = 0.26$. Compute the indirect effect by both the product and difference methods. Do they agree? Compute the proportion mediated.
(d) An analyst reports $\hat{a} = 0.40$ with $\widehat{\mathrm{SE}}(\hat{a}) = 0.08$, and $\hat{b} = 0.60$ with $\widehat{\mathrm{SE}}(\hat{b}) = 0.10$. Compute the Sobel standard error for $\hat{a}\hat{b}$ using Equation 8.12 and construct an approximate 95% confidence interval.

5. The critical role of Assumption 3. Consider the graph with an unobserved V with $V \to M$ and $V \to Y$. T is randomized.

(a) Identify all back-door paths from M to Y in this graph.
(b) Can any combination of observed variables $(T, \mathbf{X})$ block all of these paths? Explain using d-separation.
(c) Suppose an analyst fits Equation 8.10 ignoring V and obtains $\hat{b} = 0.80$. In which direction is $\hat{b}$ biased if V has positive effects on both M and Y?
(d) State the additional data structure that would be needed to identify the second-stage effect non-parametrically.

6. Front-door identification. Consider the front-door graph of Figure 8.2.

(a) Verify that the three front-door conditions hold.
(b) Walk through the three-step proof of the Front-Door Formula theorem for this graph: identify which do-calculus rule justifies each step.
(c) Add a direct edge $T \to Y$ to the graph. Which front-door condition is violated? Does formula Equation 8.13 still hold?
(d) Explain why the front-door graph does not require Assumption 3 of the Baron–Kenny framework. Which back-door path — present in the graph with unobserved V — is absent here?

7. Mediation vs. instrumental variables. A researcher studies the effect of a job training program (T) on wages (Y). She proposes two intermediate variables: (A) motivation (M_A), measured after the program starts; (B) a lottery that randomly selects applicants for admission (Z), measured before the program.

(a) For variable (A): draw the mediation DAG including M_A, T, Y, and an unobserved ability variable U. State the sequential ignorability assumption needed to identify the NIE through M_A. Explain why randomization of T does not automatically satisfy this assumption.

(b) For variable (B): draw the IV DAG with Z, T, Y, and U. State the three IV assumptions. Explain why the exclusion restriction and the "mediator inclusion" of mediation analysis are mutually incompatible conditions for the same intermediate variable.

(c) The researcher argues that M_A (motivation) and Z (lottery) are both "intermediate" variables and that the analyses are interchangeable. Write a one-paragraph critique of this argument, using the distinctions from Section 8.9.

(d) Can the front-door formula be applied if motivation M_A fully mediates the effect of T on Y and U (unobserved ability) does not directly affect M_A? State the three conditions and assess whether they hold.

Chapter 9

Estimating Equations and Influence Functions

Learning Objectives

By the end of this chapter, students should be able to:

1. Explain why identification of a causal parameter does not automatically yield a statistically reliable estimator, and articulate the distinct questions that an estimation theory must address.
2. Define an estimating equation and verify the population moment condition for standard examples (sample mean, OLS, IPW).
3. State the definition of an asymptotically linear estimator and derive the influence function from a generic first-order expansion of an estimating equation.
4. Compute the influence function for simple causal estimators when nuisance functions are known, and interpret it as the infinitesimal contribution of a single observation.
5. Describe how estimation error in nuisance functions propagates into the target estimator, and explain why this is the central statistical challenge in causal inference.
6. Use the influence function to construct a consistent variance estimator and an approximate Wald confidence interval.
7. Define efficiency in the class of regular estimators and explain the role of the efficient influence function as the foundation for doubly robust estimation in Chapter 10.

9.1 Why Estimation Needs Its Own Theory

So far the focus has been on *identification*: under what assumptions can a causal parameter be written as a functional of the observed-data distribution? Identification does not by itself provide a statistically reliable estimator. Once a causal parameter is identified, several distinct questions remain: How should the estimator be constructed from the observed sample? How does estimation of nuisance functions affect the target estimator? What is the large-sample distribution of the estimator? How can we compute valid standard errors and confidence intervals?

These questions motivate a separate theory of *estimation and inference*. In causal inference this issue is especially important because identified parameters often depend on auxiliary, or *nuisance*, functions such as the outcome regression $\mu_t(x) = \mathbb{E}(Y \mid T = t, X = x)$ or the propensity score $\pi(x) = P(T = 1 \mid X = x)$. Even when the causal parameter is identified, different estimation strategies may behave quite differently in terms of robustness, efficiency, and sensitivity to model misspecification.

The goal of this chapter is to introduce a general framework for estimation based on *estimating equations* and *influence functions*. This framework will serve as the foundation for the doubly robust and semiparametric methods developed in Chapter 10; see also Imbens and Rubin (2015) and Hernán and Robins (2020) for complementary treatments.

9.2 A Running Example: The Average Treatment Effect

Throughout this chapter we use the ATE $\psi = \mathbb{E}\{Y(1) - Y(0)\}$ as a running example. Under consistency, conditional exchangeability, and positivity, the ATE is identified by the back-door formula (Chapter 5):

$$\psi = \mathbb{E}[\mu_1(X) - \mu_0(X)], \qquad \mu_t(x) = \mathbb{E}(Y \mid T = t,\, X = x), \quad t \in \{0, 1\}.$$

This identity tells us what the target parameter is, but it does not uniquely determine how to estimate it. One may consider: a *regression-based* estimator using models for $\mu_1(x)$ and $\mu_0(x)$; a *weighting* estimator based on the propensity score $\pi(x)$ (Chapter 6); or an *augmented* estimator combining both. All of these estimators may target the same causal parameter yet differ in their statistical properties.

9.3 Estimating Equations

A broad class of estimators can be defined as solutions to *estimating equations*. Let $O_1, \ldots, O_n$ be i.i.d. observations from a distribution P, and let θ denote a finite-dimensional target parameter. Write $\mathbb{P}_n f = \frac{1}{n}\sum_{i=1}^n f(O_i)$ for the empirical average.

> Definition: Estimating Equation
>
> An **estimating equation** for a parameter θ is an equation of the form $\mathbb{P}_n\{U(O;\theta)\} = 0$, where the **population moment condition** $\mathbb{E}\{U(O;\theta_0)\} = 0$ holds at the true parameter value θ_0. The function $U(O;\theta)$ is called the **estimating function**.

> Example: Sample Mean
>
> Let $\theta = \mathbb{E}(Y)$. The sample mean $\hat{\theta} = \bar{Y}$ solves $\mathbb{P}_n(Y - \theta) = 0$. The estimating function is $U(O;\theta) = Y - \theta$.

> Example: Ordinary Least Squares
>
> Let $O = (X, Y)$ and suppose β is defined by the linear moment condition $\mathbb{E}[X\{Y - X^\top \beta\}] = 0$. Then the OLS estimator solves $\mathbb{P}_n[X\{Y - X^\top \beta\}] = 0$.

> Example: A Simple IPW Estimating Equation
>
> Suppose $\pi(X)$ is known and $\psi = \mathbb{E}\left\{\frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)}\right\}$. Then ψ solves $\mathbb{E}\left[\frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)} - \psi\right] = 0$.

9.4 From Estimating Equations to Asymptotic Linearity

Under suitable regularity conditions, an estimator solving an estimating equation can typically be approximated as:

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi(O_i) + o_p(1)$$

for some mean-zero function $\varphi(O)$.

> Definition: Asymptotic Linearity and Influence Function
>
> An estimator $\hat{\theta}$ is **asymptotically linear** with **influence function** $\varphi(O)$ if:
>
> $$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi(O_i) + o_p(1), \qquad \mathbb{E}\{\varphi(O)\} = 0.$$

This representation immediately implies asymptotic normality. When $\theta \in \mathbb{R}^p$:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(\mathbf{0},\ \Sigma)\,, \qquad \Sigma = \mathbb{E}[\varphi(O)\varphi(O)^\top].$$

Proposition: Generic First-Order Expansion

Under smoothness and regularity conditions, an estimator $\hat{\theta}$ solving $\mathbb{P}_n\{U(O;\theta)\} = 0$ admits a first-order expansion:

$$\sqrt{n}(\hat{\theta} - \theta_0) = -A^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} U(O_i;\theta_0) + o_p(1),$$

where $A = \mathbb{E}\big\{\frac{\partial}{\partial\theta^\top}U(O;\theta_0)\big\}$. Hence the influence function is $\varphi(O) = -A^{-1}U(O;\theta_0)$.

Proof

A Taylor expansion of the sample moment condition around θ_0 gives:

$$0 = \mathbb{P}_n\{U(O;\hat{\theta})\} \approx \mathbb{P}_n\{U(O;\theta_0)\} + \mathbb{P}_n\left\{\frac{\partial}{\partial\theta^\top}U(O;\theta_0)\right\}(\hat{\theta} - \theta_0).$$

By the law of large numbers, $\mathbb{P}_n\{\partial_{\theta^\top}U(O;\theta_0)\} \to A$ in probability. Rearranging and multiplying by $\sqrt{n}$ yields the stated expansion. $\square$

9.5 Influence Functions: Intuition

9.5.1 Statistical Functionals

Definition: Statistical Functional

A **statistical functional** is a map $\Psi : \mathcal{P} \longrightarrow \mathbb{R}^p$ that assigns a parameter value $\psi = \Psi(P)$ to each distribution $P \in \mathcal{P}$. The **plug-in estimator** of ψ is $\hat{\psi} = \Psi(\mathbb{P}_n)$.

The mean, OLS coefficient, and ATE are all statistical functionals. A functional Ψ is *linear* if $\Psi((1-\epsilon)P + \epsilon Q) = (1-\epsilon)\Psi(P) + \epsilon\Psi(Q)$. The mean is linear; the variance, OLS coefficient, and ATE are nonlinear, requiring local linearization — precisely what the influence function provides.

9.5.2 Influence Functions

Influence functions describe the first-order sensitivity of a statistical functional to small perturbations of the underlying distribution.

Definition: Influence Function

Let $\psi = \Psi(P)$ be a scalar parameter. A function $\varphi(O)$ is called an **influence function** for Ψ if it has mean zero under P and:

$$\hat{\psi} - \psi = \frac{1}{n}\sum_{i=1}^{n}\varphi(O_i) + o_p(n^{-1/2}).$$

The influence function is the infinitesimal contribution of a single observation to the first-order behavior of the estimator.

Remark: Formal Derivation via Pathwise Derivatives

The influence function can be defined formally through the *pathwise (Gateaux) derivative*: if $P_\epsilon = (1-\epsilon)P + \epsilon\delta_O$ is the ϵ-mixture of P with a point mass at O, then:

$$\varphi(O) = \frac{d}{d\epsilon}\Psi(P_\epsilon)\Big|_{\epsilon=0}.$$

This derivative-based definition is the starting point for semiparametric efficiency theory, developed formally in Appendix C.

Example: Influence Function of the OLS Slope

For the OLS coefficient $\beta = [\mathbb{E}(XX^\top)]^{-1}\mathbb{E}(XY)$:
Step 1. Construct the perturbed distribution $P_\epsilon = (1-\epsilon)P + \epsilon\delta_O$:

$$\Psi(P_\epsilon) = \left[(1-\epsilon)\mathbb{E}(XX^\top) + \epsilon XX^\top\right]^{-1}\left[(1-\epsilon)\mathbb{E}(XY) + \epsilon XY\right].$$

Step 2. Differentiate with respect to ϵ at $\epsilon = 0$, using $\frac{d}{d\epsilon}M(\epsilon)^{-1}\big|_0 = -[\mathbb{E}(XX^\top)]^{-1}(XX^\top - \mathbb{E}(XX^\top))[\mathbb{E}(XX^\top)]^{-1}$ and the product rule:

$$\varphi(O) = [\mathbb{E}(XX^\top)]^{-1}X(Y - X^\top\beta).$$

Verification. $\mathbb{E}[\varphi(O)] = [\mathbb{E}(XX^\top)]^{-1}\mathbb{E}[X(Y - X^\top\beta)] = 0$ by definition of β.
Connection to estimating equations: The OLS estimating function is $U(O;\beta) = X(Y - X^\top\beta)$, $A = -\mathbb{E}(XX^\top)$, and Proposition 1 gives the same $\varphi(O) = [\mathbb{E}(XX^\top)]^{-1}X(Y - X^\top\beta)$.

9.6 Influence Functions for Simple Causal Estimators

9.6.1 Known Outcome Regression

Suppose $\psi = \mathbb{E}[\mu_1(X) - \mu_0(X)]$ and the functions $\mu_1(\cdot)$ and $\mu_0(\cdot)$ are known. The natural plug-in estimator is $\widehat{\psi} = \mathbb{P}_n\{\mu_1(X) - \mu_0(X)\}$. Its influence function is:

$$\varphi(O) = \mu_1(X) - \mu_0(X) - \psi.$$

9.6.2 Known Propensity Score

Suppose $\pi(X)$ is known and $\psi = \mathbb{E}\left\{\frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)}\right\}$. The empirical analogue is the Horvitz–Thompson estimator, and its influence function is:

$$\varphi(O) = \frac{TY}{\pi(X)} - \frac{(1-T)Y}{1-\pi(X)} - \psi.$$

Remark: Nuisance Functions and the Influence Function

When nuisance functions are *known*, the influence function is simply the centered estimating function. This observation becomes much more consequential in the next section, where nuisance functions are unknown and must be estimated from data.

Remark: Estimating Function, Influence Function, and Efficient Influence Function

These three objects are closely related but not the same.

1. An *estimating function* defines an estimator through a sample moment equation $\mathbb{P}_n\{\phi(O;\theta)\} = 0$.
2. An *influence function* describes the first-order asymptotic behavior of an estimator: $\sqrt{n}(\hat{\theta} -$

$\theta_0) = n^{-1/2}\sum_i \varphi(O_i) + o_p(1)$.

3. The *efficient influence function* is the influence function with the smallest possible asymptotic variance among regular estimators in the statistical model.

In simple examples with known nuisance functions, the centered estimating function often coincides with the influence function. This is not automatic when nuisance parameters are estimated.

9.7 Z-Estimation with Nuisance Parameters

In real applications, nuisance functions must be estimated from the same data used to estimate the target parameter. Once this happens, the first-order expansion of $\hat{\psi}$ is no longer obtained by simply centering the estimating function with the nuisance held fixed. Z-estimation provides the correct framework: it treats both the target parameter and the nuisance parameter as components of a single stacked estimating equation.

9.7.1 The Stacked Estimating Equation Framework

Suppose the target parameter $\psi \in \mathbb{R}$ depends on an unknown nuisance parameter $\alpha \in \mathbb{R}^q$. Both are estimated by solving a stacked system:

$$\mathbb{P}_n\{U_1(O;\psi,\alpha)\} = 0, \qquad \mathbb{P}_n\{U_2(O;\alpha)\} = 0. \tag{9.1}$$

Write $\theta = (\psi^\top, \alpha^\top)^\top$ and $U(O;\theta) = (U_1(O;\psi,\alpha),\, U_2(O;\alpha))^\top$.

Theorem: Z-Estimation Theorem (Stacked Equations)

Suppose the true parameter value is $\theta_0 = (\psi_0, \alpha_0^\top)^\top$ and the population moment condition $\mathbb{E}\{U(O;\theta_0)\} = 0$ holds, the Jacobian $\mathcal{A} = \mathbb{E}\{\partial_{\theta^\top} U(O;\theta_0)\}$ is invertible, and regularity conditions hold. Then $\hat{\theta}$ is consistent for θ_0 and:

$$\sqrt{n}(\hat{\theta} - \theta_0) = -\mathcal{A}^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} U(O_i;\theta_0) + o_p(1).$$

The influence function of $\hat{\psi}$ is:

$$\varphi(O) = -A_{11}^{-1}\, U_1(O;\psi_0,\alpha_0) + A_{11}^{-1}A_{12}A_{22}^{-1}\, U_2(O;\alpha_0), \tag{9.2}$$

where $A_{11} = \mathbb{E}\{\partial_\psi U_1\}$, $A_{12} = \mathbb{E}\{\partial_{\alpha^\top} U_1\}$, $A_{22} = \mathbb{E}\{\partial_{\alpha^\top} U_2\}$.

Proof

Partitioning $\mathcal{A}$ conformably with (ψ,α):

$$\mathcal{A} = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}, \qquad \mathcal{A}^{-1} = \begin{pmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{pmatrix}.$$

The lower-left block is zero because U_2 does not depend on ψ. Reading off the first row of $-\mathcal{A}^{-1}U(O;\theta_0)$ gives Equation 9.2. □

Remark: Interpreting the Correction Term

Equation 9.2 decomposes the influence function into two parts. The first term, $-A_{11}^{-1}U_1(O;\psi_0,\alpha_0)$, is exactly what we would obtain if α_0 were known. The second term, $A_{11}^{-1}A_{12}A_{22}^{-1}U_2(O;\alpha_0)$, is a *nuisance correction* that accounts for the additional variability introduced by estimating α. If $A_{12} = 0$, the target estimating function U_1 does not depend on α at α_0, and the correction vanishes. This is the key condition exploited by *doubly robust* estimators in Chapter 10.

9.7.2 Working Example: IPW with Estimated Propensity Score

Assume a logistic regression model $\pi(X;\alpha) = \operatorname{expit}(X^\top\alpha)$. The IPW estimating equation for ψ is $U_1(O;\psi,\alpha) = \frac{TY}{\pi(X;\alpha)} - \frac{(1-T)Y}{1-\pi(X;\alpha)} - \psi$, and the score for the logistic regression is $U_2(O;\alpha) = X\{T - \pi(X;\alpha)\}$.

Computing the Jacobian blocks: $A_{11} = -1$, $A_{22} = -\Sigma_\pi := -\mathbb{E}\{\pi(X;\alpha_0)(1-\pi(X;\alpha_0))XX^\top\}$. After computing A_{12}, the corrected influence function is:

$$\varphi(O) = U_1(O;\psi_0,\alpha_0) + A_{12}\,\Sigma_\pi^{-1}\,U_2(O;\alpha_0). \tag{9.3}$$

Example: Variance Reduction from ML Estimation of Propensity Score

We show that estimating α by maximum likelihood can only *reduce* the asymptotic variance relative to using the true α_0.
Write $\sigma_1^2 = \mathbb{E}[U_1^2]$ for the variance of the naive IPW estimating function. Two Bartlett-type identities are the key:
Identity 1 (Bartlett for ML score): $\mathbb{E}[U_2U_2^\top] = \Sigma_\pi$.
Identity 2 (cross-identity): $A_{12} = -\mathbb{E}[U_1U_2^\top]$.
Substituting into $\mathbb{E}[\varphi(O)^2] = \sigma_1^2 + 2A_{12}\Sigma_\pi^{-1}\mathbb{E}[U_1U_2] + A_{12}\Sigma_\pi^{-1}\mathbb{E}[U_2U_2^\top]\Sigma_\pi^{-1}A_{12}^\top$:

$$\mathbb{E}[\varphi(O)^2] = \sigma_1^2 - A_{12}\Sigma_\pi^{-1}A_{12}^\top \le \sigma_1^2.$$

Since Σ_π is positive definite, the variance is reduced. *ML estimation of the propensity score parameters reduces the asymptotic variance of the IPW estimator* — the *estimated propensity score paradox* (Robins 1986).

Remark: Sandwich Variance Estimator

Equation 9.2 provides the basis for the *sandwich variance estimator*: $\hat V = n^{-2}\sum_{i=1}^n \hat\varphi_i^2$, where $\hat\varphi_i$ is $\varphi(O_i)$ evaluated at $(\hat\psi,\hat\alpha)$. This estimator automatically accounts for nuisance estimation uncertainty. Standard errors from a logistic regression routine that ignores the link between $\hat\alpha$ and $\hat\psi$ will in general be *incorrect*.

The Central Challenge in Causal Estimation

The primary difficulty in practice is often not identifying the target parameter, but *correctly accounting for the effect of nuisance estimation* on the asymptotic distribution of the final estimator. Treating $\hat\alpha$ as if it were the true α_0 yields correct point estimates under a correctly specified propensity score model but *incorrect standard errors*. The Z-estimation framework resolves this variance-accounting issue whenever the nuisance model is correctly specified and the nuisance estimator converges at rate $n^{-1/2}$.

Remark: Machine Learning for Nuisance Estimation

The Z-estimation theorem as stated requires $\hat\alpha$ to converge at rate $n^{-1/2}$, which holds for finite-dimensional parametric models but fails for nonparametric or machine-learning estimators. When flexible methods are used for $\pi(x)$ or $\mu_t(x)$, the correction term in Equation 9.2 no longer has the simple form derived here. This is the starting point for the doubly robust and debiased machine-learning estimators introduced in Chapter 11.

9.8 Variance Estimation and Confidence Intervals

Once an estimator admits the asymptotic linear representation, its asymptotic variance is $\operatorname{Var}(\hat\psi) \approx n^{-1}\operatorname{Var}\{\varphi(O)\}$. A natural estimator is:

$$\hat V = \frac{1}{n^2}\sum_{i=1}^n \hat\varphi_i^2 = \frac{1}{n(n-1)}\sum_{i=1}^n (\hat\varphi_i - \bar\varphi)^2. \tag{9.4}$$

This yields the approximate Wald confidence interval $\hat{\psi} \pm z_{1-\alpha/2}\sqrt{\hat{V}}$.

Proposition: Variance from the Influence Function

If $\sqrt{n}(\hat{\psi} - \psi) = n^{-1/2}\sum_{i=1}^{n}\varphi(O_i) + o_p(1)$ with $\mathbb{E}\{\varphi(O)\} = 0$ and $\mathbb{E}\{\varphi(O)^2\} < \infty$, then:

$$\sqrt{n}(\hat{\psi} - \psi) \xrightarrow{d} N(0,\, \mathbb{E}[\varphi(O)^2])\,, \qquad \mathrm{Var}(\hat{\psi}) = \frac{1}{n}\,\mathbb{E}[\varphi(O)^2] + o(n^{-1}).$$

9.9 Toward Efficiency: Semiparametric Models and the EIF

Different regular estimators of the same parameter may have different large-sample variances. What is the smallest achievable asymptotic variance among all regular estimators? Answering that question leads to semiparametric efficiency theory. The formal machinery — regular parametric submodels, scores, tangent spaces, and the characterization of the EIF as a canonical gradient — is developed in Appendix C.

9.9.1 Semiparametric Models

Definition: Semiparametric Model

A **semiparametric model** is a statistical model $\mathcal{P}$ for the distribution P of the observed data in which (i) the **parameter of interest** $\psi = \Psi(P) \in \mathbb{R}^p$ is finite-dimensional, and (ii) the remaining aspects of P — collectively called the **nuisance parameter** — are infinite-dimensional.

The canonical example is the nonparametric model $\mathcal{P} = \{P : 0 < \pi(x) < 1\ \forall x\}$ with target parameter $\psi = \mathbb{E}[Y(1) - Y(0)]$. The nuisance consists of $\mu_t(x)$, $\pi(x)$, and the marginal distribution of X.

9.9.2 Regular Estimators

Definition: Regular Estimator (informal)

An estimator $\hat{\psi}$ is called **regular** at P_0 if, along every smooth parametric submodel $\{P_t\}$ with $P_0 = P_{t=0}$, the limiting distribution of $\sqrt{n}(\hat{\psi} - \psi_t)$ under $P_t = P_{1/\sqrt{n}}$ does not depend on the submodel.

Regularity rules out pathological "superefficient" estimators. All estimators in this course — regression, IPW, and their augmented variants — are regular.

9.9.3 The Semiparametric Efficiency Bound and the EIF

Theorem: Semiparametric Convolution Theorem

(Bickel et al. 1993) In a semiparametric model $\mathcal{P}$, the asymptotic distribution of any regular estimator $\hat{\psi}$ of ψ_0 satisfies:

$$\sqrt{n}(\hat{\psi} - \psi_0) \xrightarrow{d} N(0, V^*) * M$$

for some distribution M, where $V^* = \mathbb{E}_P[\varphi^*(O)\,\varphi^*(O)^\top]$ is the **semiparametric efficiency bound**. The function $\varphi^*(O)$ with $\mathbb{E}[\varphi^*(O)] = 0$ and $\mathbb{E}[\varphi^*(O)^2] = V^*$ is called the **efficient influence function** (EIF). An estimator achieves V^* if and only if it is asymptotically linear with influence function $\varphi^*(O)$.

Definition: Semiparametrically Efficient Estimator

A regular estimator $\hat{\psi}$ is **semiparametrically efficient** at $P_0 \in \mathcal{P}$ if $\sqrt{n}(\hat{\psi} - \psi_0) \xrightarrow{d} N(0, V^*)$. Equivalently, $\hat{\psi}$ is asymptotically linear with influence function $\varphi^*(O)$.

Remark: How the EIF is Computed

The EIF $\varphi^*(O)$ can be derived as the pathwise derivative of $\Psi(P)$ in the *least favorable* direction. In practice: (i) compute the pathwise derivative $\varphi(O; P_\epsilon)$ for an arbitrary parametric submodel; (ii) project onto the tangent space of $\mathcal{P}$ at P_0. For nonparametric or large semiparametric models, the tangent space is rich enough that this projection is often the identity, and the EIF equals the ordinary pathwise derivative. See Appendix C for the formal treatment.

9.9.4 The Efficient Influence Function for the ATE

Remark: EIF for the ATE

For the ATE $\psi = \mathbb{E}\{Y(1) - Y(0)\}$ in the nonparametric observational-data model under strong ignorability and positivity, the efficient influence function is:

$$\varphi^*(O) = \frac{T\{Y - \mu_1(X)\}}{\pi(X)} - \frac{(1-T)\{Y - \mu_0(X)\}}{1 - \pi(X)} + \mu_1(X) - \mu_0(X) - \psi. \tag{9.5}$$

This expression has a natural decomposition: the first two terms are an IPW-style residual correction, and the last two are the regression estimator centered at ψ. The EIF depends on *both* nuisance functions $\pi(X)$ and $\mu_t(X)$, and an estimator that plugs in consistent estimates of both achieves the efficiency bound $V^* = \mathbb{E}[\varphi^*(O)^2]$.

Remark: Bridge to Chapter 10

Two important properties of $\varphi^*(O)$ will be central in Chapter 10:

(i) **Double robustness.** The estimator obtained by replacing $\pi(X)$ and $\mu_t(X)$ by estimates and averaging $\varphi^*(O_i)$ is consistent for ψ if *either* $\hat{\pi}$ or $\hat{\mu}_t$ is consistent — not necessarily both.

(ii) **Semiparametric efficiency.** When both nuisance estimators are consistent and converge at suitable rates, the resulting estimator achieves the efficiency bound V^*.

The *augmented IPW* (AIPW) estimator is the principal tool for exploiting these two properties simultaneously, and will be derived and analyzed in Chapter 10.

9.10 Chapter Summary

Object	**Role**
Estimating function $U(O; \theta)$	Defines the estimator via $\mathbb{P}_n\{U(O; \hat{\theta})\} = 0$
Influence function $\varphi(O)$	First-order term in the expansion of $\hat{\psi} - \psi$; used for CLT and variance estimation
Asymptotic variance $\mathbb{E}[\varphi(O)^2]/n$	Governs precision; the target for efficiency comparisons
Efficient influence function $\varphi^*(O)$	Achieves the semiparametric efficiency bound; basis for doubly robust estimators (Chapter 10)

1. **Identification is not estimation.** Identification provides a population formula, but not automatically a satisfactory estimator.
2. **Estimating equations.** Many estimators are defined as solutions to sample moment conditions $\mathbb{P}_n\{U(O; \theta)\} = 0$, with the population condition $\mathbb{E}\{U(O; \theta_0)\} = 0$ ensuring consistency.

3. **Asymptotic linearity.** Estimating equations often yield a first-order expansion, immediately implying asymptotic normality via the CLT.
4. **Influence functions.** The function $\varphi(O)$ describes the first-order sensitivity of the estimator to a single observation. It is the key object for both asymptotic theory and variance estimation.
5. **Nuisance estimation.** In causal inference, nuisance functions must be estimated, and this estimation error propagates into the target estimator. The Z-estimation framework corrects for this propagation.
6. **Variance from the influence function.** The asymptotic variance is $n^{-1}\mathbb{E}[\varphi(O)^2]$, consistently estimated by $n^{-2}\sum_i \hat{\varphi}_i^2$.
7. **Efficiency.** Among regular estimators, the one with the smallest asymptotic variance is efficient. The EIF characterizes this lower bound and will guide estimator construction in Chapter 10.

9.11 Problems

1. Estimating equations and moment conditions.

(a) Let $O = (X, Y)$ and define $\theta = \mathrm{Cov}(X, Y)/\mathrm{Var}(X)$. Write down an estimating equation for θ and verify that the population moment condition holds at θ_0.
(b) Let $\theta = F^{-1}(0.5)$ be the population median. Propose an estimating function $U(O; \theta)$ and verify the population moment condition. (*Hint*: consider $U(O; \theta) = \mathbf{1}(Y \leq \theta) - 0.5$.)
(c) Show that the OLS estimator and the IPW estimator of the ATE are both special cases of the general M-estimator framework.

2. Asymptotic linearity and the CLT.

(a) Let $\hat{\psi} = \bar{Y}$ be the sample mean with $\mathbb{E}Y = \psi$ and $\mathrm{Var}(Y) = \sigma^2 < \infty$. Write down the influence function, state its asymptotic distribution, and give a consistent estimator of $\mathrm{Var}(\hat{\psi})$.
(b) Suppose $\hat{\psi}_1$ and $\hat{\psi}_2$ are two asymptotically linear estimators with influence functions φ_1 and φ_2. Show that $\hat{\psi}_\lambda = \lambda\hat{\psi}_1 + (1-\lambda)\hat{\psi}_2$ is also asymptotically linear for any fixed $\lambda \in [0, 1]$, and find its influence function.
(c) Use part (b) to explain why, among linear combinations of two estimators, the one minimizing asymptotic variance is generally not the simple average ($\lambda = 1/2$).

3. Influence functions for causal estimators. Consider the ATE $\psi = \mathbb{E}\{Y(1) - Y(0)\}$ identified under strong ignorability.

(a) Assume $\pi(X)$ and $\mu_t(X)$ are both known. Compute the influence functions of: (i) the regression estimator $\hat{\psi}_{\mathrm{reg}} = \mathbb{P}_n\{\mu_1(X) - \mu_0(X)\}$; (ii) the IPW estimator $\hat{\psi}_{\mathrm{IPW}} = \mathbb{P}_n\{TY/\pi(X) - (1-T)Y/(1-\pi(X))\}$.
(b) Under what conditions on the data-generating process will $\hat{\psi}_{\mathrm{reg}}$ have a smaller asymptotic variance than $\hat{\psi}_{\mathrm{IPW}}$?
(c) Using the EIF Equation 9.5, verify that it has mean zero under P by computing $\mathbb{E}[\varphi^*(O)]$.

4. Variance estimation.

(a) Let $\hat{\varphi}_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) - \hat{\psi}_{\mathrm{reg}}$ be the estimated influence values for the regression estimator. Write down the plug-in variance estimator $\hat{V}$ and simplify. Under what conditions is $\hat{V}$ consistent for $\mathrm{Var}(\hat{\psi}_{\mathrm{reg}})$?
(b) A researcher reports $\hat{\psi} = 2.4$ and $\hat{V} = 0.09$ based on $n = 400$ observations. Compute a 95% Wald confidence interval. Is the treatment effect statistically distinguishable from zero at the 5% level?
(c) Explain why simply plugging in $\hat{\mu}_t$ and $\hat{\pi}$ into the influence function formula may yield an inconsistent variance estimator when these nuisance estimators converge at a slower rate than $n^{-1/2}$.

5. Efficiency comparisons.

(a) Suppose φ_{reg} and φ_{IPW} denote the influence functions of the regression and IPW estimators. Without calculation, explain why neither estimator is always more efficient than the other.
(b) The semiparametric efficiency bound for the ATE is $\mathbb{E}[\varphi^*(O)^2]$, where φ^* is given in Equation 9.5. Show that $\mathbb{E}[\varphi^*(O)^2] \leq \mathbb{E}[\varphi_{\mathrm{IPW}}(O)^2]$ by expanding the squared EIF. (*Hint*: use the law of iterated expectations to show that the cross-terms cancel.)

(c) Give one practical reason why an efficient estimator based on the EIF may not always be preferred over a simpler, less efficient estimator.

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Appendix A

Graphical Intuition for Conditional Independence and d-Separation

This appendix provides a gentle introduction to the probabilistic and graphical ideas that underlie Chapters 2 and 3. Our goal is not to give a complete treatment of graphical models, but rather to develop enough intuition so that the formal machinery of d-separation, back-door adjustment, and intervention graphs does not appear abruptly.

A directed acyclic graph (DAG) is more than a picture. It is a compact language for expressing assumptions about how variables are related. Once those assumptions are represented graphically, the graph tells us which variables may be associated, which paths transmit dependence, and which variables should or should not be conditioned on.

The key pedagogical idea of this appendix is simple: before learning the full d-separation criterion, it is helpful to understand three elementary three-node patterns. Those local patterns explain most of what happens later in larger graphs.

A.1 Conditional Independence: The Probabilistic Language Behind Graphs

Before introducing graphs, we first recall the probabilistic notion that graphs are designed to encode.

Definition: Conditional Independence

Let X, Y, and Z be random variables. We say that X and Y are *conditionally independent given* Z, written $X \perp\!\!\!\perp Y \mid Z$, if

$$p(x, y \mid z) = p(x \mid z)\, p(y \mid z)$$

for all z with positive probability.

Equivalently, once Z is known, learning X gives no additional information about Y. In terms of densities, conditional independence admits the following equivalent characterizations:

$$X \perp\!\!\!\perp Y \mid Z \iff f(x, y, z)\, f(z) = f(x, z)\, f(y, z) \iff \exists\, a, b\colon f(x, y, z) = a(x, z)\, b(y, z).$$

The last form is especially useful: it says that the joint density factors into one piece depending on (x, z) and another depending on (y, z), with no cross-term in x and y.

Proposition: Fundamental Properties of Conditional Independence

(Dawid 1979, 1980) For random variables X, Y, Z, and W, the following hold:

- **(C1) Symmetry.** $X \perp\!\!\!\perp Y \mid Z \;\Rightarrow\; Y \perp\!\!\!\perp X \mid Z$.
- **(C2) Decomposition.** $X \perp\!\!\!\perp (Y, W) \mid Z \;\Rightarrow\; X \perp\!\!\!\perp Y \mid Z$.

- **(C3) Weak Union.** $X \perp\!\!\!\perp (Y, W) \mid Z \;\Rightarrow\; X \perp\!\!\!\perp Y \mid (Z, W)$.
- **(C4) Contraction.** $X \perp\!\!\!\perp Y \mid Z$ and $X \perp\!\!\!\perp W \mid (Y, Z) \;\Rightarrow\; X \perp\!\!\!\perp (Y, W) \mid Z$.
- **(C5) Intersection.** If $f(x, y, z, w) > 0$ for all (x, y, z, w), then $X \perp\!\!\!\perp Y \mid (Z, W)$ and $X \perp\!\!\!\perp Z \mid (Y, W) \;\Rightarrow\; X \perp\!\!\!\perp (Y, Z) \mid W$.

Properties (C1)–(C4) hold for any probability distribution and are known as the *semigraphoid axioms*. Property (C5) additionally requires the joint density to be strictly positive; together with (C1)–(C4) it forms the *graphoid axioms*. These properties are used implicitly throughout the course whenever conditional independence statements are combined or simplified.

Conditional independence is not the same as marginal independence. Two variables may be dependent marginally but independent after conditioning on a third variable. Conversely, two variables may be independent marginally but become dependent after conditioning. Both phenomena occur repeatedly in causal inference.

Example: Ice Cream, Drowning, and Season

Let X = ice cream sales, Y = drowning incidents, and Z = season. Marginally, X and Y are positively associated because both tend to be higher in summer. This does not mean that ice cream sales cause drowning. A more plausible explanation is that season is a common cause of both variables. Once season is fixed, the association largely disappears: $X \perp\!\!\!\perp Y \mid Z$.

This example illustrates a recurring theme in causal inference: an observed association may be induced by a third variable, and conditioning on that variable can remove the spurious dependence.

Remark: The Central Question

At this stage, the main question for the reader is: *Does conditioning on a variable remove association, preserve it, or create it?* Graphs provide a systematic answer to exactly this question.

A.2 Three Basic Motifs: Chain, Fork, and Collider

Every path in a DAG is built from local three-node configurations. There are three fundamental types: a *chain*, a *fork*, and a *collider*. Their behavior under conditioning is the foundation of d-separation.

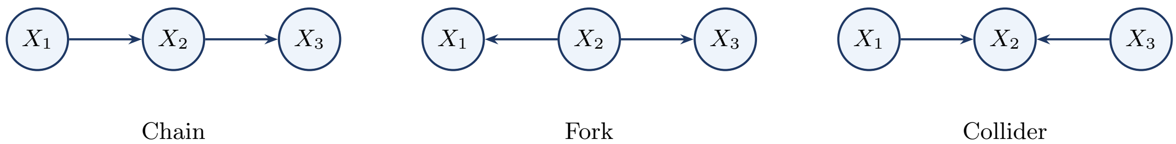

Figure A.1: The three fundamental three-node motifs. Chains and forks are blocked by conditioning on the middle node. Colliders are blocked by default but opened by conditioning on the middle node or one of its descendants.

A.2.1 Chain

Consider the pattern $X_1 \to X_2 \to X_3$, where the middle node X_2 lies on a directed pathway from X_1 to X_3.

Example: Exercise, Body Weight, and Blood Pressure

Let X_1 = exercise, X_2 = body weight, and X_3 = blood pressure. Exercise may affect blood pressure partly through its effect on body weight, so the path $X_1 \to X_2 \to X_3$ transmits association from X_1 to X_3. Marginally, exercise and blood pressure are associated; but if we condition on body weight, that particular pathway is blocked: $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

Once the middle variable is fixed, the chain no longer transmits additional information from X_1 to X_3.

A.2.2 Fork

Consider the pattern $X_1 \leftarrow X_2 \rightarrow X_3$, where the middle node X_2 is a common cause of X_1 and X_3.

Example: Ice Cream, Season, and Drowning

With X_1 = ice cream sales, X_2 = season, and X_3 = drowning incidents, season affects both variables, so X_1 and X_3 are associated even though neither causes the other. Conditioning on season blocks this path: $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

A fork is the simplest graphical form of confounding: the middle node creates association, and conditioning on it blocks the path.

A.2.3 Collider

Consider the pattern $X_1 \rightarrow X_2 \leftarrow X_3$, where the middle node X_2 is a common effect of X_1 and X_3.

Example: Talent, Legacy Status, and College Admission

Let X_1 = academic talent, X_3 = legacy status, and X_2 = admission to an elite university. Talent and legacy status may be unrelated in the general applicant pool. But among admitted students they can become statistically associated: low legacy status makes unusually high talent more likely among admitted applicants, and vice versa. Symbolically, $X_1 \perp\!\!\!\perp X_3$ but $X_1 \not\!\perp\!\!\!\perp X_3 \mid X_2$.

Unlike chains and forks, a collider *blocks* the path by default. Conditioning on the collider opens the path and may induce association that was not present marginally.

A.2.4 Summary of the Three Motifs

A chain $(X_1 \rightarrow X_2 \rightarrow X_3)$ and a fork $(X_1 \leftarrow X_2 \rightarrow X_3)$ are each open by default and blocked by conditioning on the middle node X_2. A collider $(X_1 \rightarrow X_2 \leftarrow X_3)$ is blocked by default and opened by conditioning on the middle node or any of its descendants.

Conditioning Is Not Always Beneficial

Many adjustment mistakes arise from forgetting this asymmetry. Conditioning on a collider can create bias rather than remove it. The informal rule "control for more variables" is unsafe in causal inference; d-separation teaches a more precise lesson: condition on the *right* variables, not simply on many variables.

Chapter 2 develops these three motifs into the full d-separation criterion; see in particular Section 2.3 for the blocking rules and Section 2.4 for an extended treatment of collider bias.

A.3 d-Separation: When Does Conditioning Block a Path?

The three-node motifs explain what happens locally on a path. The next step is to extend this logic to a general DAG, where two variables may be connected by many paths.

Definition: Path

A *path* between two nodes is a sequence of distinct nodes such that each consecutive pair is connected by an edge, regardless of edge direction.

Definition: Blocked Path and d-Separation

A path is *blocked* by a conditioning set S if at least one of the following holds:

1. The path contains a chain or fork node that belongs to S, or
2. The path contains a collider such that neither the collider nor any of its descendants belongs

to S.
A path is *open* given S if it is not blocked. Two nodes X and Y are *d-separated* by a set S if every path between X and Y is blocked by S.

A.3.1 A Confounding Example

Consider the DAG with edges $X \to T$, $T \to Y$, and $X \to Y$. Here T is the treatment, Y is the outcome, and X is a pre-treatment covariate that affects both. There are two paths from T to Y: the directed causal path $T \to Y$, and the back-door path $T \leftarrow X \to Y$.

Without conditioning, the back-door path is open, so the observed association between T and Y mixes the causal effect with confounding. Conditioning on X blocks the fork $T \leftarrow X \to Y$, thereby isolating the causal path.

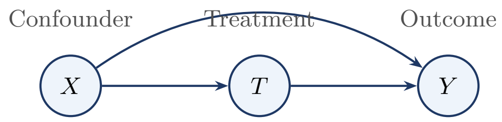

Figure A.2: A simple confounding graph. The path $T \leftarrow X \to Y$ is a back-door path from treatment to outcome. Conditioning on X blocks this noncausal path.

This confounding graph anticipates the back-door criterion of Chapter 3: the graphical condition that makes adjustment valid is precisely that all back-door paths are blocked.

A.3.2 A Collider Warning

Now consider the DAG with edges $T \to C$, $U \to C$, and $U \to Y$. Here C is a collider on the path $T \to C \leftarrow U \to Y$. Without conditioning on C, the path is blocked at the collider. Conditioning on C opens the path, creating a spurious association between T and Y through U.

Even more subtly, conditioning on a *descendant* of C can also open the path. If additionally $C \to D$, then conditioning on D may also induce association between T and Y through the collider at C.

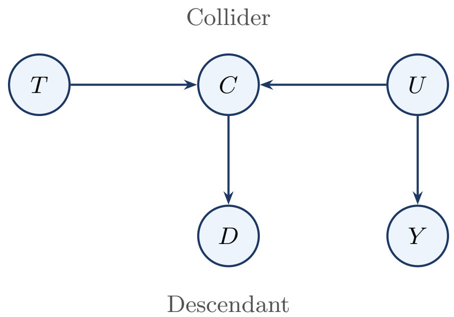

Figure A.3: Collider bias with a descendant. The path $T \to C \leftarrow U \to Y$ is blocked by default at the collider C. Conditioning on C, or on its descendant D, can open the path and induce a spurious association between T and Y.

A.3.3 A Practical Checklist

To decide whether X and Y are d-separated by S, proceed as follows. First, list all paths between X and Y. On each path, classify each interior node as part of a chain, fork, or collider. Then check whether each path is blocked by S. Finally, conclude that X and Y are d-separated if and only if every path is blocked.

For beginners, this path-by-path procedure is often more useful than starting with the most compact formal definition.

A.4 DAG Factorization and the Markov Property

Up to this point, we have used DAGs qualitatively, to decide which paths are open or blocked. We now connect the graph to probability algebra: the same parent structure that governs d-separation also determines how the joint distribution factorizes.

Definition: DAG Factorization

Let $\mathcal{G}$ be a DAG with nodes $V_1, \dots, V_p$. A joint distribution $p(v_1, \dots, v_p)$ is said to *factorize according to* $\mathcal{G}$ if

$$p(v_1, \dots, v_p) = \prod_{j=1}^{p} p\big(v_j \mid \mathrm{Pa}(V_j)\big),$$

where $\mathrm{Pa}(V_j)$ denotes the set of parents of V_j in $\mathcal{G}$.

Example: A Simple Confounding Graph

For the DAG with edges $X \to T$, $T \to Y$, and $X \to Y$, the joint density factorizes as

$$p(x, t, y) = p(x)\, p(t \mid x)\, p(y \mid t, x).$$

Definition: Local Markov Property

A distribution P satisfies the *local Markov property* with respect to $\mathcal{G}$ if, for every node $V_i \in V$,

$$V_i \perp\!\!\!\perp (\mathrm{Nd}(V_i) \setminus \mathrm{Pa}(V_i)) \mid \mathrm{Pa}(V_i).$$

That is, once we condition on the direct causes of a node, that node is independent of all variables that are neither its descendants nor its parents.

Remark: Global Markov Property

The global Markov property is the statement that every d-separation in $\mathcal{G}$ implies a conditional independence in P. Thus, if two sets of variables are d-separated by a set S in the graph, then they are conditionally independent given S in the distribution. This is studied in detail in Chapter 2.

Remark: Equivalence for DAGs

For DAGs, the local and global Markov properties are equivalent under the usual graphical-model assumptions. The local property implies the global one via the Markov factorization, which is what is actually used in identification arguments.

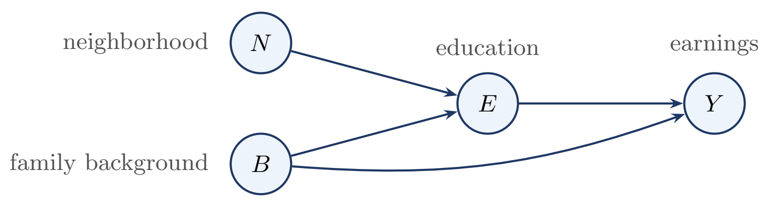

Figure A.4: Education–earnings DAG used to illustrate factorization and the local and global Markov properties.

Example: Education–Earnings Graph

Consider the DAG in Figure A.4, with edges $N \to E$, $B \to E$, $E \to Y$, and $B \to Y$. The corresponding factorization is

$$p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

The parent set of Y is $\mathrm{Pa}(Y) = \{E, B\}$, so the local Markov property gives $Y \perp\!\!\!\perp N \mid \{E, B\}$. The same conclusion follows from the global viewpoint: the only path from N to Y is $N \to E \to Y$, which is blocked by conditioning on E, and conditioning additionally on B opens no new path.

Remark: Edges as Scientific Claims

In a causal DAG, an arrow is typically drawn only when a direct dependence-generating relation is believed to be present. This explains why edges are not included gratuitously: every arrow represents a substantive scientific claim. Minimality and faithfulness are discussed in Chapter 2.

A.5 Optional: Moralization as an Alternative Criterion

Note to Reader

This section may be skipped on a first reading.

There is an alternative graph-theoretic way to check d-separation based on constructing an undirected graph called the *moral graph*. The method is elegant and useful in more advanced graphical-model theory, but it is not necessary for understanding the main causal ideas of Chapters 2 and 3. For most students, the path-by-path method using chains, forks, and colliders is more intuitive on first exposure.

The procedure is as follows. First, restrict attention to the ancestors of the variables of interest. Second, connect any two parents of a common child by an undirected edge. Third, drop all arrow directions. Finally, check ordinary graph separation in the resulting undirected graph. This procedure yields a criterion equivalent to d-separation.

Example: Moral Graph of a Collider

Suppose a DAG contains the pattern $A \to C \leftarrow B$. In the moral graph, the two parents A and B are connected by an undirected edge, producing the triangle $A - C - B$ together with $A - B$. This added edge reflects the special role of colliders in d-separation.

A.6 Summary

This appendix introduced the graphical ideas that support Chapters 2 and 3. Conditional independence is the probabilistic language that graphs are designed to encode. The three local motifs — chain, fork, and collider — determine how conditioning affects association along a path. D-separation extends these local rules to arbitrary graphs: two variables are d-separated by S if every path between them is blocked by S. The most important application in causal inference is to distinguish confounding paths from causal paths and to identify valid adjustment sets. Finally, the Markov property gives a probabilistic interpretation to the graph by linking graphical structure to a factorization of the joint distribution.

Appendix B

Single World Intervention Graphs

This appendix develops single world intervention graphs (SWIGs), a graphical device introduced by Richardson and Robins (2014) and given an accessible practical treatment by Bezuidenhout et al. (2025). A SWIG provides a formal bridge between the potential outcomes framework and the do-calculus by encoding both in a single diagram. A reader familiar with Chapters 2–4 will find that SWIGs require no new conceptual ingredients: they are ordinary DAGs in which the treatment node is split into two pieces, making potential outcomes and identifiability assumptions graphically explicit rather than leaving them implicit.

Chapter 4 used the back-door criterion and the adjustment formula to identify causal effects. SWIGs make that identification argument graphically explicit: potential outcomes appear as labeled nodes inside the graph, so ignorability becomes a routine d-separation statement rather than a free-standing probabilistic assumption. Section B.5 completes the picture by establishing the biconditional equivalence between ignorability and the back-door criterion that was only stated in one direction in Chapter 4.

B.1 The SWIG Construction

In the original DAG $\mathcal{G}$, the node T represents the treatment as it naturally occurs. Under do($T{=}t$), however, the treatment is externally fixed. The SWIG separates these two roles by *splitting* T into two pieces displayed side by side in the graph:

- a **random half** (left side, capital letter T): represents the treatment value that would naturally occur, and retains all incoming edges from T's causal parents;
- a **fixed half** (right side, lowercase letter t): represents the externally imposed intervention value, carries all outgoing edges that formerly left T, and has *no* edge to or from the random half.

> Definition: Single World Intervention Graph
>
> (Richardson and Robins 2014) The *SWIG* $\mathcal{G}(t)$ is constructed from $\mathcal{G}$ by:
> 1. Replacing T with two nodes: random half T (all incoming edges retained) and fixed half t (all outgoing edges retained).
> 2. Severing the direct connection between the two halves.
> 3. Redrawing causal arrows at the split node: *all arrows into the split node land on the random half*; *all arrows departing the split node leave from the fixed half.*
> 4. Relabeling every descendant V of T in $\mathcal{G}$ as $V(t)$, indicating it is a potential outcome under do($T{=}t$).

Figure B.1 illustrates the construction for the simple confounded-treatment graph.

B.2 The Fixed Half as a Source Node

After the SWIG $\mathcal{G}(t)$ is constructed, d-separation among its nodes is evaluated using the standard rules of Chapter 2. The key structural fact that governs all path analysis in a SWIG is:

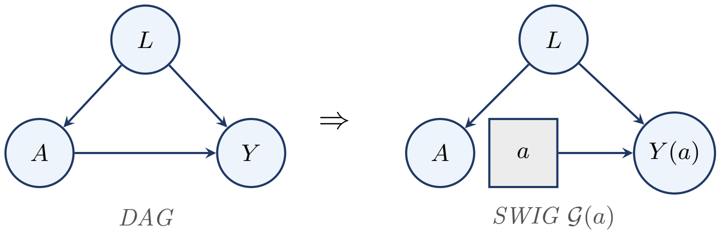

Figure B.1: Converting the confounded-treatment DAG (L is a confounder) into the SWIG for do(A=a). The treatment node splits: the random half A (circle) retains the incoming arrow from L; the fixed half a (square) carries the outgoing arrow to $Y(a)$. D-separation in the SWIG gives $A \perp\!\!\!\perp Y(a) \mid L$ (conditional ignorability).

The Fixed Half Is a Source Node

In $\mathcal{G}(t)$, the fixed half t has **no incoming edges**. Step 2 of the construction severs the connection from the random half T, and no other node in $\mathcal{G}$ has a direct arrow into t. Three consequences follow immediately.

1. **No path arrives at t from another node.** The edge $T \to t$ does not exist in $\mathcal{G}(t)$. In particular, the path $T \to t \to Y(t)$ *cannot be formed*: it is not "blocked" by a d-separation rule — it simply does not exist as a path in the graph.
2. **t cannot lie strictly between two other nodes on a path.** Because t has no parents, it is never a transit node; it can only be an *origin* of paths leading to descendants $V(t)$.
3. **The random half T and the fixed half t are d-separated by the empty set.** With no edge between them and no common ancestor, $T \perp\!\!\!\perp t$ holds unconditionally in $\mathcal{G}(t)$.

This is not an additional d-separation rule: it is a direct consequence of the SWIG construction. The fixed half t represents an externally stipulated intervention — it is not *caused* by anything within the system, so it has no parents.

Path analysis in $\mathcal{G}(t)$. With t as a source node, every path from the random half T to a potential outcome $V(t)$ must travel *away* from T through T's parents in $\mathcal{G}$ — that is, every such path is a back-door path. This is the structural fact that makes the back-door criterion readable directly from the SWIG.

Recovering ignorability from the SWIG. In Figure B.1, the only paths from the random half A to $Y(a)$ are back-door paths through A's parents. The path $A \leftarrow L \to Y(a)$ is open, but blocked by conditioning on L. No other path connects A to $Y(a)$. Conditioning on L achieves d-separation:

$$A \perp\!\!\!\perp Y(a) \,\big|\, L \quad \text{in } \mathcal{G}(a),$$

which is exactly the conditional ignorability of the Strong Ignorability definition in Chapter 4, now *derived* as a graph-structural consequence rather than asserted as an assumption.

B.3 When Ignorability Fails: Hidden Confounding

The SWIG is equally useful for diagnosing *failures* of ignorability. Figure B.2 shows the SWIG when a hidden common cause U affects both A and Y.

In the SWIG $\mathcal{G}(a)$ with hidden U, the paths from A to $Y(a)$ are: (1) $A \leftarrow L \to Y(a)$: blocked by conditioning on L; (2) $A \leftarrow U \to Y(a)$: open regardless of L, since U is unobserved. Even after conditioning on L, the path through U remains open, so $A \not\perp\!\!\!\perp Y(a) \mid L$ in $\mathcal{G}(a)$, and ignorability fails. The failure is *immediately visible* from the SWIG: an unblocked dashed arrow entering the random half A signals that selection into treatment carries information about the potential outcome that no set of observed covariates can remove.

Remark

This failure motivates the instrumental variable designs of Chapter 7. When a valid instrument Z is available — one that affects A but has no direct effect on Y and is independent of U — the IV

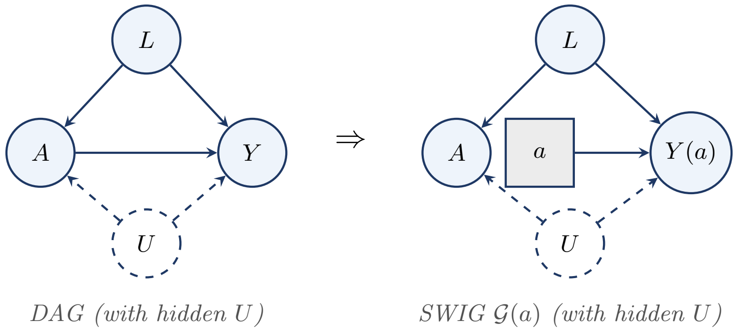

Figure B.2: Adding a hidden confounder U (dashed circle) to the confounded-treatment DAG and its SWIG. In $\mathcal{G}(a)$, two back-door paths connect A to $Y(a)$: the observed path $A \leftarrow L \rightarrow Y(a)$, which can be blocked by conditioning on L, and the hidden path $A \leftarrow U \rightarrow Y(a)$, which cannot. Ignorability fails.

strategy sidesteps the unblocked U-path rather than attempting to block it by covariate adjustment.

B.4 What SWIGs Achieve

1. Potential outcomes as random variables in a DAG. In $\mathcal{G}(t)$, the variable $Y(t)$ is an ordinary node with well-defined parents: the fixed half t and any other causes of Y retained from $\mathcal{G}$. Its marginal distribution equals the interventional distribution $f(y \mid \mathrm{do}(T{=}t))$. $Y(t)$ is the counterfactual random variable, and $f(y \mid \mathrm{do}(T{=}t))$ is its induced interventional distribution; the SWIG makes both explicit in the same graph.

2. The mutilated graph as a special case. Removing the random half T from $\mathcal{G}(t)$ yields exactly the mutilated graph $\mathcal{G}_{\overline{T}}$ of Chapter 1. The SWIG is a more informative version that retains the random half alongside the fixed half, making it possible to ask — and answer by d-separation — questions about the relationship between the natural treatment T and the potential outcomes $V(t)$.

3. Unconfoundedness as a d-separation statement. The ignorability condition $(Y(t) \perp\!\!\!\perp T \mid X)$ is a standard d-separation statement *in the SWIG* $\mathcal{G}(t)$, verifiable by path-tracing. This gives the analyst a concrete graphical check: draw the SWIG, identify all paths from the random half T to $Y(t)$, and verify that X blocks all of them.

4. A single graph for both frameworks. The SWIG simultaneously hosts the natural treatment T (random half), the intervention value t (fixed half), the potential outcomes $V(t)$, and all d-separation relationships among them.

Cross-World Independence and Its Limits

Consider $Y(1) \perp\!\!\!\perp Y(0)$: $Y(1)$ lives in SWIG $\mathcal{G}(1)$ and $Y(0)$ lives in SWIG $\mathcal{G}(0)$. No unit is ever observed in both worlds simultaneously, so no data can directly test this independence. More generally, statements such as $Y(t) \perp\!\!\!\perp T(t') \mid X$ for $t \neq t'$ involve variables from *different SWIGs*. SWIGs make such assumptions explicit: the analyst must specify a joint distribution across multiple SWIGs — a commitment that potential-outcome notation alone does not force.

Remark

More than one node may be split in a single SWIG. In longitudinal settings with time-varying treatments (A_0, A_1), splitting both nodes yields the sequential-randomization SWIG, from which the two *sequential exchangeability* conditions needed to identify the g-formula follow directly by d-separation.

B.5 Biconditional Equivalence under Faithfulness

Proposition 4.1 in Chapter 4 established the practically important direction: if X satisfies the back-door criterion, then unconfoundedness holds. That proof required only the Markov property. The converse — that every ignorability statement corresponds to a back-door condition — additionally requires faithfulness and follows naturally from the SWIG construction.

Definition: Faithfulness

A distribution P is **faithful** to a DAG $\mathcal{G}$ if every conditional independence that holds in P is entailed by d-separation in $\mathcal{G}$: that is, $X \perp\!\!\!\perp Y \mid \mathbf{Z}$ in P only if X and Y are d-separated by $\mathbf{Z}$ in $\mathcal{G}$. Equivalently, faithfulness rules out *accidental* cancellations of causal paths. Together, the Markov property and faithfulness make d-separation a *complete* characterization of the conditional independence structure of P. Faithfulness fails only on a measure-zero set of parameter values, so it is considered a mild regularity condition in practice.

Theorem: Ignorability and the Back-Door Criterion Are Equivalent

Under the Markov and faithfulness assumptions for $\mathcal{G}$, unconfoundedness $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ holds if and only if X satisfies the back-door criterion for the effect of T on Y in $\mathcal{G}$.

Proof

($\Rightarrow$) This is Proposition 4.1 in Chapter 4, which uses only the Markov property.
($\Leftarrow$) Suppose $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ holds. By faithfulness, every conditional independence in P corresponds to a d-separation in $\mathcal{G}$. Consider the SWIG $\mathcal{G}(t)$. Because the fixed half t has no incoming edges (Section B.2), every path from the random half T to a potential outcome $Y(t)$ must pass through T's parents in $\mathcal{G}$ — that is, every such path is a back-door path. The d-separation $(Y(t) \perp\!\!\!\perp T \mid X)_{\mathcal{G}(t)}$ therefore requires X to block all back-door paths from T to Y. If any member of X were a descendant of T, conditioning on it could open a previously blocked collider path, contradicting the assumed independence; hence no node in X is a descendant of T. Both conditions of the back-door criterion are therefore satisfied. $\square$

The SWIG thus provides a *unified* perspective: ignorability is neither an assumption of the potential outcomes framework alone nor of the do-calculus in isolation, but a graph-structural property that both frameworks share. The back-door criterion and conditional exchangeability are two names for the same identifying condition, seen from two different representational languages.

B.6 Problems

1. SWIG construction. Consider the DAG: $X \to T$, $X \to Y$, $T \to Y$, with X fully observed.

(a) Construct the SWIG $\mathcal{G}(t)$ by splitting T into its random and fixed halves. Draw the result, labeling the random half, fixed half, and the potential outcome $Y(t)$.
(b) In $\mathcal{G}(t)$, identify all paths between the random half T and $Y(t)$. Apply d-separation to determine which are blocked and which are open before conditioning.
(c) Use d-separation in $\mathcal{G}(t)$ to verify that $(Y(t) \perp\!\!\!\perp T \mid X)_{\mathcal{G}(t)}$ holds. Conclude that X satisfies the back-door criterion, confirming Theorem B.1 above.
(d) Now add a hidden common cause $U \to T$, $U \to Y$. Draw the revised SWIG. Does the ignorability argument still hold? State the correct conclusion and identify what additional structure (if any) would be needed for identification.

2. Sequential exchangeability in a longitudinal SWIG. Suppose the treatment is time-varying: T_0 is assigned at baseline and T_1 at a second time point. The DAG has edges $X \to T_0 \to T_1 \to Y$, $X \to Y$, and $T_0 \to Y$.

(a) Construct the SWIG $\mathcal{G}(t_0, t_1)$ by splitting both T_0 and T_1. Label all random halves, fixed halves, and potential outcomes.

(b) List all back-door paths from the random half T_0 to $Y(t_0, t_1)$ and from T_1 to $Y(t_0, t_1)$. For each, determine whether conditioning on X and the history up to that point is sufficient to achieve d-separation.

(c) State the two sequential exchangeability conditions that the SWIG makes graphically transparent, and explain what they say about treatment assignment at each time point.

Appendix C

Pathwise Differentiability and Efficient Influence Functions

Section 9.9 in Chapter 9 introduced the efficient influence function of the average treatment effect functional and stated, without proof, that this object is the *canonical gradient* of Ψ relative to the nonparametric tangent space. This appendix supplies the geometric machinery behind that statement. The treatment follows Bickel et al. (1993) and Tsiatis (2006), with notation aligned to Chapter 9.

The reader should think of this appendix as the formal counterpart of Sections 9.3–9.7: those sections showed how an influence function determines the asymptotic distribution of an estimator; here we develop the parallel notion of an influence function as a *derivative of a functional*, and characterize the unique influence function that achieves the semiparametric efficiency bound.

The applied consequences — the AIPW estimator, double robustness, Neyman orthogonality, cross-fitting, and rate conditions — are developed in Chapters 10 and 11. This appendix supplies only the foundational geometry on which those chapters rest.

C.1 Regular Parametric Submodels and Scores

Let $\mathcal{P}$ be a statistical model for the distribution of the observed data O on a sample space $\mathcal{O}$, with true distribution $P \in \mathcal{P}$. All densities below are taken with respect to a common dominating measure μ. The parameter of interest is a smooth real-valued functional $\Psi : \mathcal{P} \to \mathbb{R}$, $\psi = \Psi(P)$.

To define what it means for Ψ to be "smooth," we need a notion of a path through $\mathcal{P}$ that begins at P.

> Definition: Regular Parametric Submodel
>
> A **regular parametric submodel** of $\mathcal{P}$ passing through P is a one-parameter family $\{P_\varepsilon : \varepsilon \in (-\delta, \delta)\} \subset \mathcal{P}$ with $P_0 = P$, with densities p_ε such that $\varepsilon \mapsto \log p_\varepsilon(O)$ is differentiable at $\varepsilon = 0$ in $L_2(P)$, with **score**:
>
> $$S(O) = \frac{\partial}{\partial \varepsilon} \log p_\varepsilon(O)\Big|_{\varepsilon=0} \;\in\; L_2(P).$$

The L_2-differentiability requirement is a mild smoothness condition that ensures $\int p_\varepsilon \, d\mu$ can be differentiated under the integral. For a rigorous formulation, see Vaart (1998, sec. 7.2).

Because $\int p_\varepsilon \, d\mu = 1$ for all ε, differentiating under the integral gives:

$$\mathbb{E}_P[S(O)] = 0, \qquad \mathbb{E}_P[S(O)^2] < \infty, \tag{C.1}$$

so every score lies in $L_2^0(P) := \{f \in L_2(P) : \mathbb{E}_P[f] = 0\}$.

Remark: Why Submodels Rather Than Point-Mass Contamination

A heuristic device sometimes used for "probing" a functional is the contamination path $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_o$, which yields the classical Hampel influence function $\Psi(\delta_o) - \Psi(P)$. Contamination is intuitive but does not yield a regular submodel: when P is continuous, the formal "score" of such a path is $\delta_o/p - 1$, which is not in $L_2(P)$. All formal results below use the definition above, while the contamination heuristic remains useful for guessing the form of an influence function in concrete examples.

C.2 Pathwise Differentiability

The functional Ψ is differentiable along a submodel if the map $\varepsilon \mapsto \Psi(P_\varepsilon)$ is differentiable at $\varepsilon = 0$. Pathwise differentiability asserts that this derivative can be represented as an inner product between a fixed function and the score, uniformly over submodels.

Definition: Pathwise Differentiability and Influence Function

The functional $\Psi : \mathcal{P} \to \mathbb{R}$ is **pathwise differentiable** at P if there exists a function $\varphi \in L_2^0(P)$ such that, for every regular parametric submodel $\{P_\varepsilon\}$ with score S:

$$\frac{\partial}{\partial\varepsilon}\Psi(P_\varepsilon)\bigg|_{\varepsilon=0} = \mathbb{E}_P[\varphi(O)\,S(O)]. \tag{C.2}$$

Any such φ is called an **influence function** (or **gradient**) of Ψ at P. The set of influence functions is denoted $\mathrm{IF}(\Psi, P)$.

Equation C.2 is the defining identity of semiparametric theory. It says the perturbation of Ψ along a submodel is fully encoded by the $L_2(P)$ inner product of a fixed function φ with the score S. Geometrically, φ is a representer for the linear functional $S \mapsto \partial_\varepsilon \Psi(P_\varepsilon)|_0$ restricted to the space of scores.

Example: Mean Functional

Let $O = Y$ with $\mathbb{E}_P|Y| < \infty$ and $\Psi(P) = \mathbb{E}_P[Y]$. For any regular submodel $\{P_\varepsilon\}$ with score S:

$$\frac{\partial}{\partial\varepsilon}\int y\,p_\varepsilon(y)\,d\mu(y)\bigg|_0 = \mathbb{E}_P[Y \cdot S(O)].$$

Since $\mathbb{E}_P[S] = 0$, subtracting $\mathbb{E}_P[Y]\cdot\mathbb{E}_P[S] = 0$ gives $\partial_\varepsilon\Psi(P_\varepsilon)|_0 = \mathbb{E}_P[(Y - \mathbb{E}_P Y)\,S(O)]$, so $\varphi(O) = Y - \Psi(P)$ is an influence function of the mean functional. Under the nonparametric model, this is in fact the *unique* influence function, and hence the efficient influence function.

Remark: Connection to Estimation

If an estimator $\hat\psi$ is asymptotically linear at P with influence function φ in the sense of Chapter 9, and if $\hat\psi$ is regular along every submodel, then φ satisfies Equation C.2 and is therefore a gradient of Ψ. Pathwise differentiability is thus a necessary condition for the existence of a regular asymptotically linear estimator of Ψ (Tsiatis 2006, Theorem 3.1).

C.3 Non-Uniqueness of Influence Functions

Definition Section C.2 does not produce a unique influence function. If φ satisfies Equation C.2 and $h \in L_2^0(P)$ is orthogonal to every score in $L_2(P)$, then $\varphi + h$ also satisfies Equation C.2, since $\mathbb{E}_P[(\varphi+h)S] = \mathbb{E}_P[\varphi S] + \mathbb{E}_P[hS] = \mathbb{E}_P[\varphi S]$. Whether φ is unique depends on how rich the collection of scores is — a point made precise through the notion of the tangent space.

The terminological distinction can now be made precise:

- An **estimating function** is any function $U(O;\theta)$ whose root defines an estimator (Chapter 9, Section 9.3).
- An **asymptotic influence function** of an estimator is the function φ in its asymptotic expansion (Chapter 9, Section 9.4).
- A **(pathwise) influence function** of a functional is a $\varphi \in L_2^0(P)$ satisfying Equation C.2.

For a regular asymptotically linear estimator of a pathwise-differentiable functional, all three notions coincide. The third depends crucially on $\mathcal{P}$ through the collection of admissible submodels, which is what makes $\mathrm{IF}(\Psi, P)$ generally non-unique in semiparametric models.

C.4 The Tangent Space

Definition: Tangent Space

The **tangent space** of $\mathcal{P}$ at P, denoted $T_P(\mathcal{P})$ or simply T, is the closed linear span in $L_2^0(P)$ of all scores of regular parametric submodels of $\mathcal{P}$ passing through P.

The tangent space is a closed subspace of the Hilbert space $L_2^0(P)$. Two extreme cases are illustrative.

Nonparametric model. If $\mathcal{P}$ contains all distributions satisfying mild regularity, then for any bounded $h \in L_2^0(P)$, the submodel $p_\varepsilon(o) \propto (1 + \varepsilon\, h(o))p(o)$ is regular with score $S = h$. Since bounded mean-zero functions are dense in $L_2^0(P)$, taking closure gives $T = L_2^0(P)$.

Fully parametric model. If $\mathcal{P} = \{P_\theta : \theta \in \Theta \subset \mathbb{R}^k\}$ with score S_θ at θ_0, then $T = \mathrm{span}\{S_{\theta_0,1}, \dots, S_{\theta_0,k}\}$ is k-dimensional.

Most causal models lie between these extremes: the parameter of interest ψ is finite-dimensional, but the rest of P is unrestricted, making T infinite-dimensional but not all of $L_2^0(P)$.

Definition: Nuisance Tangent Space

A submodel $\{P_\varepsilon\}$ is a **nuisance submodel** at P if $\Psi(P_\varepsilon) = \Psi(P)$ for all ε, i.e., the parameter of interest is held fixed along the path. The **nuisance tangent space**, denoted T_η, is the closed linear span of scores of all nuisance submodels.

The nuisance tangent space is a closed subspace of T, and hence of $L_2^0(P)$. The orthogonal complement is:

$$T_\eta^\perp := \{f \in L_2^0(P) : \mathbb{E}_P[fg] = 0 \text{ for all } g \in T_\eta\}. \tag{C.3}$$

By the projection theorem, the Hilbert-space decomposition $L_2^0(P) = T_\eta \oplus T_\eta^\perp$ holds automatically.

Proposition: Every Influence Function Lies in $T_\eta^\perp$

Let Ψ be pathwise differentiable at P. Then every influence function $\varphi \in \mathrm{IF}(\Psi, P)$ satisfies $\varphi \in T_\eta^\perp$.

Proof

For any nuisance submodel $\{P_\varepsilon\}$ with score S_η, the definition of nuisance submodel gives $\Psi(P_\varepsilon) = \Psi(P)$ for all ε, hence $\partial_\varepsilon \Psi(P_\varepsilon)|_0 = 0$. Combined with Equation C.2:

$$0 = \frac{\partial}{\partial\varepsilon}\Psi(P_\varepsilon)\bigg|_0 = \mathbb{E}_P[\varphi\, S_\eta].$$

This holds for every score S_η of a nuisance submodel; by linearity and L_2-continuity it extends to every element of the closed linear span T_η. Hence $\varphi \perp T_\eta$, i.e., $\varphi \in T_\eta^\perp$. □

Remark: Causal Example

For the ATE under the nonparametric model on $O = (X, T, Y)$, $T = L_2^0(P)$ and T_η is the closed linear span of all scores of submodels along which $\mathbb{E}_P[Y(1) - Y(0)]$ is constant. A standard calculation (Tsiatis 2006, sec. 7.4) identifies $T_\eta^\perp$ as a one-dimensional subspace spanned by the EIF φ^* of Chapter 9, Equation 9.5; this is the formal content of the claim that φ^* is uniquely determined.

C.5 The Canonical Gradient and the Efficiency Bound

Among all influence functions, there is a unique one lying in the full tangent space T, and it is the one with smallest variance.

Theorem: Canonical Gradient

Let Ψ be pathwise differentiable at P with non-empty influence function set $\mathrm{IF}(\Psi, P)$. Then:
(i) Any two influence functions differ by an element of $T^\perp$: for $\varphi_1, \varphi_2 \in \mathrm{IF}(\Psi, P)$, $\varphi_1 - \varphi_2 \in T^\perp$.
(ii) There exists a unique $\varphi^* \in \mathrm{IF}(\Psi, P)$ with $\varphi^* \in T$, namely $\varphi^* = \Pi[\varphi \mid T]$ for any $\varphi \in \mathrm{IF}(\Psi, P)$, where $\Pi[\cdot \mid T]$ denotes orthogonal projection in $L_2^0(P)$ onto T.
(iii) For every $\varphi \in \mathrm{IF}(\Psi, P)$, $\mathbb{E}_P[\varphi(O)^2] \geq \mathbb{E}_P[\varphi^*(O)^2]$, with equality iff $\varphi = \varphi^*$ in $L_2(P)$.

Proof

(i) For $\varphi_1, \varphi_2 \in \mathrm{IF}(\Psi, P)$ and every score S of a regular submodel, $\mathbb{E}_P[(\varphi_1 - \varphi_2)S] = \partial_\varepsilon \Psi|_0 - \partial_\varepsilon \Psi|_0 = 0$. By linearity and L_2-continuity, $\mathbb{E}_P[(\varphi_1 - \varphi_2)g] = 0$ for every $g \in T$, i.e., $\varphi_1 - \varphi_2 \in T^\perp$.
(ii) Fix any $\varphi \in \mathrm{IF}(\Psi, P)$ and set $\varphi^* := \Pi[\varphi \mid T]$, so $\varphi - \varphi^* \in T^\perp$. For any score $S \in T$:

$$\mathbb{E}_P[\varphi^* S] = \mathbb{E}_P[\varphi S] - \mathbb{E}_P[(\varphi - \varphi^*)S] = \mathbb{E}_P[\varphi S] = \frac{\partial}{\partial \varepsilon} \Psi(P_\varepsilon)\Big|_0,$$

so $\varphi^* \in \mathrm{IF}(\Psi, P)$. Uniqueness: if $\tilde{\varphi}^* \in \mathrm{IF}(\Psi, P) \cap T$, then by (i), $\varphi^* - \tilde{\varphi}^* \in T^\perp \cap T = \{0\}$.
(iii) Write $\varphi = \varphi^* + (\varphi - \varphi^*)$ with the two summands orthogonal in $L_2(P)$ (since $\varphi^* \in T$ and $\varphi - \varphi^* \in T^\perp$). The Pythagorean identity gives:

$$\mathbb{E}_P[\varphi^2] = \mathbb{E}_P[(\varphi^*)^2] + \mathbb{E}_P[(\varphi - \varphi^*)^2] \geq \mathbb{E}_P[(\varphi^*)^2],$$

with equality iff $\varphi = \varphi^*$. □

Definition: Efficient Influence Function and Efficiency Bound

The unique element $\varphi^* \in \mathrm{IF}(\Psi, P) \cap T$ is called the **efficient influence function** (EIF), or **canonical gradient**, of Ψ at P. Its variance:

$$V^*(\Psi, P) = \mathbb{E}_P[\varphi^*(O)^2]$$

is the **semiparametric efficiency bound** for estimating $\Psi(P)$ in the model $\mathcal{P}$.

Remark: Role of the Nuisance Tangent Space

Since every influence function already lies in $T_\eta^\perp$, the EIF can equivalently be described as the unique influence function in $T \cap T_\eta^\perp$. This intersection contains exactly the directions in T that are orthogonal to nuisance perturbations — directions along which ψ genuinely moves. In practice, the EIF is often constructed by taking a naive influence function (such as the IPW form) and subtracting its projection onto T_η to remove nuisance components; this is the projection view exploited in Chapter 10.

Theorem: Asymptotic Efficiency Bound

Let $\hat{\psi}_n$ be a regular asymptotically linear estimator of $\Psi(P)$ in the model $\mathcal{P}$. Then its asymptotic variance satisfies:

$$\mathrm{AVar}(\sqrt{n}(\hat{\psi}_n - \psi)) \geq V^*(\Psi, P),$$

and the bound is achieved if and only if the influence function of $\hat{\psi}_n$ is φ^* in $L_2(P)$.

This theorem is a corollary of the Hájek–Le Cam convolution theorem, which states the stronger result that $\sqrt{n}(\hat{\psi}_n - \psi) \rightsquigarrow Z + W$, where $Z \sim N(0, V^*)$ and W is independent of Z. A precise statement requires the local asymptotic normality framework of Vaart (1998, secs. 25.3–25.6).

Remark: Why the EIF Appears in Concrete Formulas

The Canonical Gradient Theorem explains why specific objects such as the AIPW influence function in Chapter 9 look “constructed” rather than guessed. Given a naive influence function φ (e.g., the IPW form $TY/\pi - (1-T)Y/(1-\pi) - \psi$), it may fail to lie in T; its projection $\varphi^* = \Pi[\varphi \mid T]$ is the EIF. In Chapter 10 this projection is computed explicitly for the ATE under ignorability, producing the AIPW formula. The construction in Chapter 10 should therefore be read as an instance of the Canonical Gradient Theorem applied to a specific causal functional.

Bibliographic Notes

The modern formulation of pathwise differentiability and tangent spaces is developed in the monographs of Bickel et al. (1993) and Vaart (1998, chap. 25); the latter is the standard reference for the convolution theorem and local asymptotic normality underlying the Asymptotic Efficiency Bound theorem. Tsiatis (2006) gives a treatment oriented specifically toward missing data and causal inference, and is a natural companion to the material developed here.